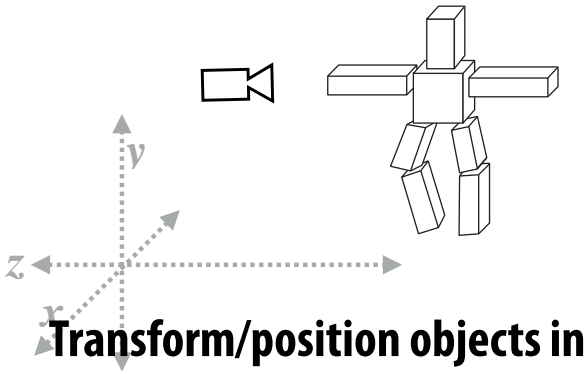
Lecture 6:

# Texture Mapping

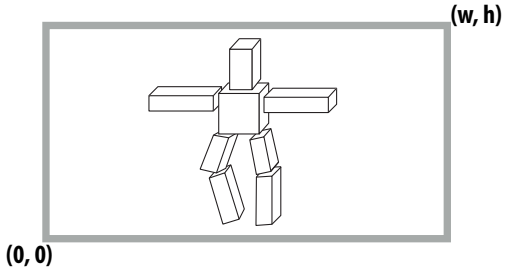
Computer Graphics 2026

Fuzhou University - Computer Science

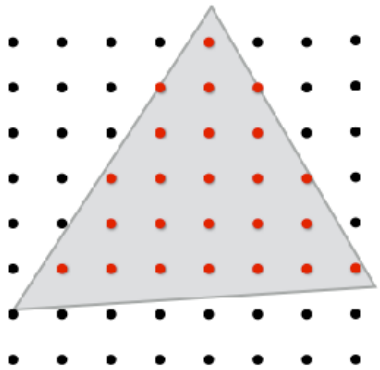
# What We've Covered So Far



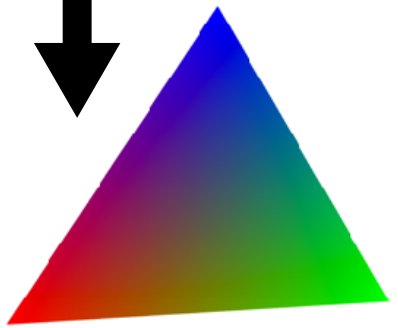

Transform/position objects in the world




Project objects onto the screen



Sample triangle coverage



Interpolate triangle attributes at covered samples



Sample texture maps / evaluate shaders



Combine samples into final image (depth, alpha, …)

Rasterization Pipeline

# Today's Topics

What is Texture Mapping?

Applying Textures

Texture Filtering

# Drawing Surface with Complex Details

# Texture Coordinate Mappings

# Think Chocolate Wrappers

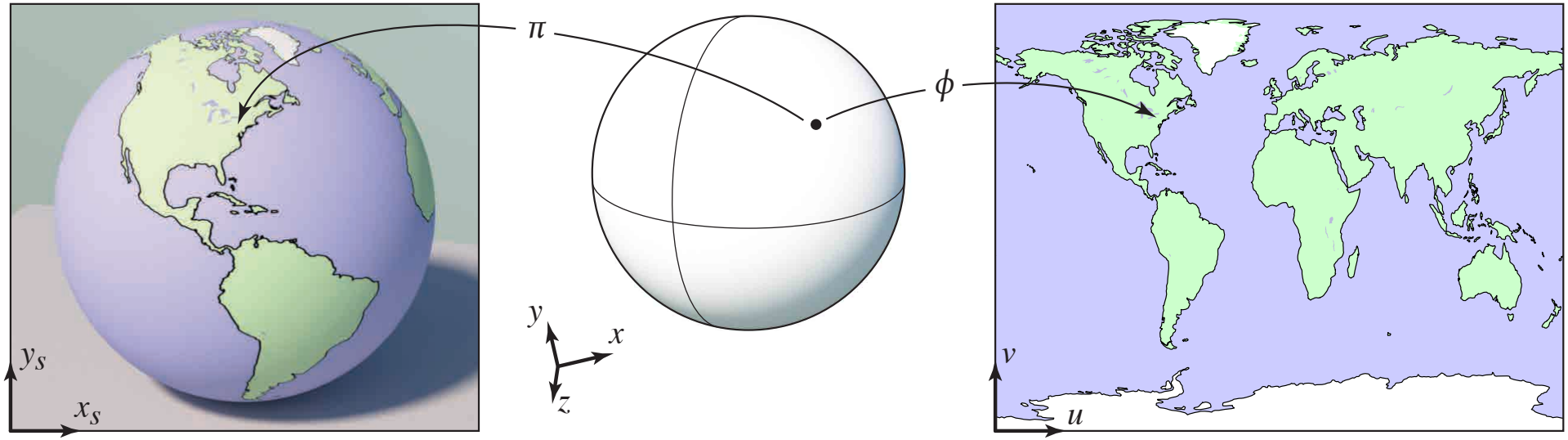
Texture image

# Surfaces are 2D

三维物体的表面是二维的（可撕扯、摊开成二维图像）

Surface lives in 3D world space

Every 3D surface point also has a place where it goes in the 2D image and in the 2D texture.

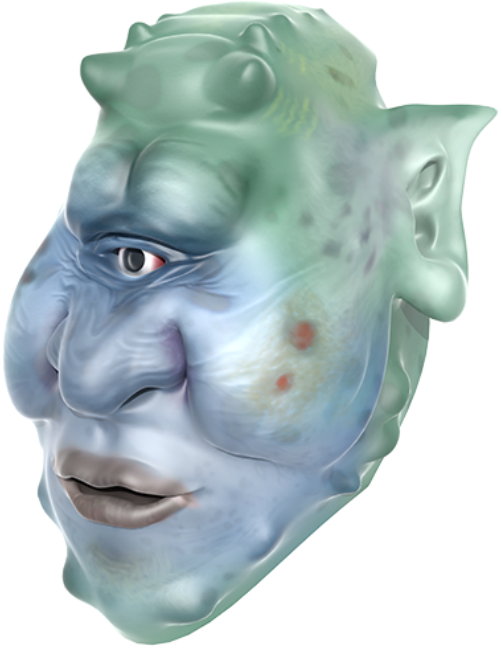



Screen space　　World space　　Texture space

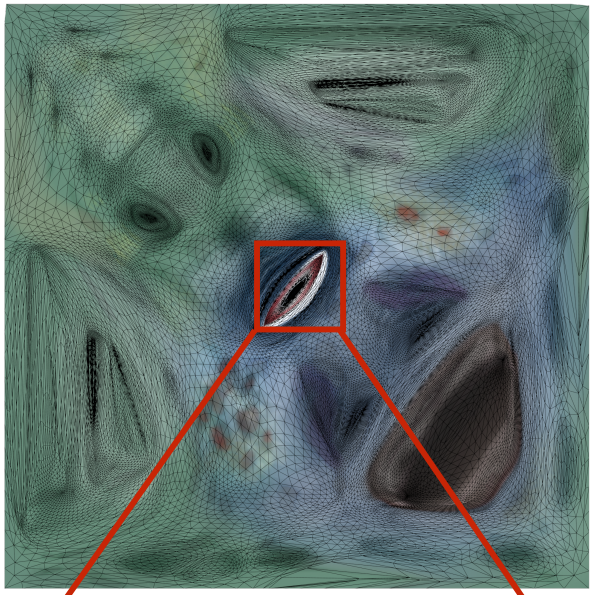
# Texture Applied to Surface

**Rendering without texture**

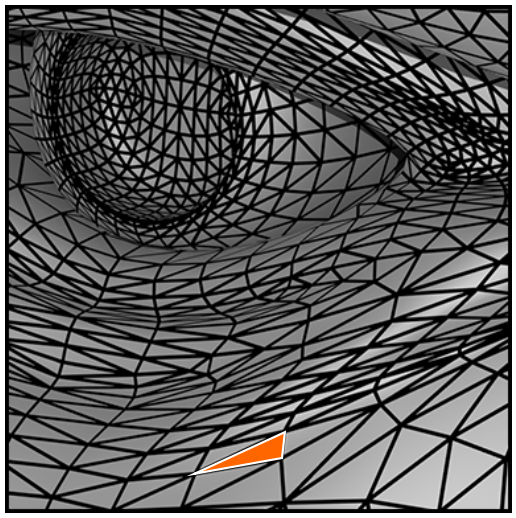
**Rendering with texture**

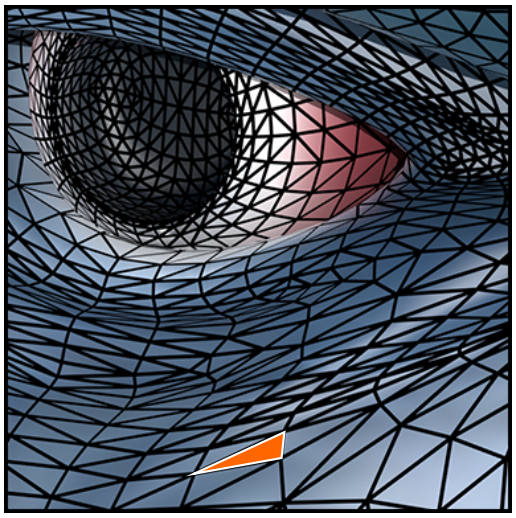


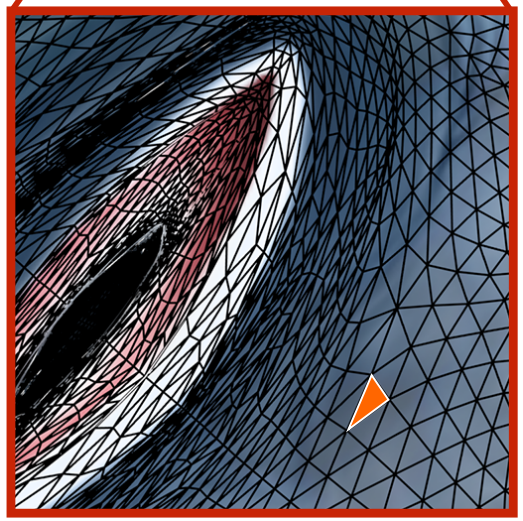
**Texture image**



Zoom





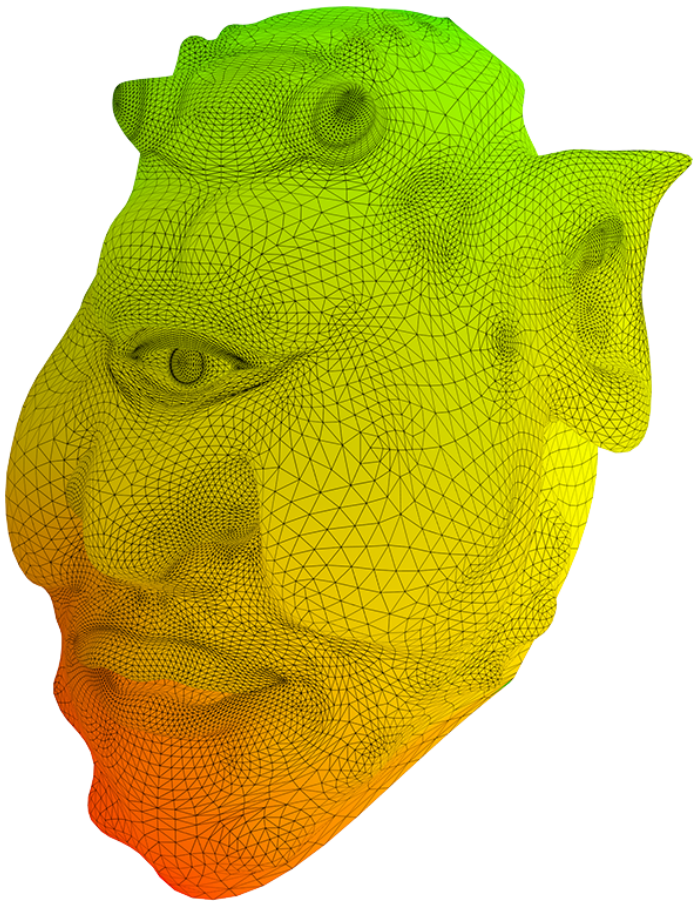


**Each triangle "copies" a piece of the texture image back to the surface.**

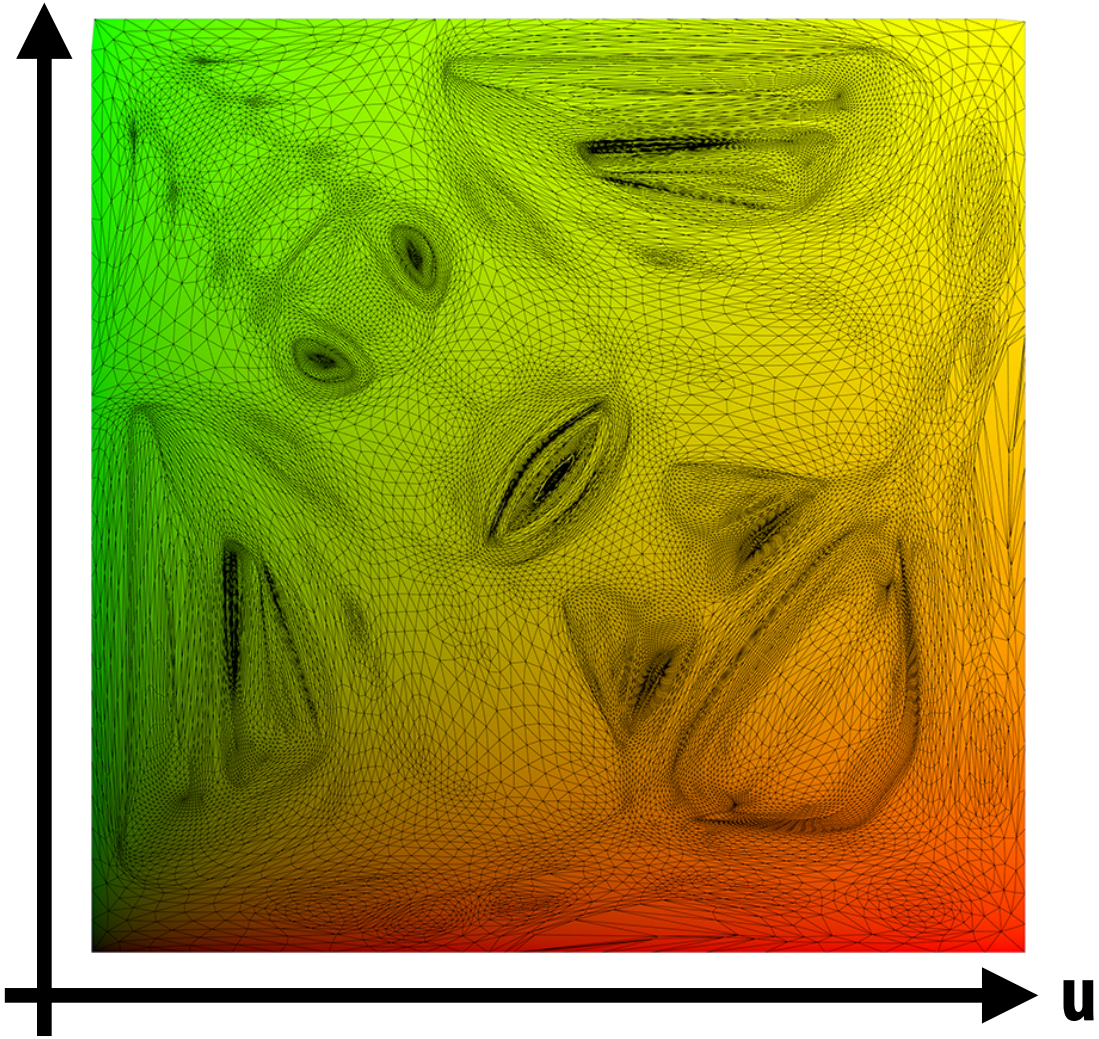
# Visualization of Texture Coordinates

## Each surface point is assigned a texture coordinate (u,v)

**Visualization of texture coordinates**



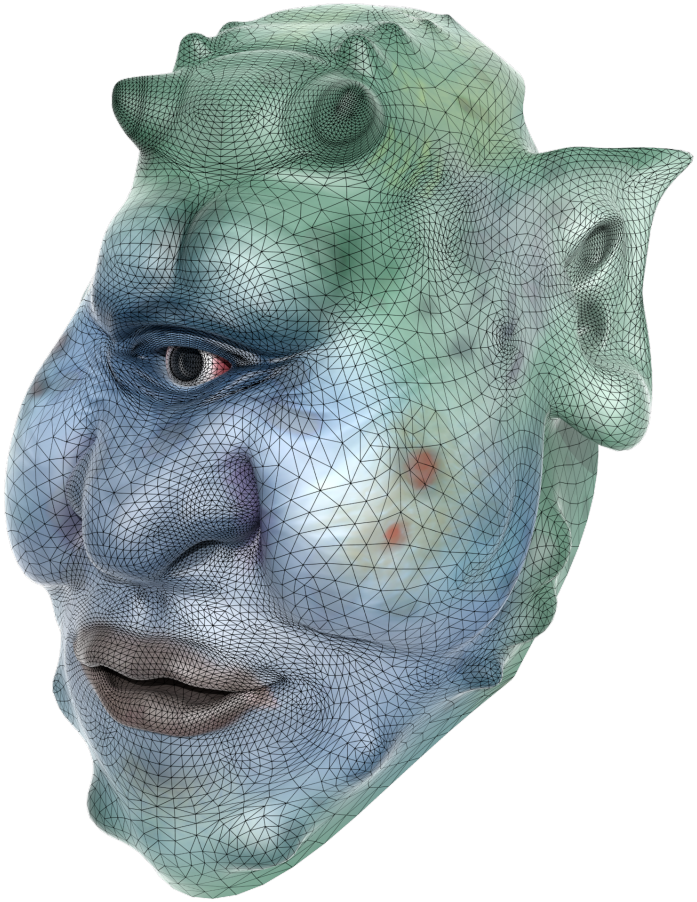
**Triangle vertices in texture space**



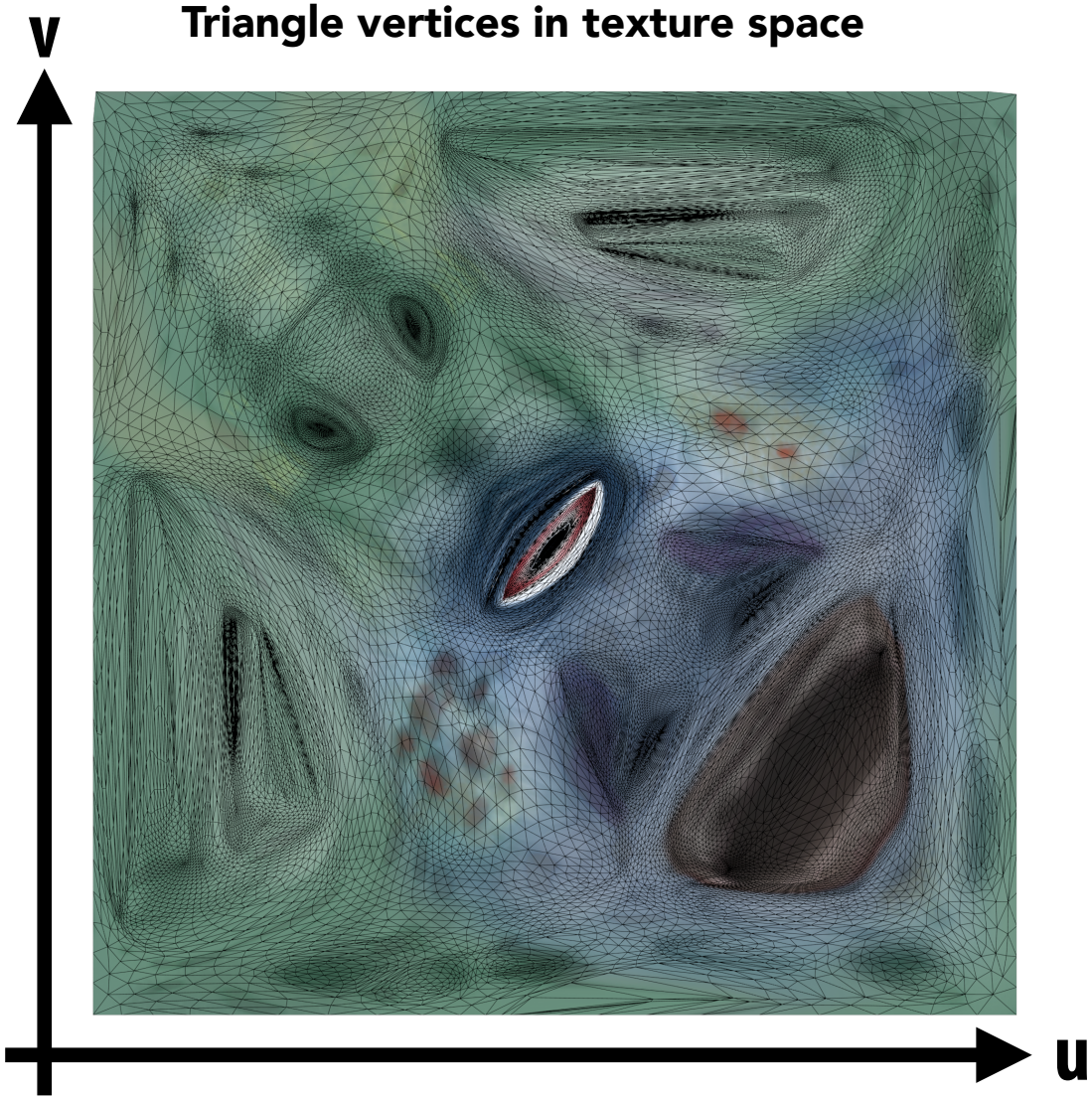
# Image Texture Applied to Surface

## Each surface point is assigned a texture coordinate (u,v)

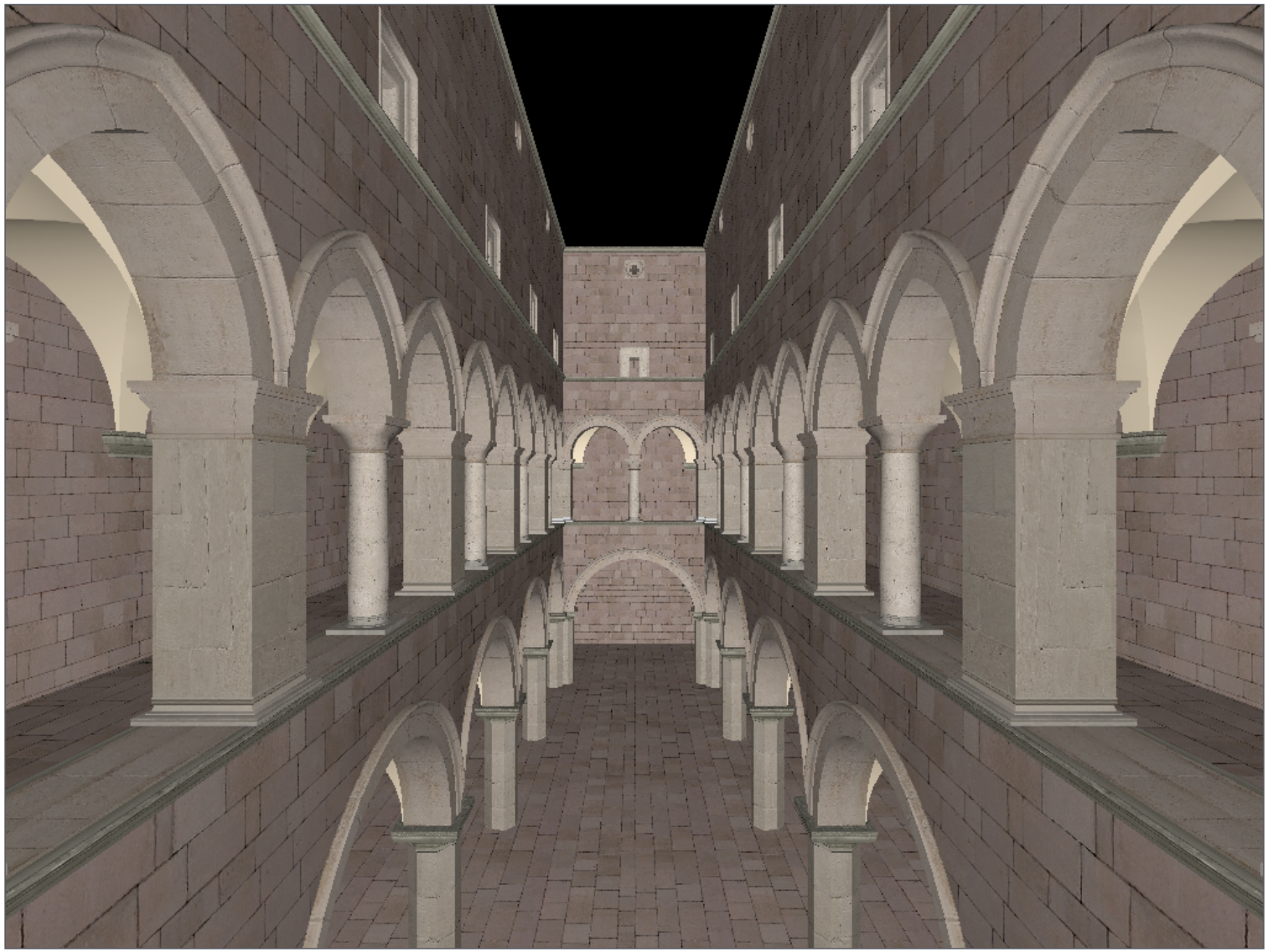
**Rendered result**



**Triangle vertices in texture space**

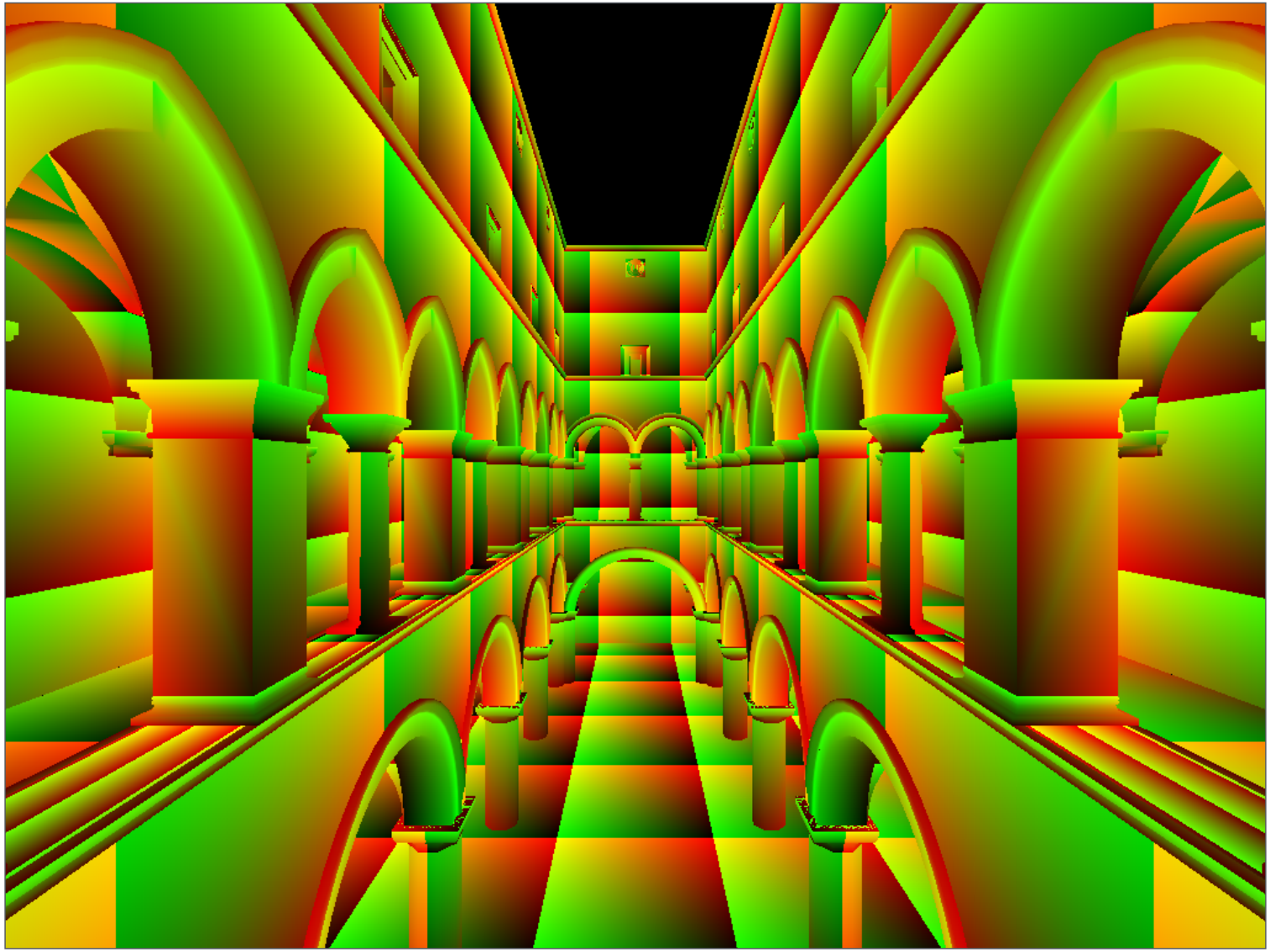


# Sponza Palace Model



Textures applied to surfaces

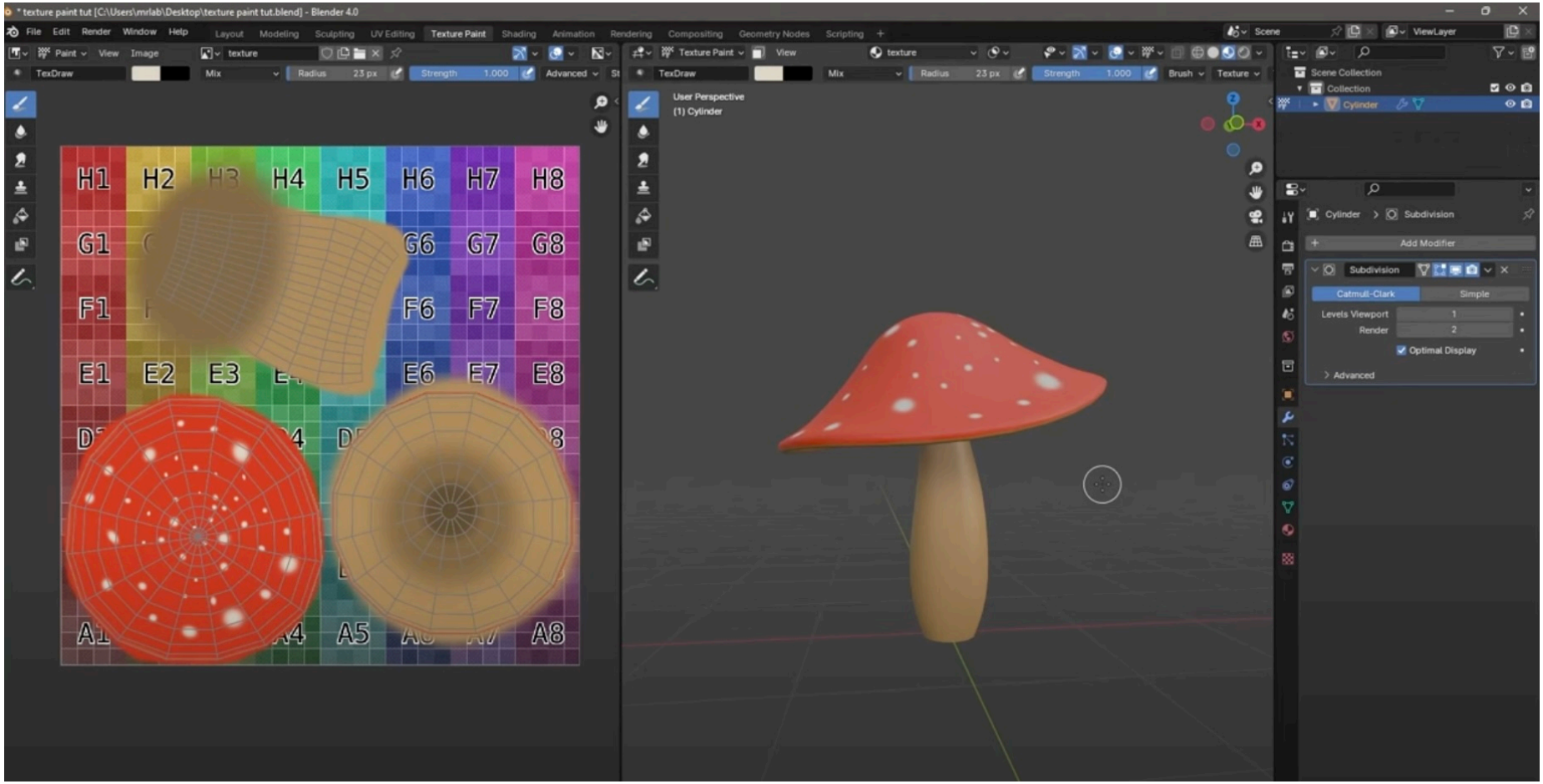
# Sponza Palace Model



Visualization of texture coordinates

# UV Unwrapping

Before a texture image is mapped onto the surface of a 3D object, UV unwrapping is required to flatten the surface into 2D space, and generate UV coordinates for each vertex of the object.

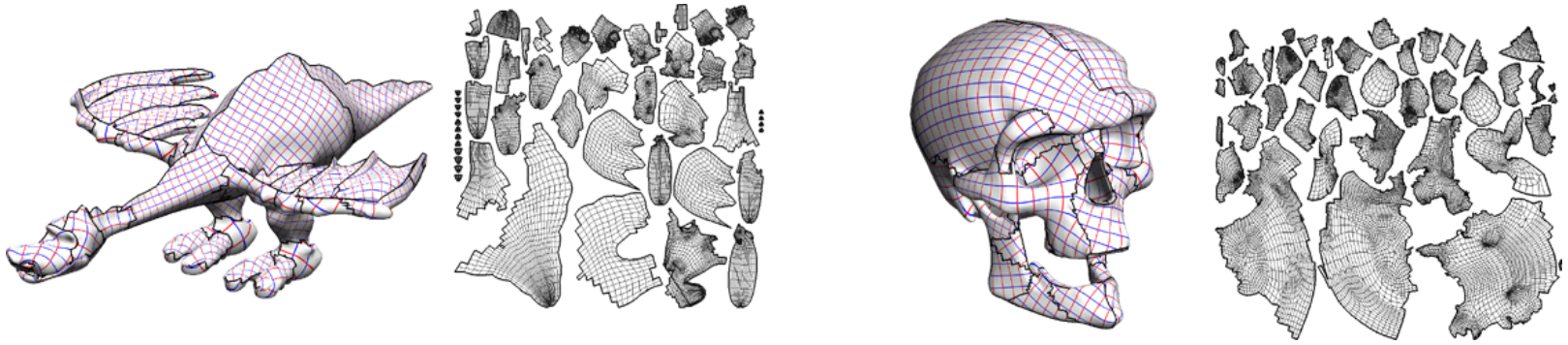



https://www.youtube.com/watch?v=-sPGZprp5JU

# Creating Good Surface Coordinates is Hard

## Finding cuts

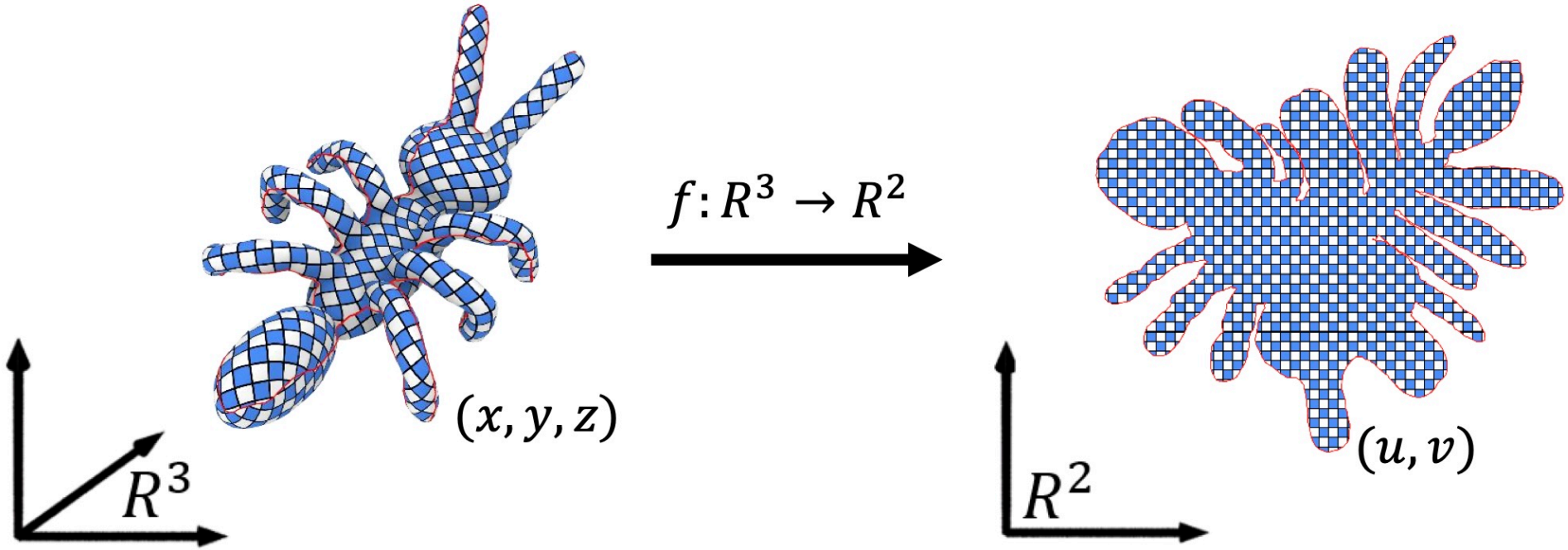
## Texture atlases



**Levy et al: Least Squares Conformal Maps**
**for Automatic Texture Atlas Generation, SIGGRAPH, 2002**

# 曲面参数化

参考GAMES301课程



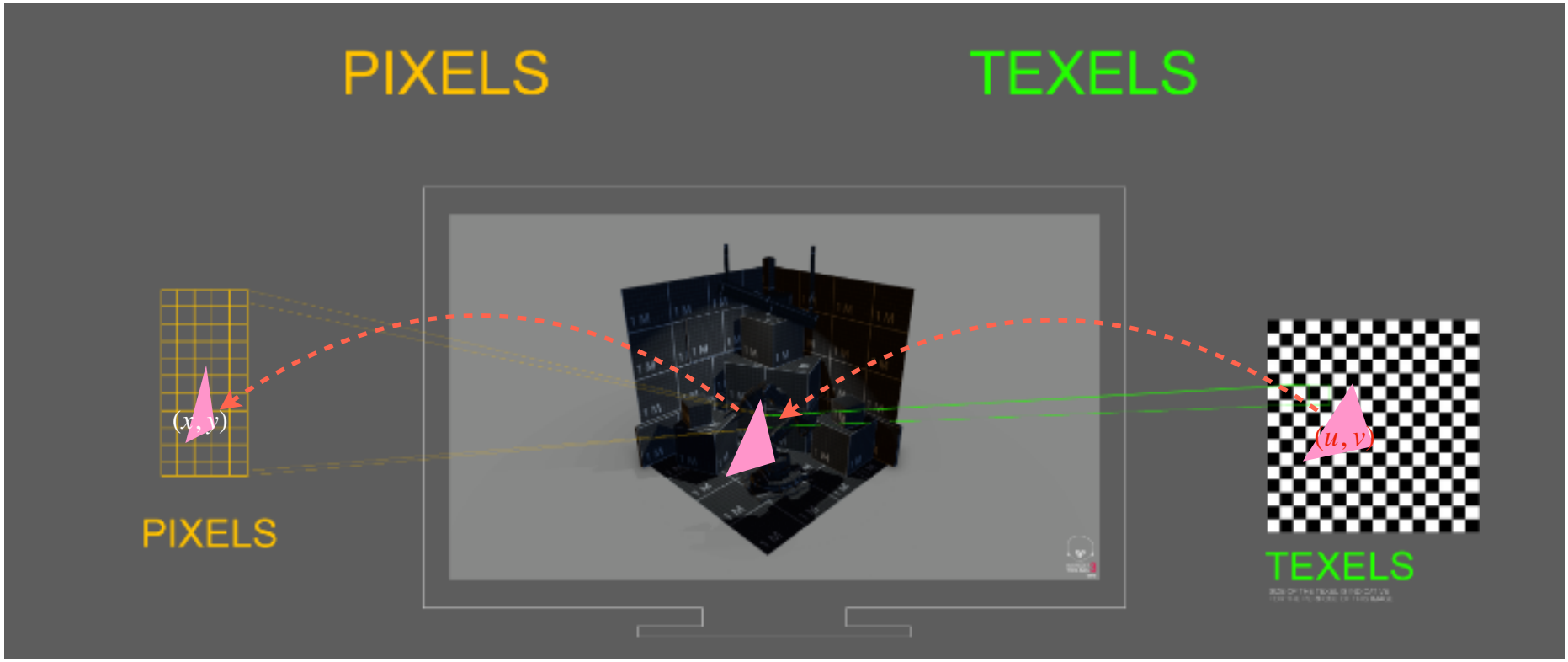
# Today's Topics

What is Texture Mapping?

Applying Textures

Texture Filtering

# Simple Texture Mapping Operation



# Simple Texture Mapping Operation

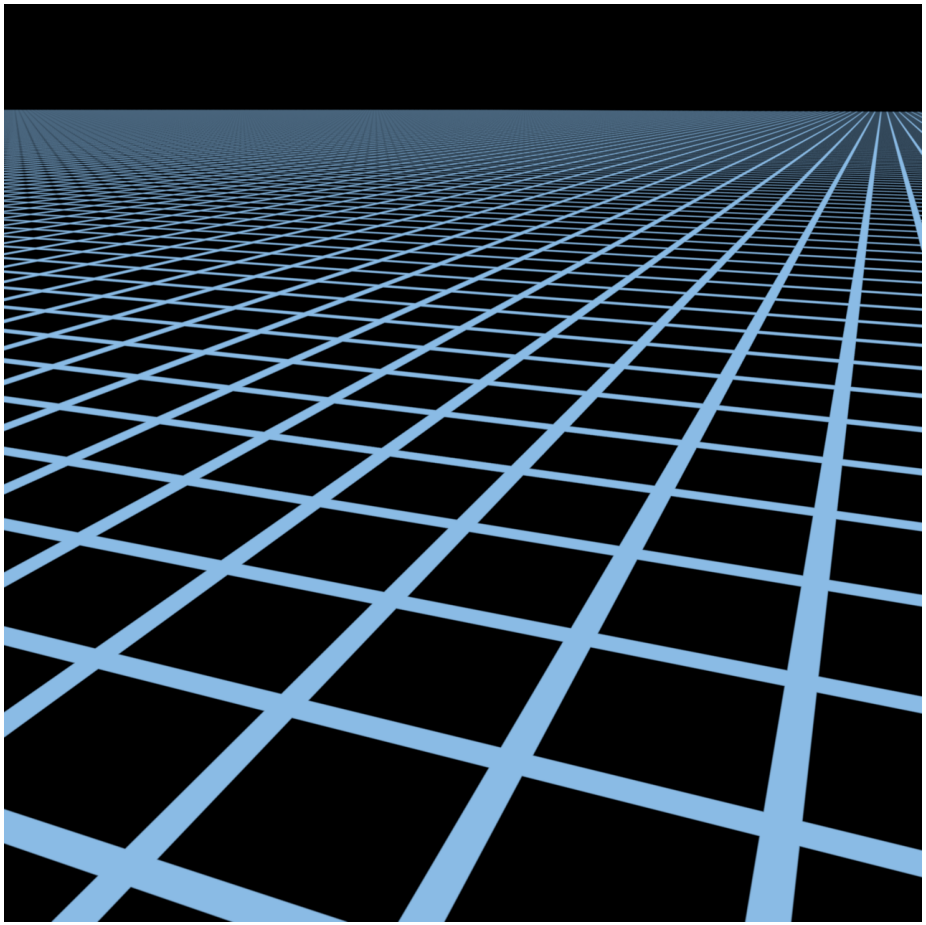
```
for each rasterized screen sample (x,y):
    (u,v) = evaluate texcoord value at (x,y)
    float3 texcolor = texture.sample(u,v);
    set sample's color to texcolor;
```

Interpolation using barycentric coordinates!

Usually the diffuse coefficient $k_d$

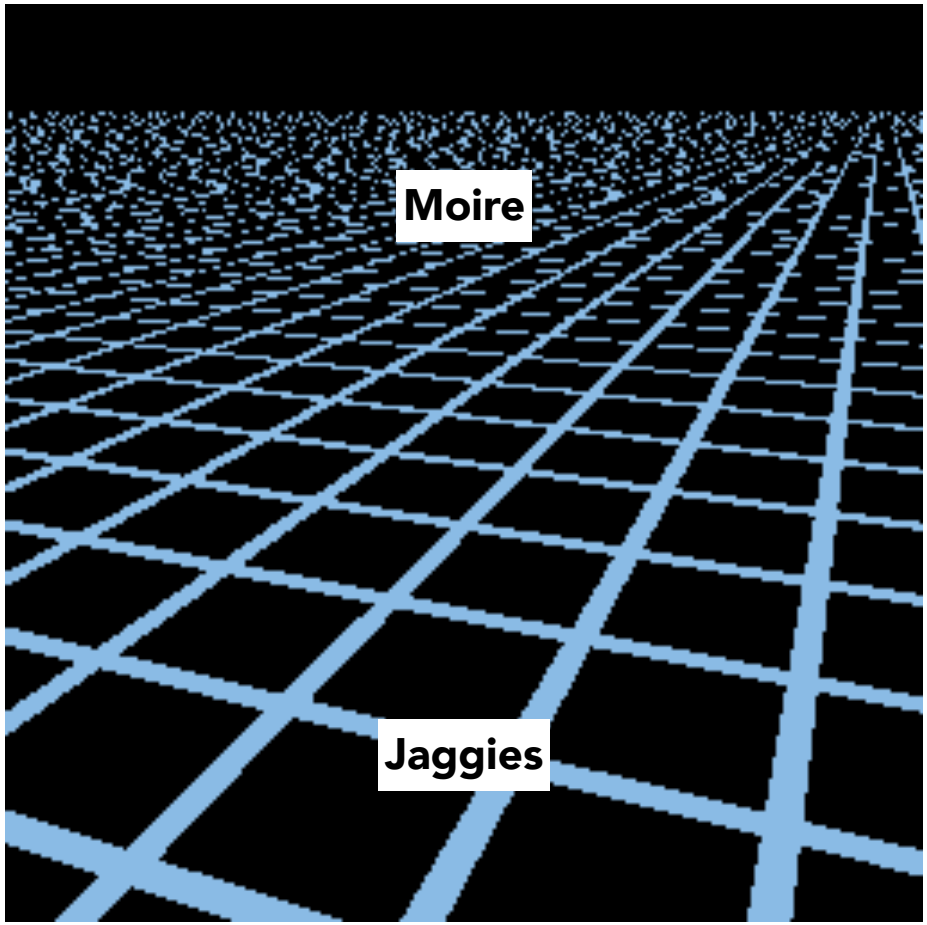
(recall the Blinn-Phong reflectance model)

# Point Sampling Textures



High-res reference

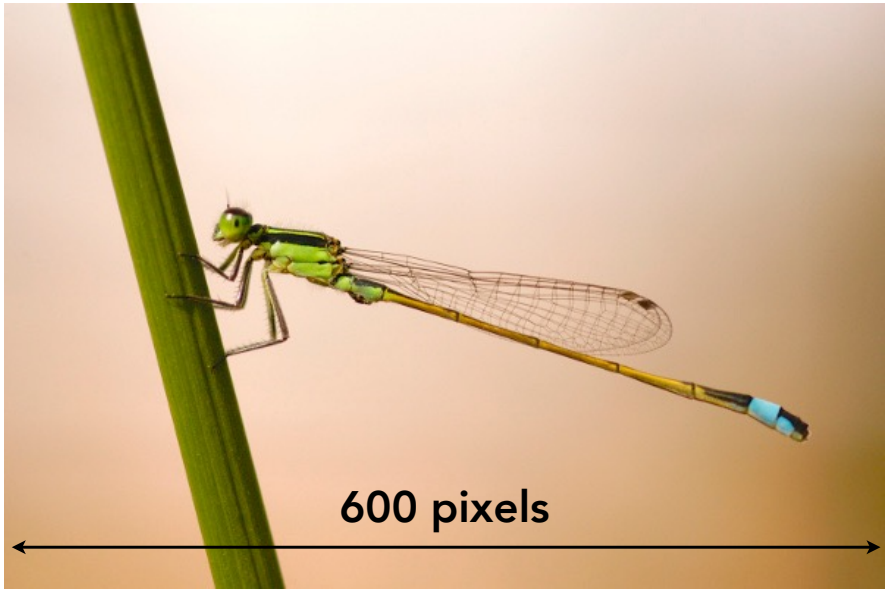
**Source image: 1280x1280 pixels**

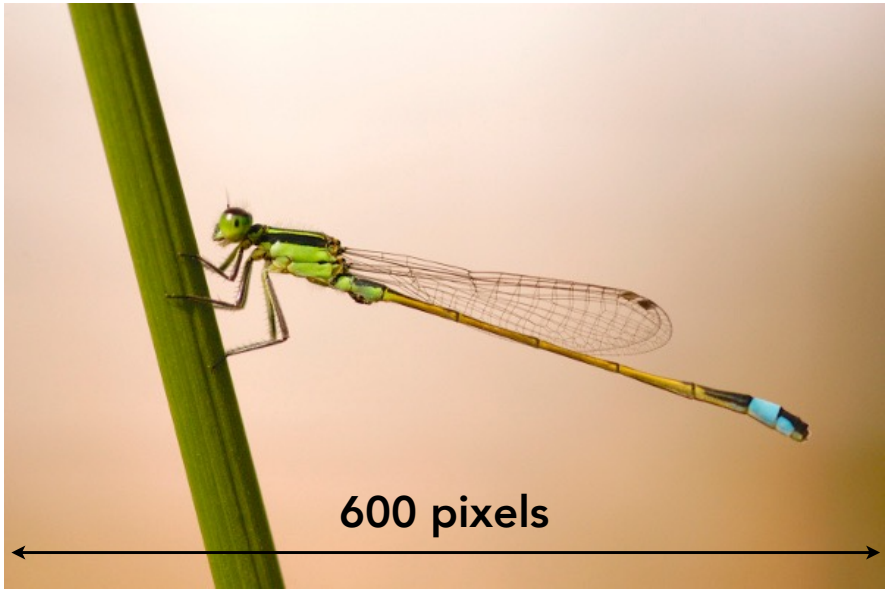


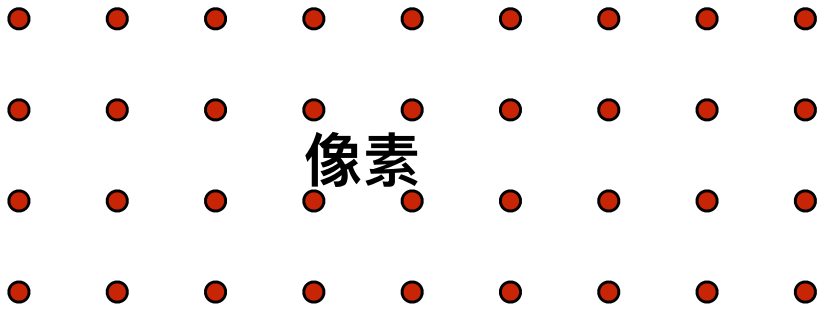

Point sampling

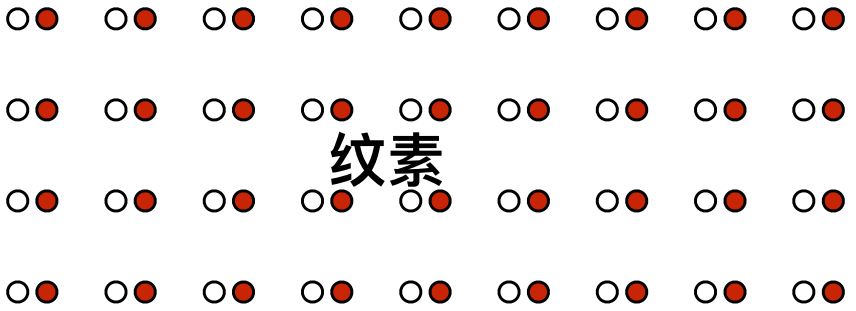
**256x256 pixels**

# Sampling Rate on Screen vs Texture



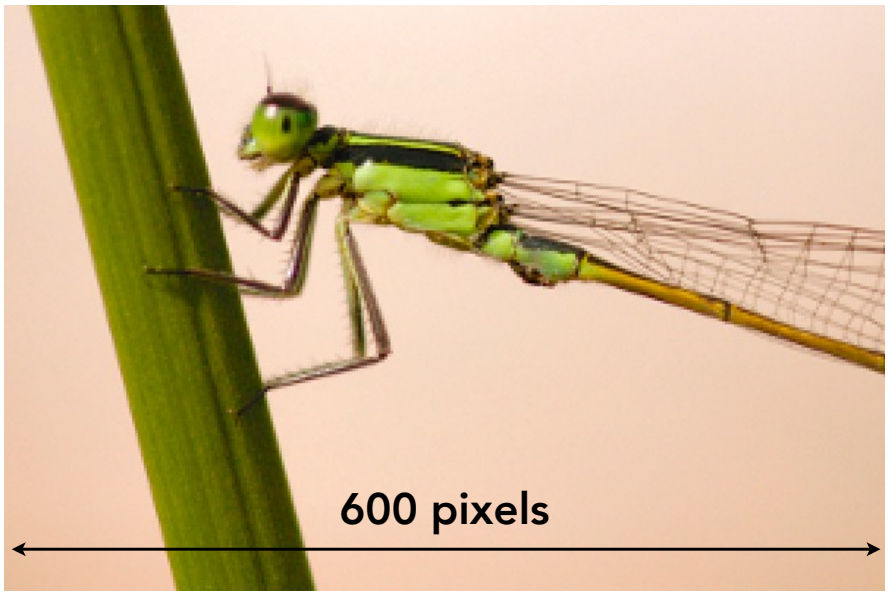



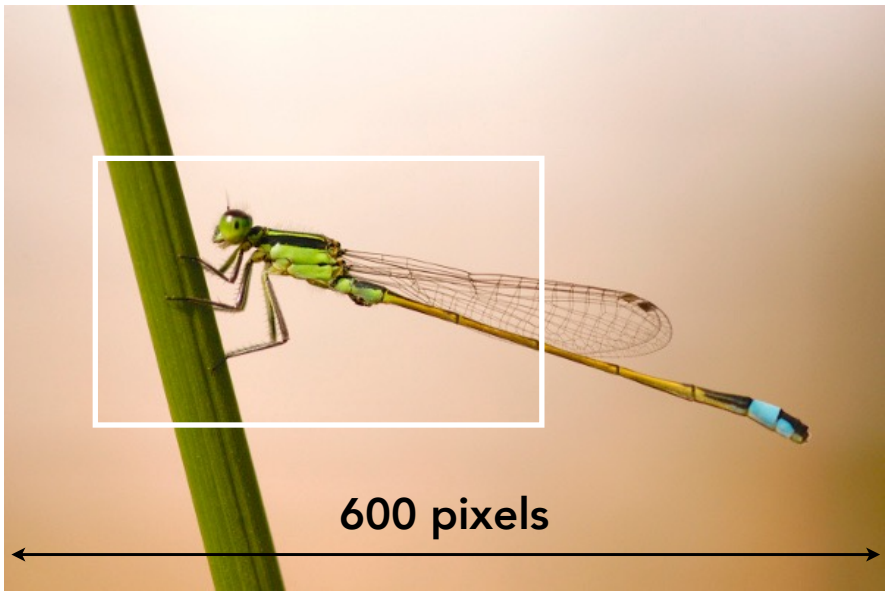

Screen space (x,y)

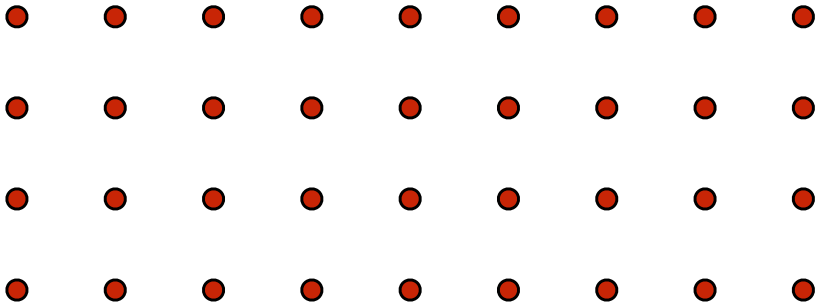
Texture space (u,v)

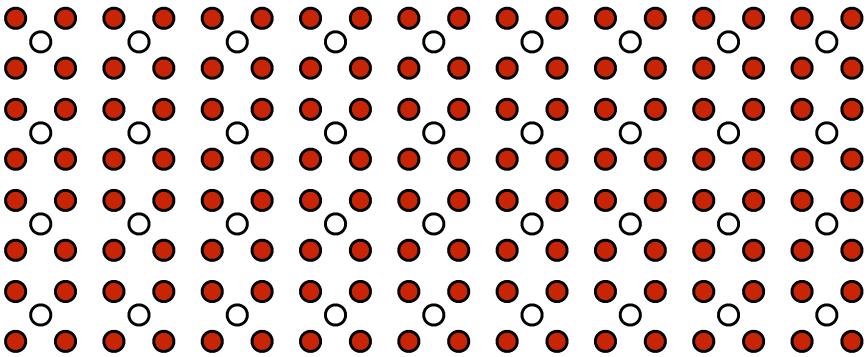
1:1 mapping

# Sampling Rate on Screen vs Texture



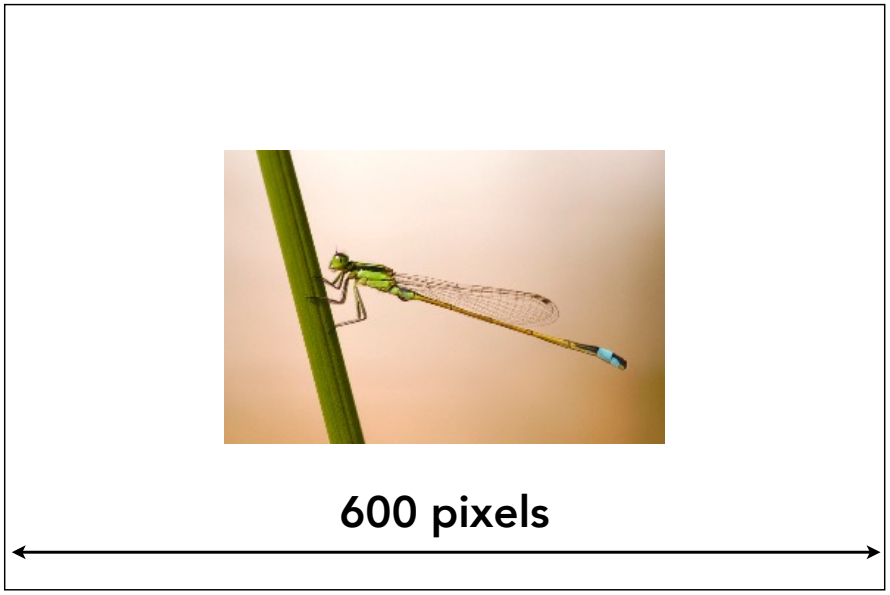


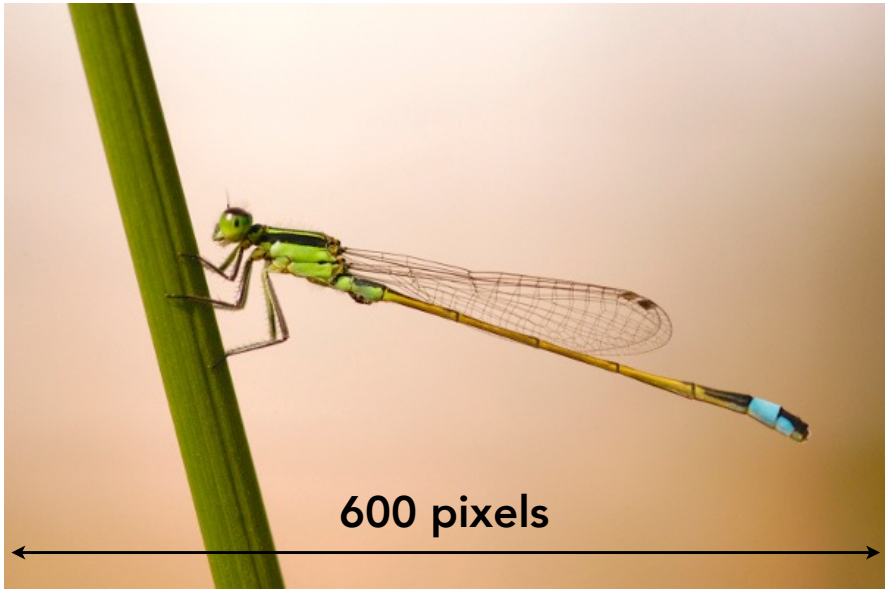
Screen space (x,y)

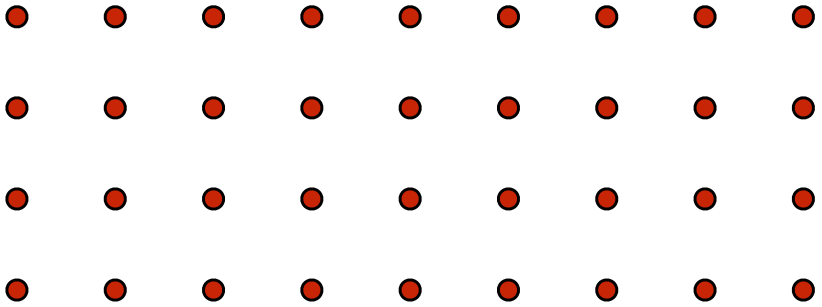
Texture space (u,v)

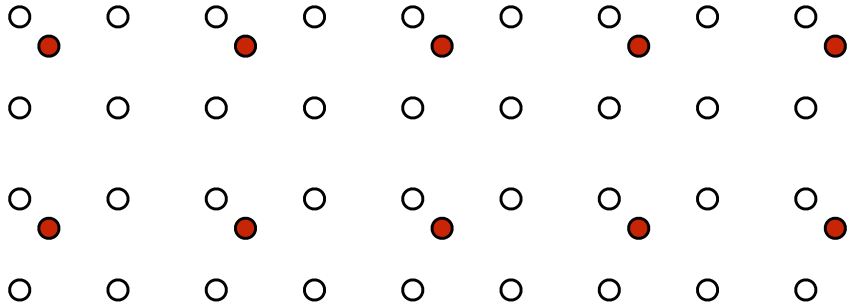
Magnified

# Sampling Rate on Screen vs Texture





Screen space (x,y)

Texture space (u,v)

"Minified"

# Screen Pixel Area vs Texel Area

At optimal viewing size:

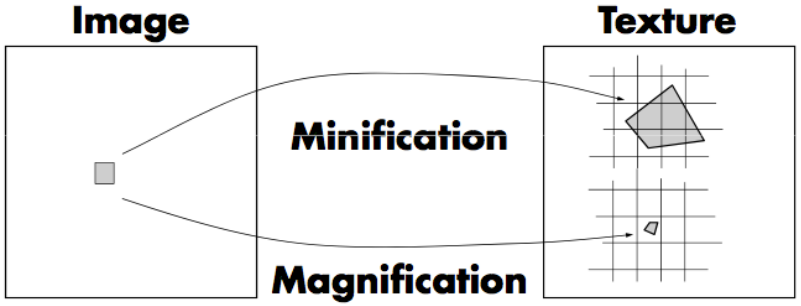
- 1:1 mapping between pixel sampling rate and texel sampling rate
- Dependent on texture resolution!  e.g. 512x512

When larger (magnification)

- Multiple pixel samples per texel sample

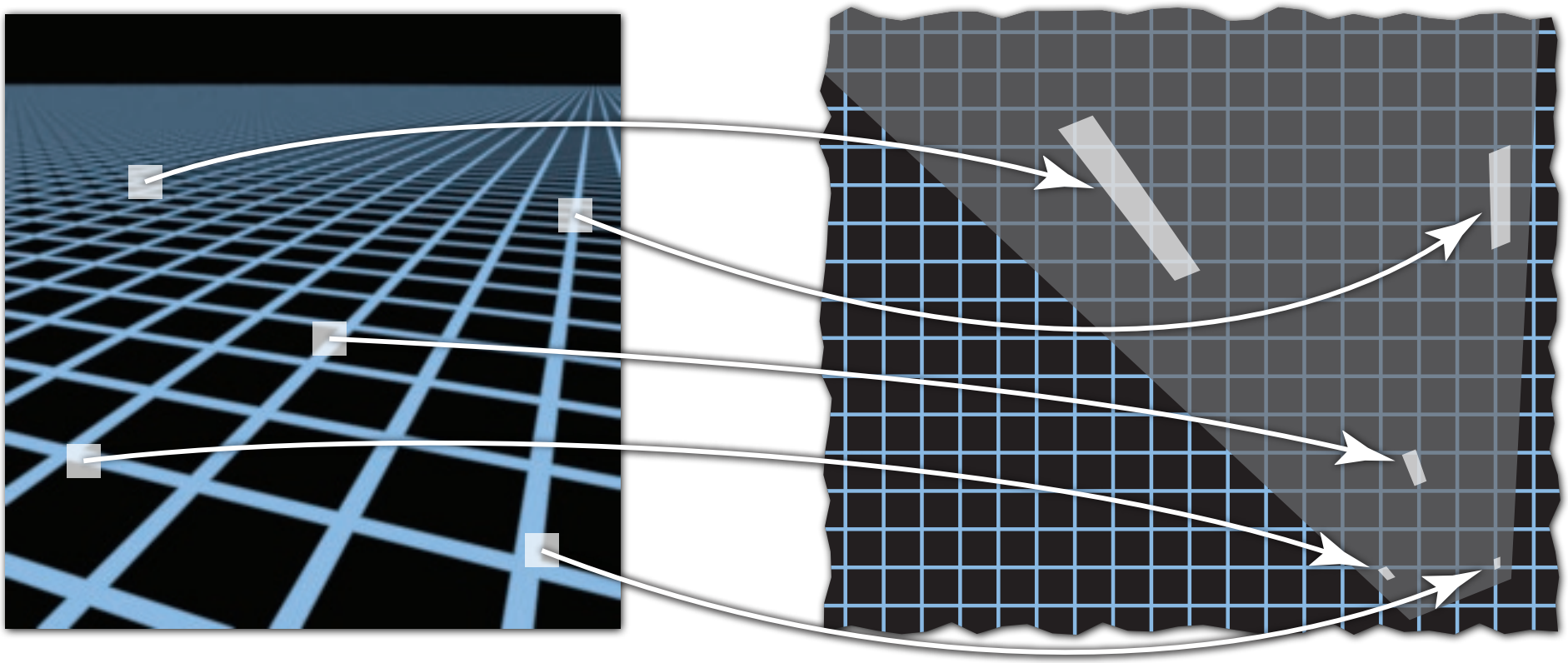
When smaller (minification)

- One pixel sample per multiple texel samples



# Screen Pixel Footprint in Texture

Pixel Footprint：一个像素在真实场景、纹理空间或 3D 空间中所覆盖的实际区域大小



Screen space

Texture space

NB: texture sampling pattern not rectilinear or isotropic

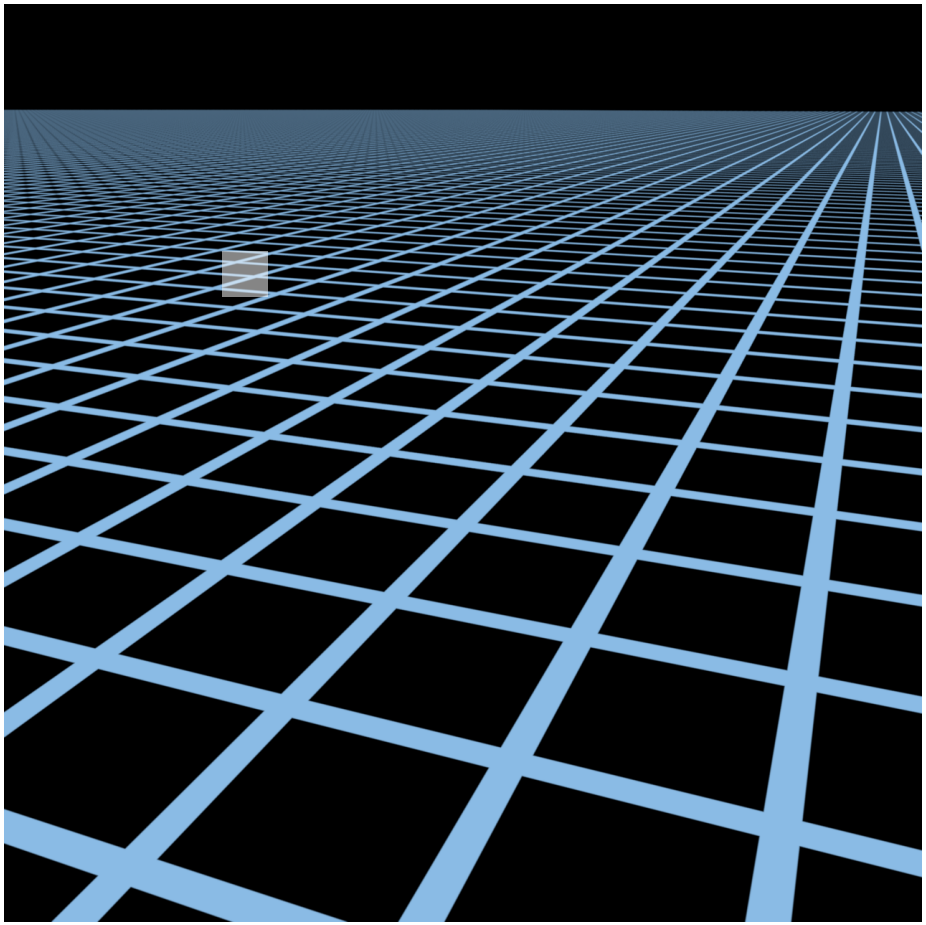
# Today's Topics

What is Texture Mapping?

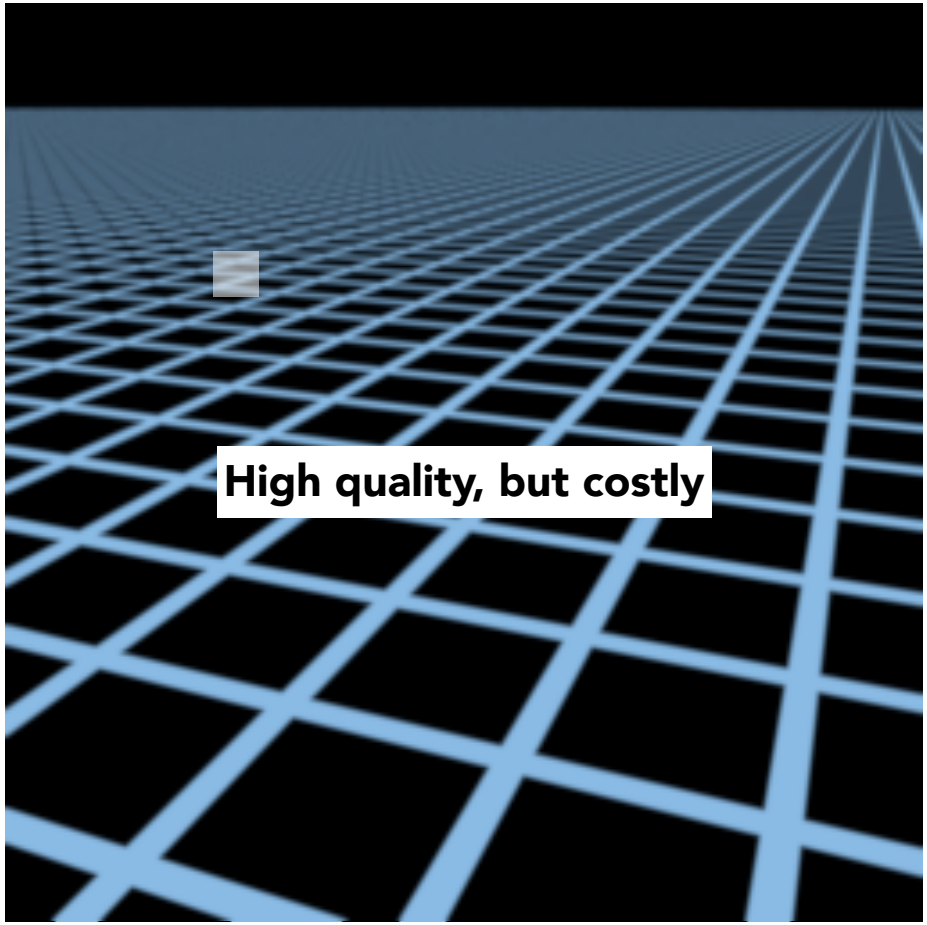
Applying Textures

Texture Filtering

# Will Supersampling Antialias?



High-res reference




512x supersampling

# Texture Antialiasing

Will supersampling work?

- Yes, high quality, but costly
- When highly minified, many texels in pixel footprint

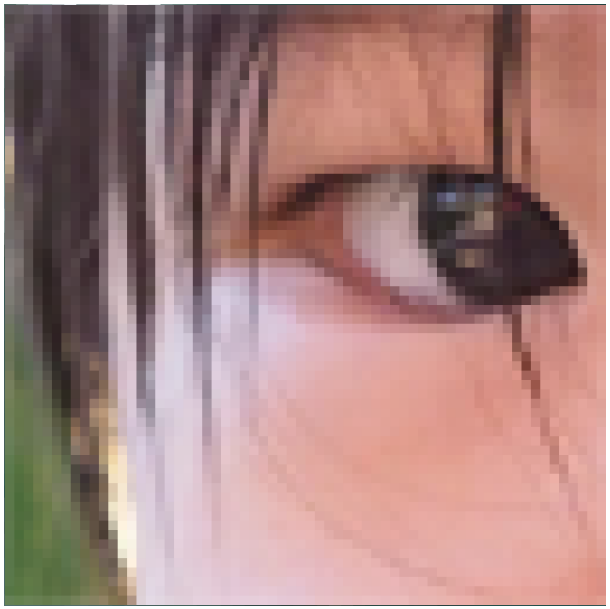
Goal: efficient texture antialiasing

- Want antialiasing with one/few texels per pixel
- How? Antialiasing = filtering before sampling!

# Texture Magnification - Easy Case

(Generally don't want this — insufficient resolution)

This is image interpolation (will see kernel function)

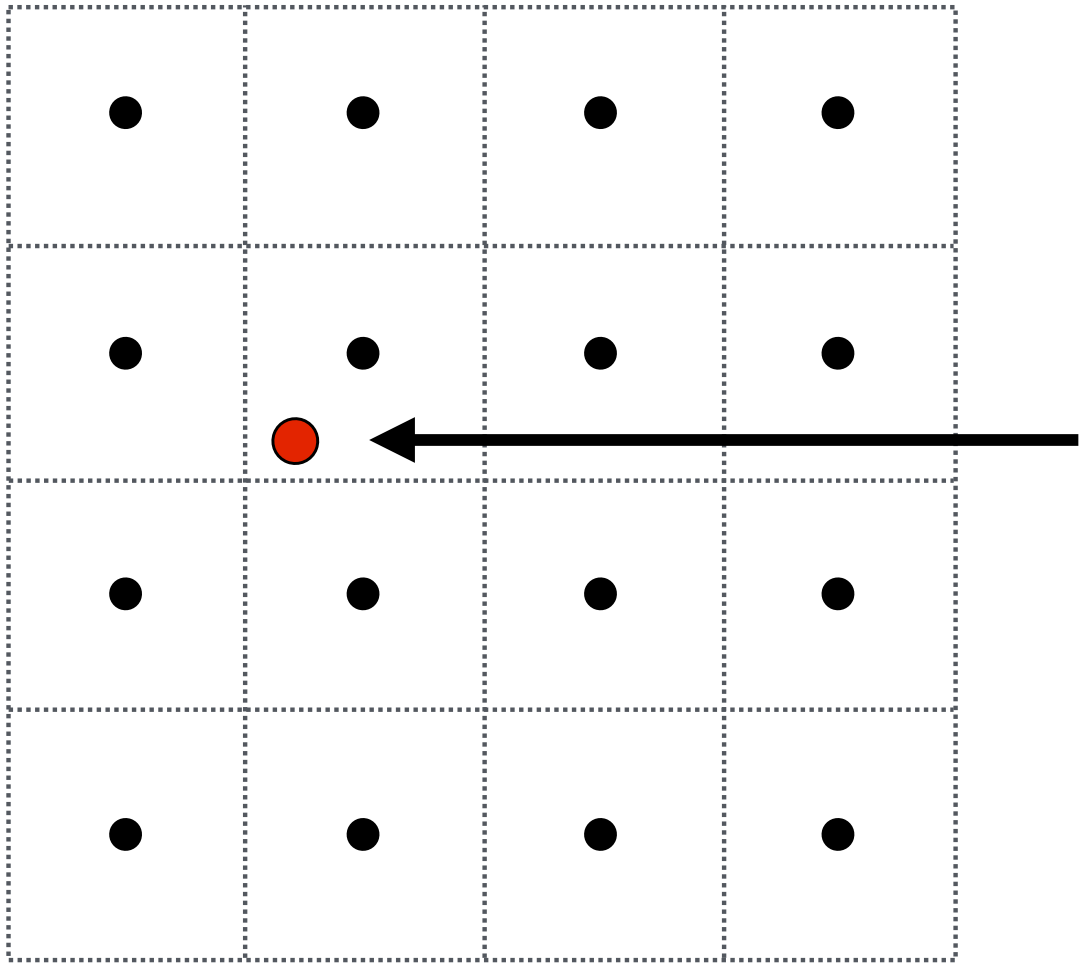


Nearest

Bilinear

Bicubic

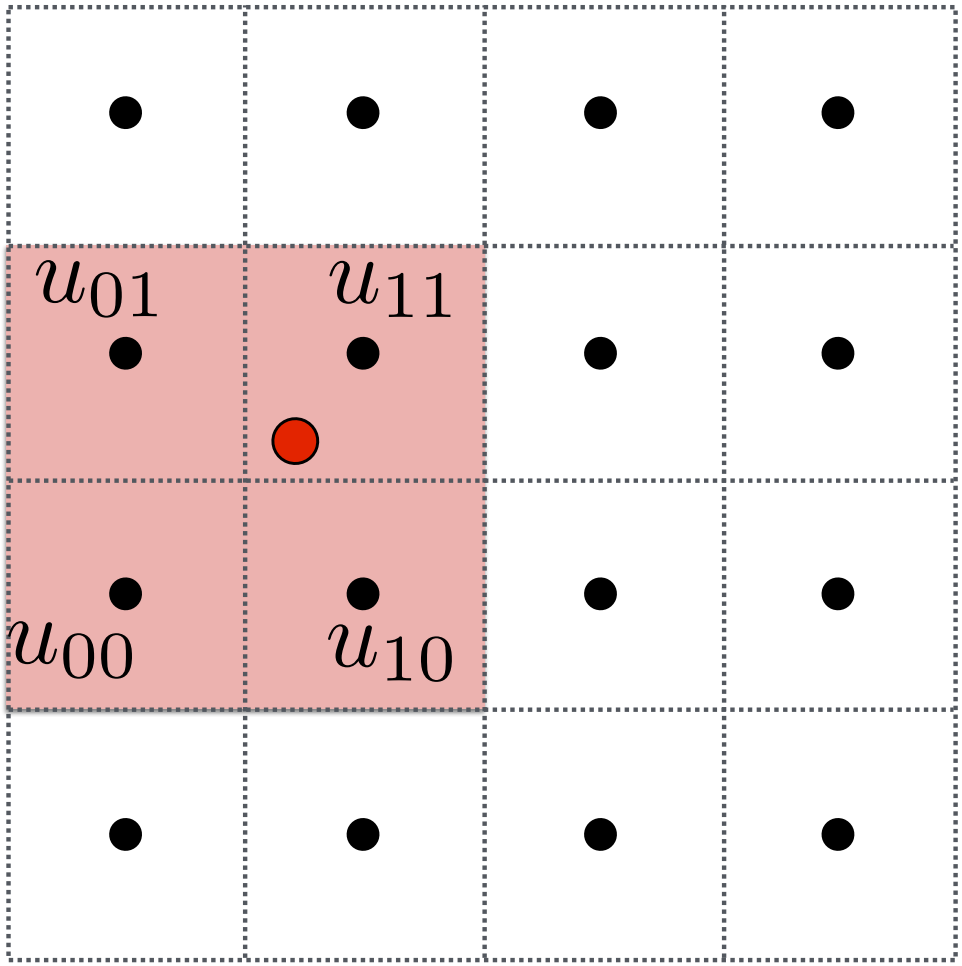
# Bilinear Filtering



Want to sample texture value f(u,v) at red point

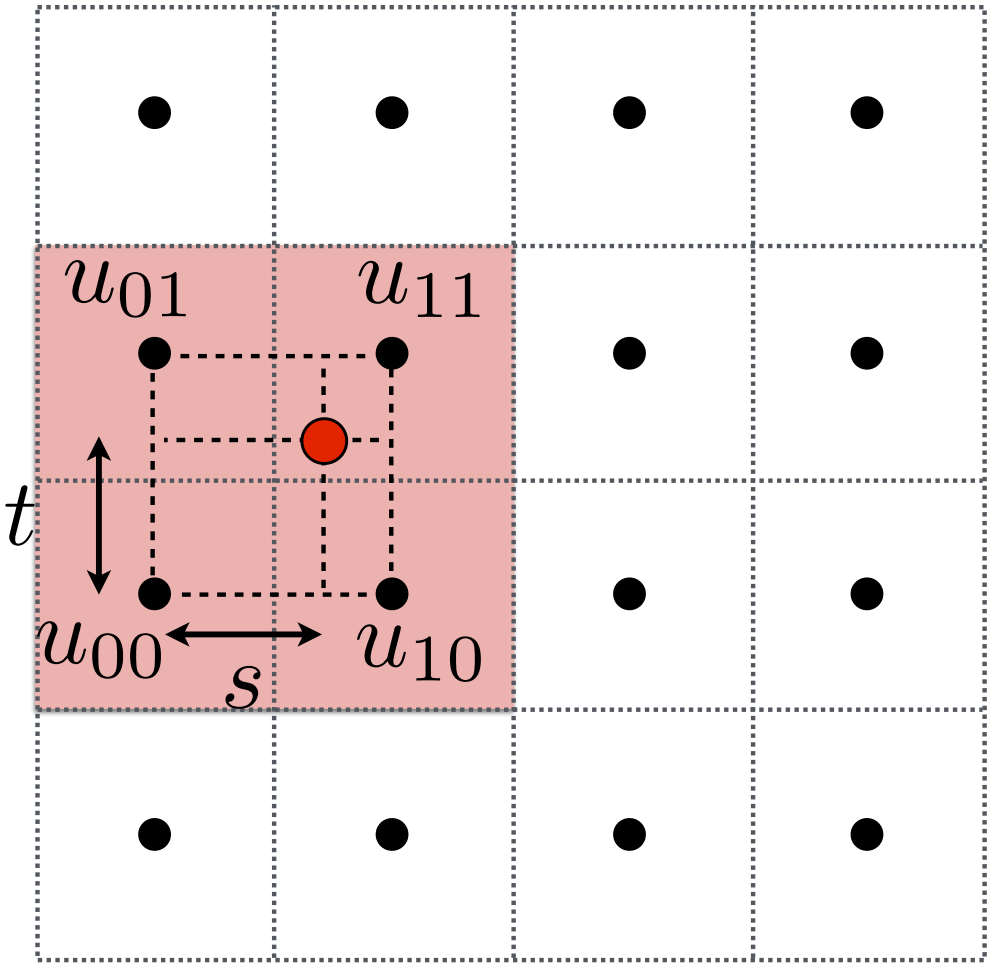
Black points indicate texture sample locations

# Bilinear Filtering



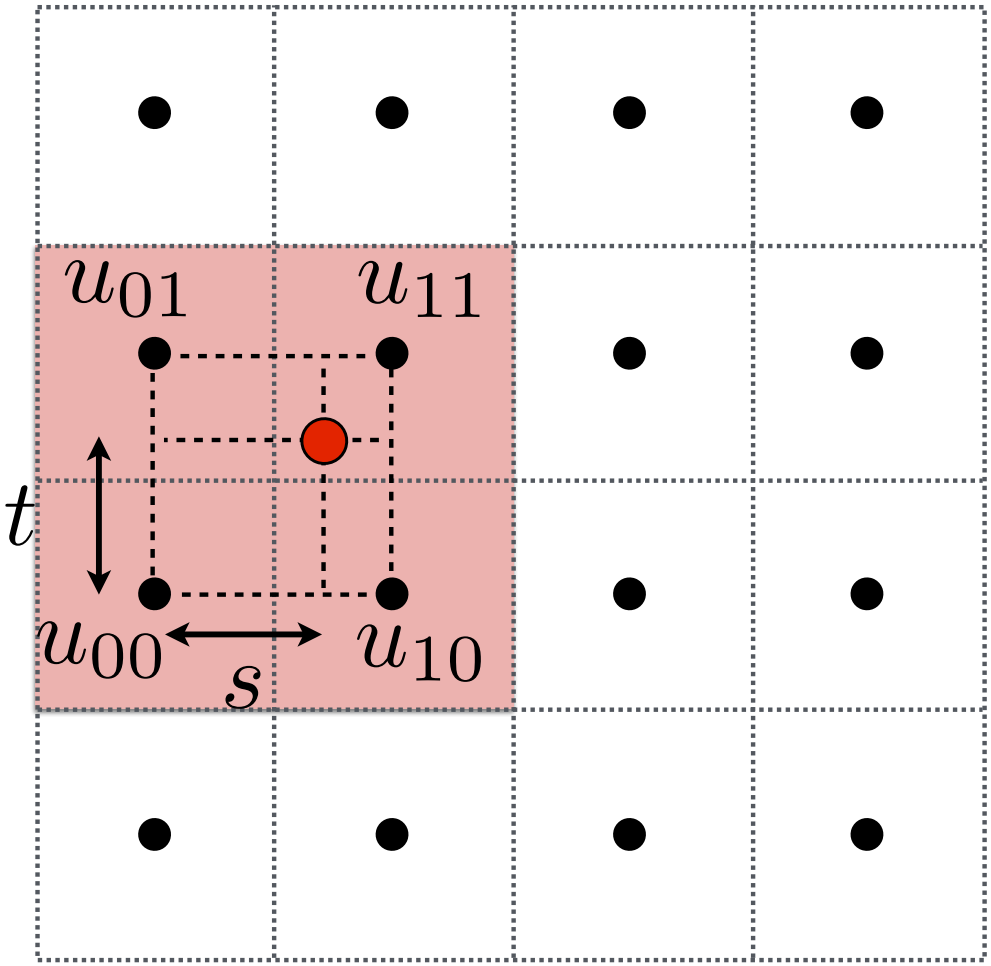

Take 4 nearest sample locations, with texture values as labeled.

# Bilinear Filtering




And fractional offsets, (s,t) as shown

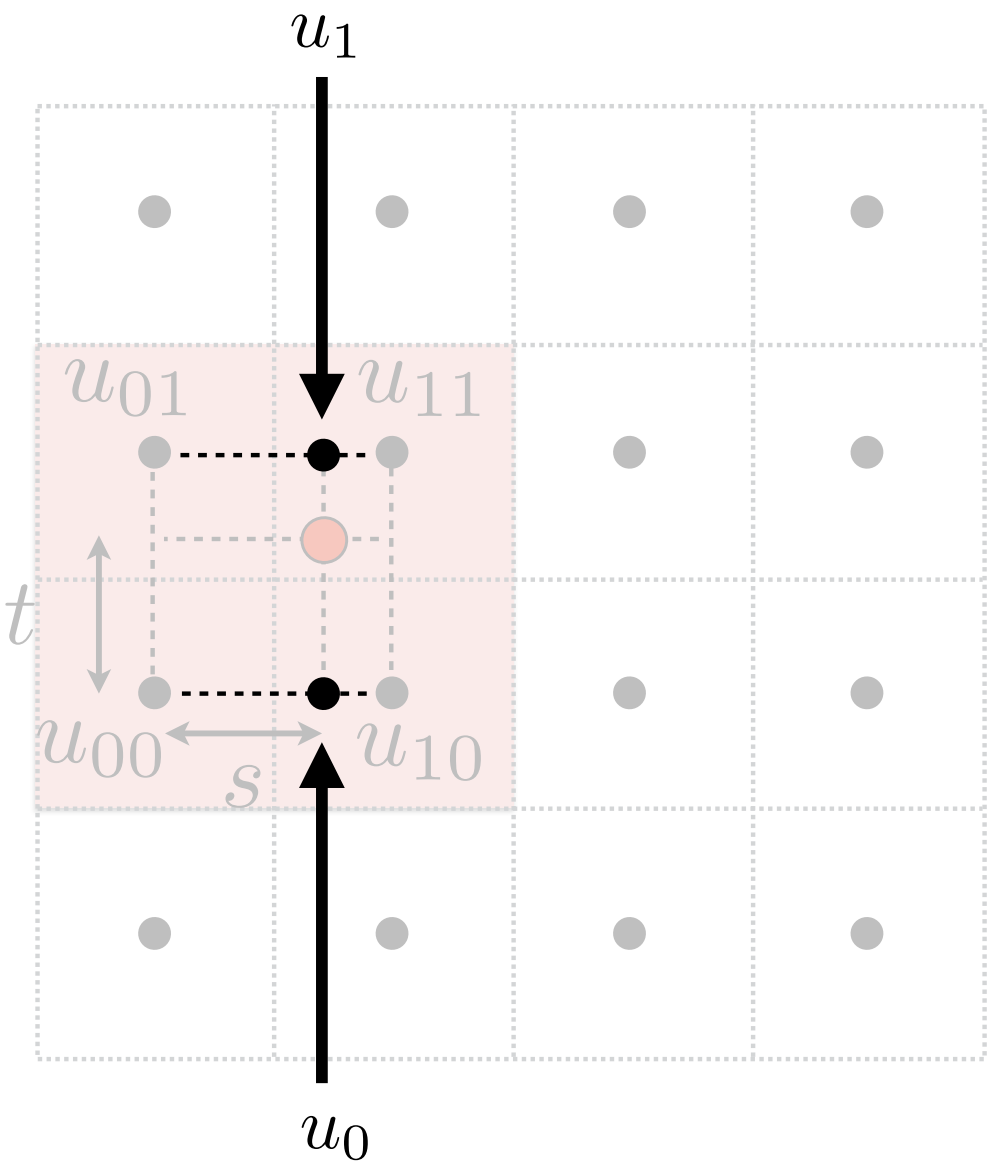
# Bilinear Filtering




**Linear interpolation (1D)**

$$\text{lerp}(x, v_0, v_1) = v_0 + x(v_1 - v_0)$$

# Bilinear Filtering




**Linear interpolation (1D)**

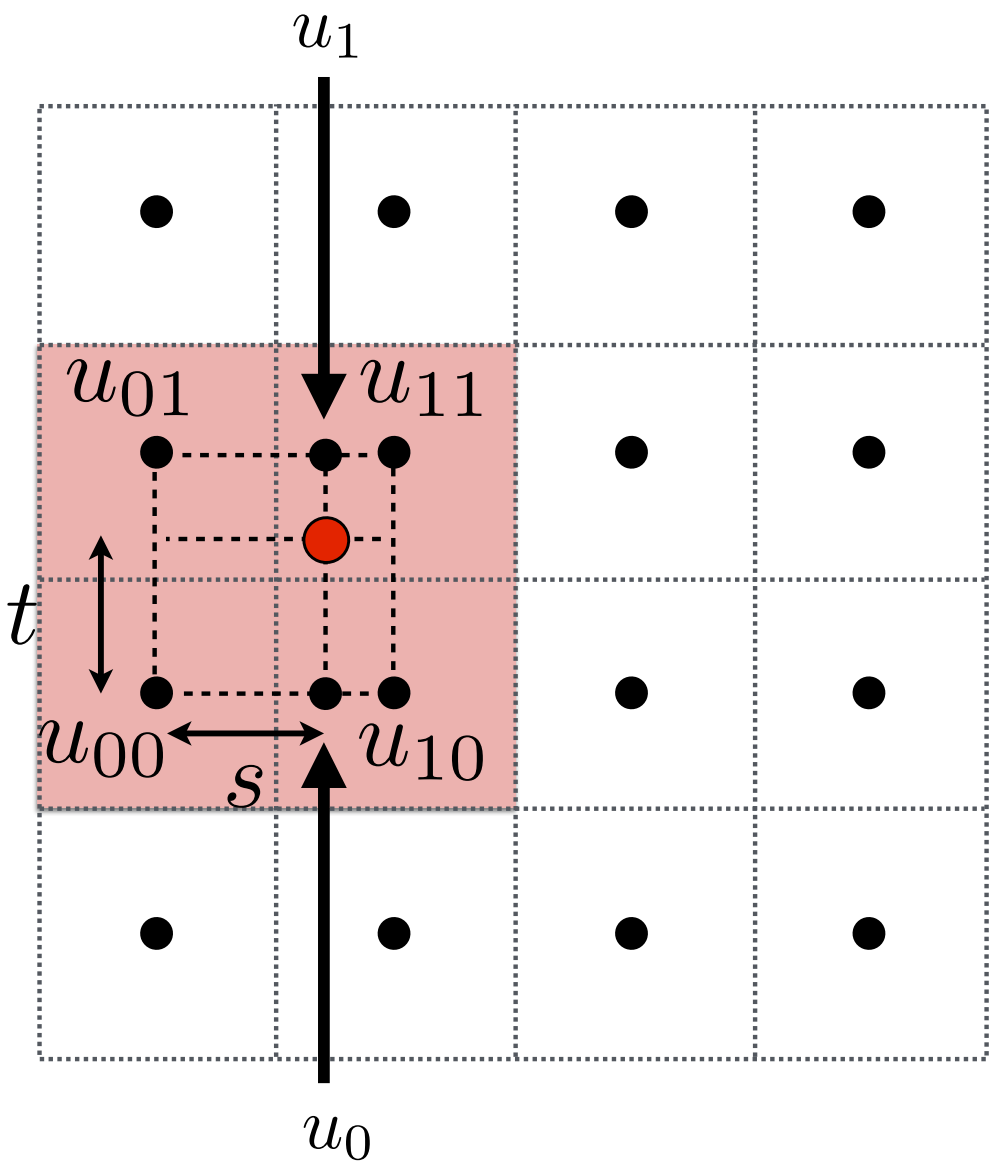
$$\text{lerp}(x, v_0, v_1) = v_0 + x(v_1 - v_0)$$

**Two helper lerps (horizontal)**

$$u_0 = \text{lerp}(s, u_{00}, u_{10})$$

$$u_1 = \text{lerp}(s, u_{01}, u_{11})$$

# Bilinear Filtering




**Linear interpolation (1D)**

$$\text{lerp}(x, v_0, v_1) = v_0 + x(v_1 - v_0)$$

**Two helper lerps**

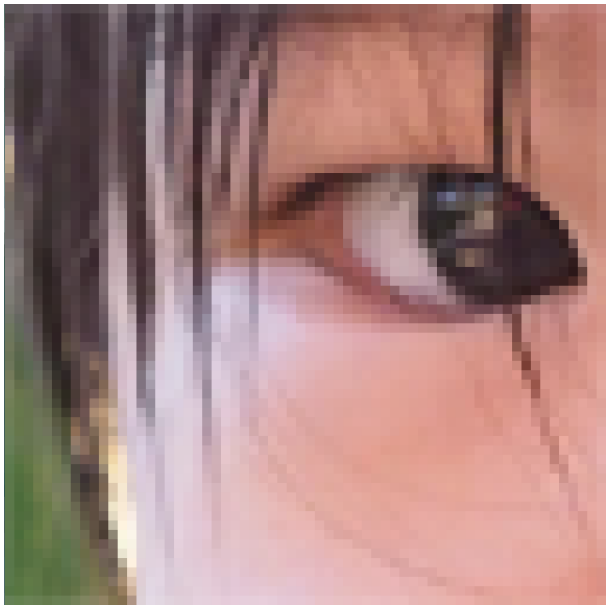
$$u_0 = \text{lerp}(s, u_{00}, u_{10})$$

$$u_1 = \text{lerp}(s, u_{01}, u_{11})$$

**Final vertical lerp, to get result:**

$$f(x, y) = \text{lerp}(t, u_0, u_1)$$

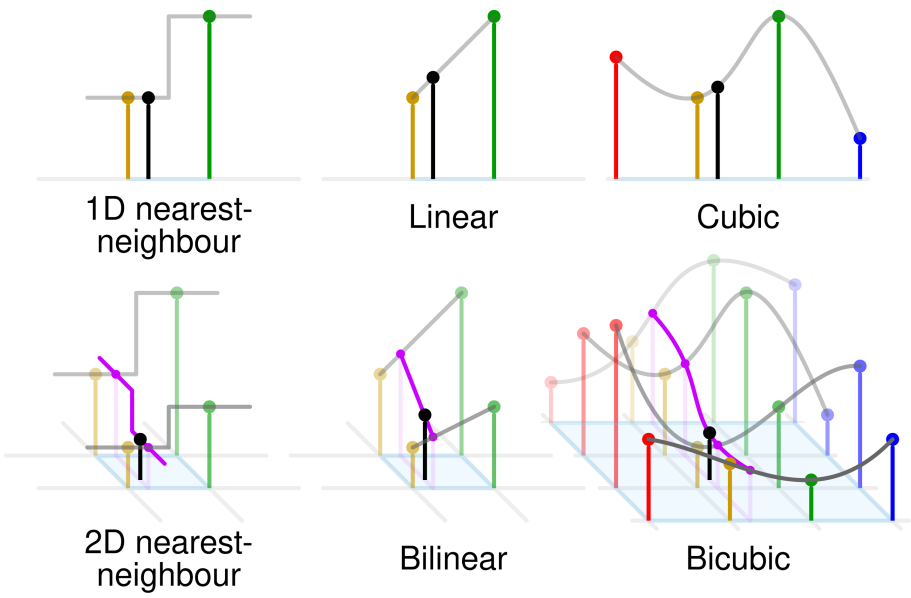
# Texture Magnification - Easy Case



Nearest

Bilinear

Bicubic



# Texture Minification - Hard Case

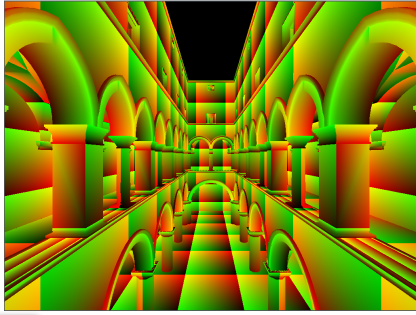
Challenging

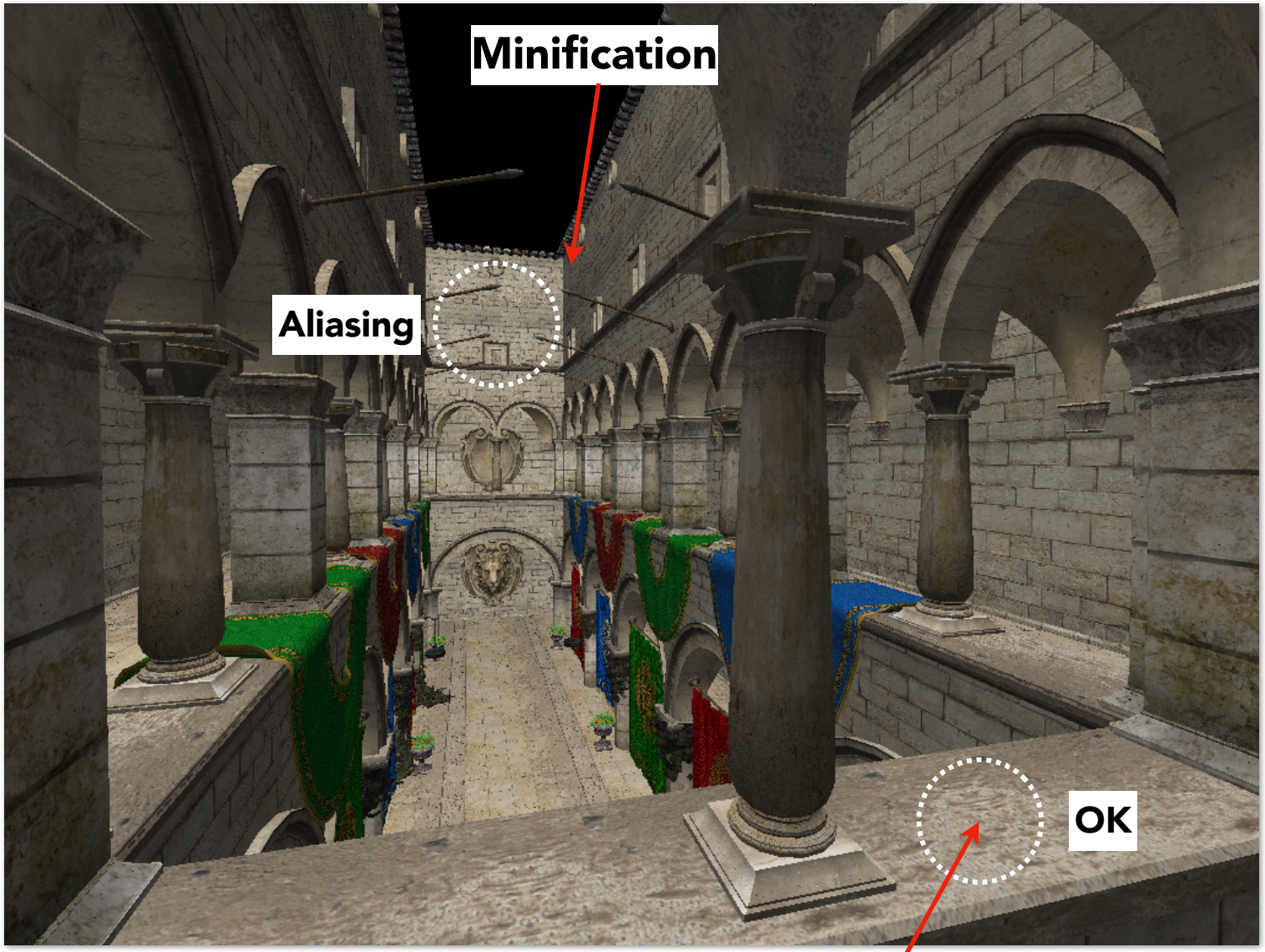
- Many texels can contribute to pixel footprint
- Shape of pixel footprint can be complex

Idea:

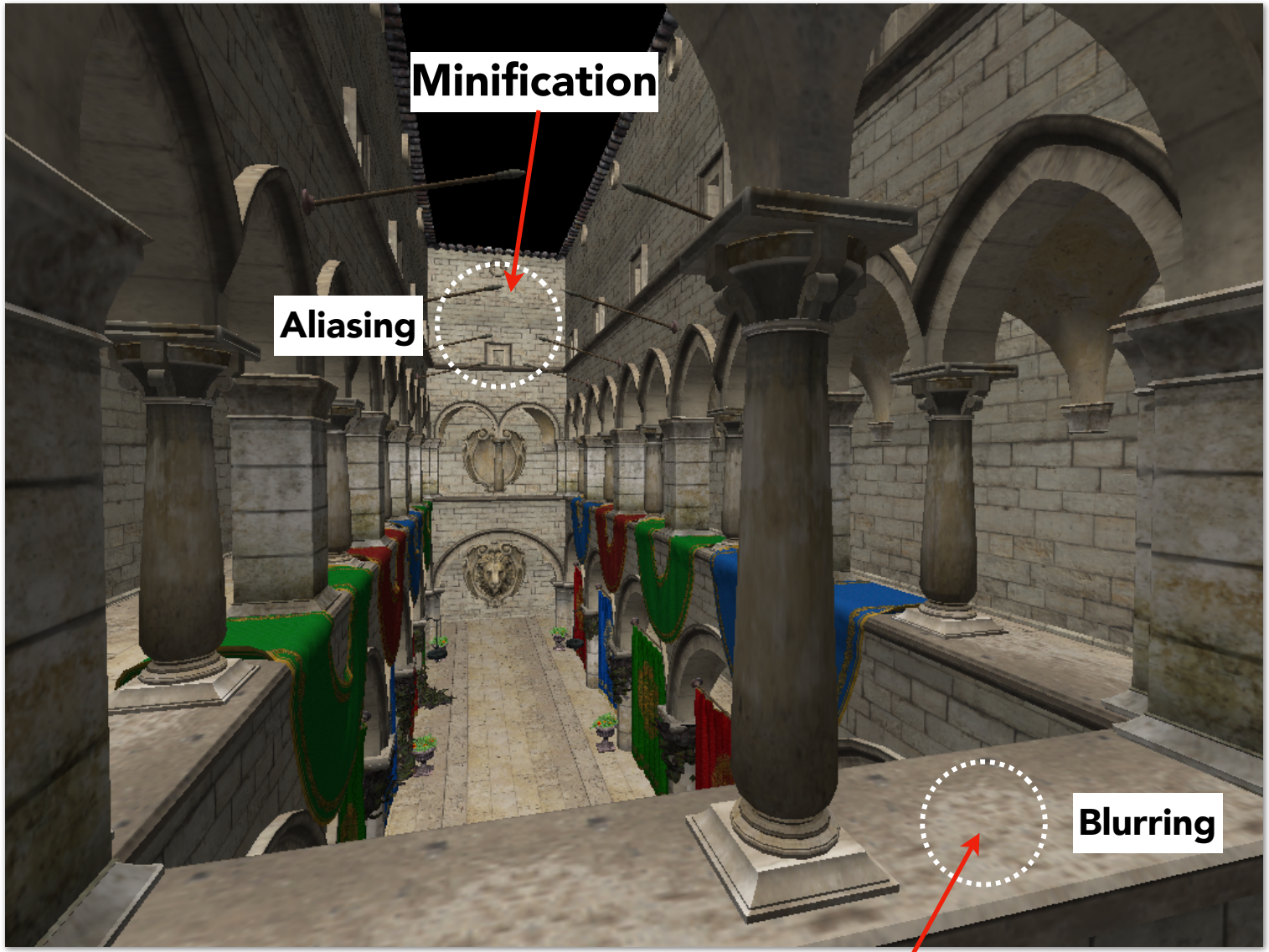
- Take texture image file, then low-pass filter it (i.e. filter out high frequencies) and downsample it (i.e. sample at a lower resolution) texture file.  Do this recursively, and store successively lower resolution, each with successively lower maximum signal frequency.
- For each sample, use the texture file whose resolution approximates the screen sampling rate

# Level 0 - Full Resolution Texture

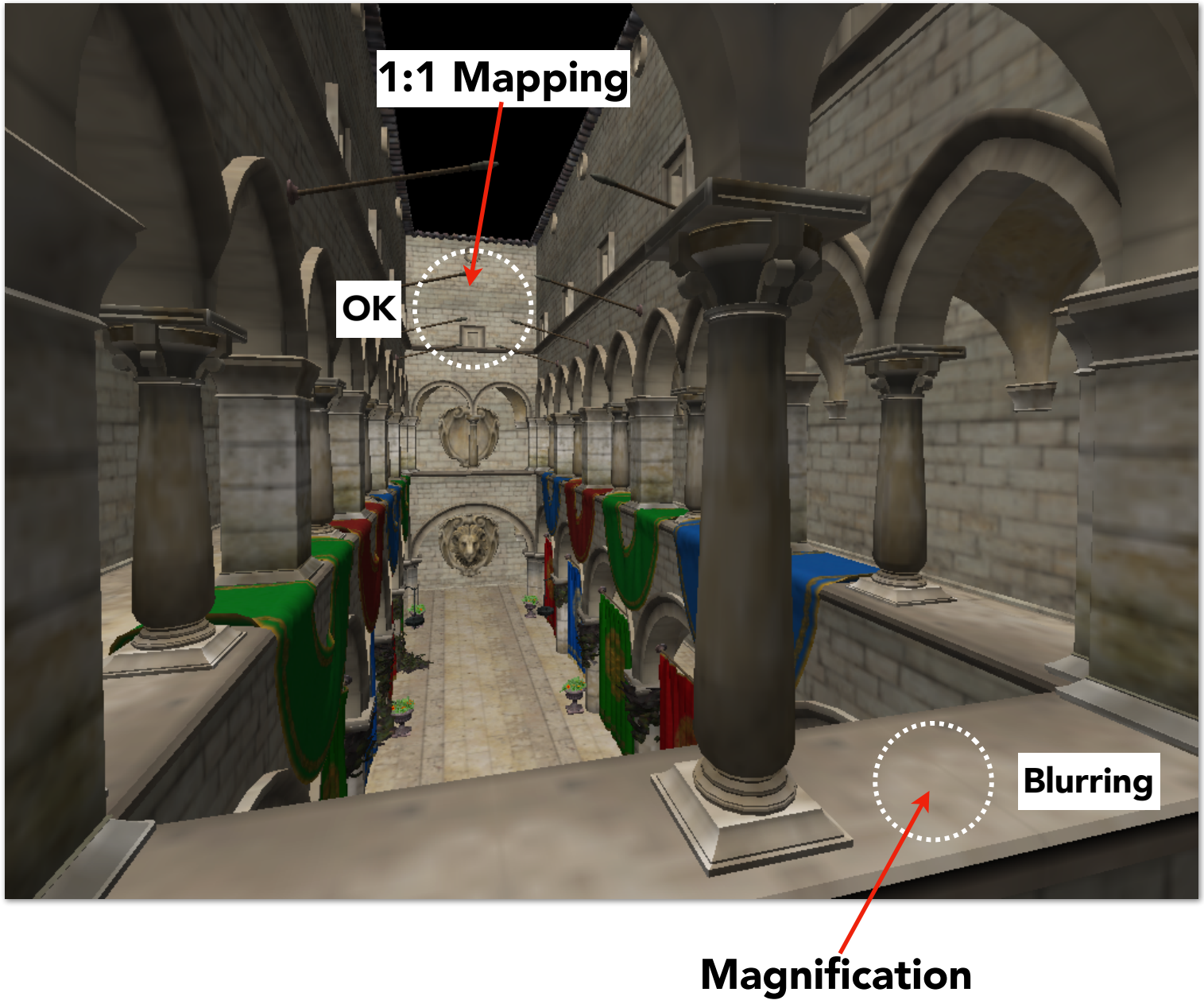




# Level 2 - Downsample 4x4 (bilinear resampling)

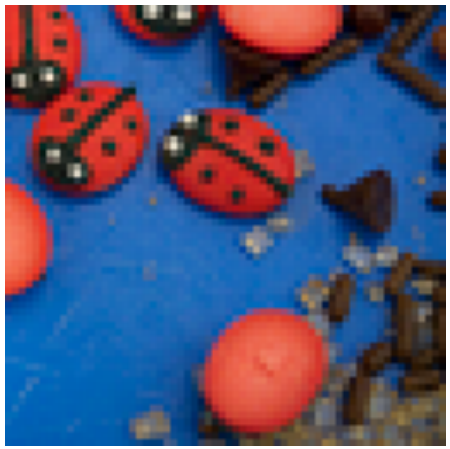


# Level 4 - Downsample 16x16 (bilinear resampling)



# Mipmap (L. Williams 83)

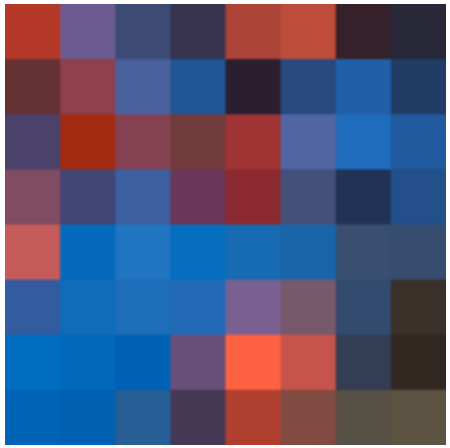
Level 0 = 128x128



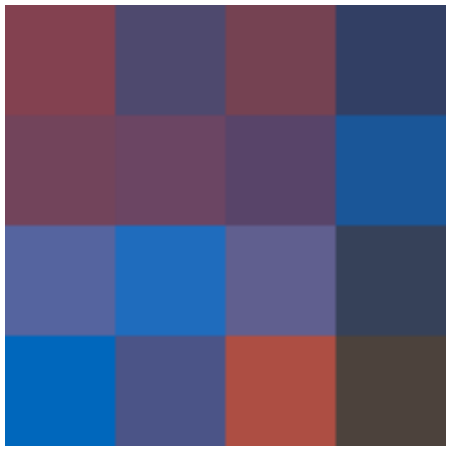
Level 1 = 64x64

Level 2 = 32x32

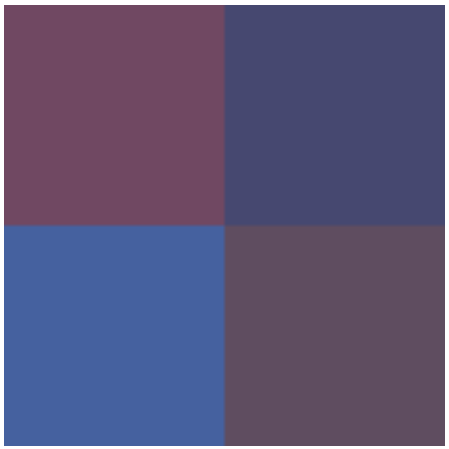
Level 3 = 16x16



Level 4 = 8x8



Level 5 = 4x4



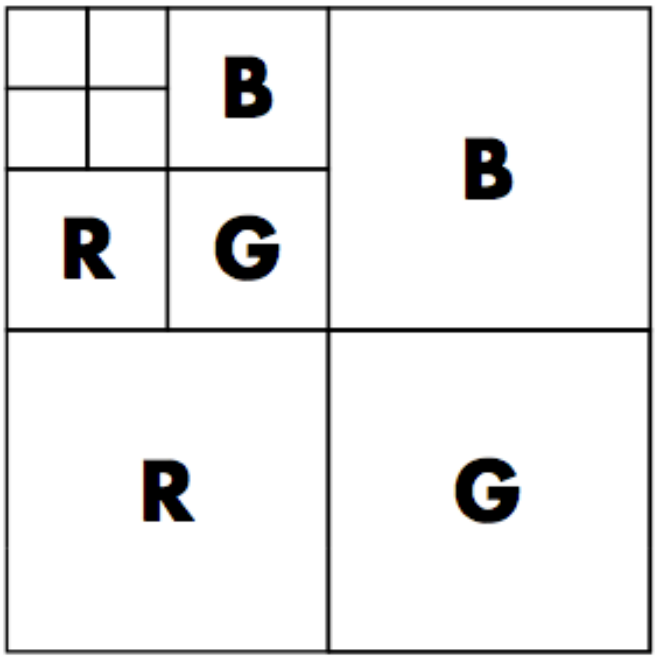
Level 6 = 2x2

Level 7 = 1x1

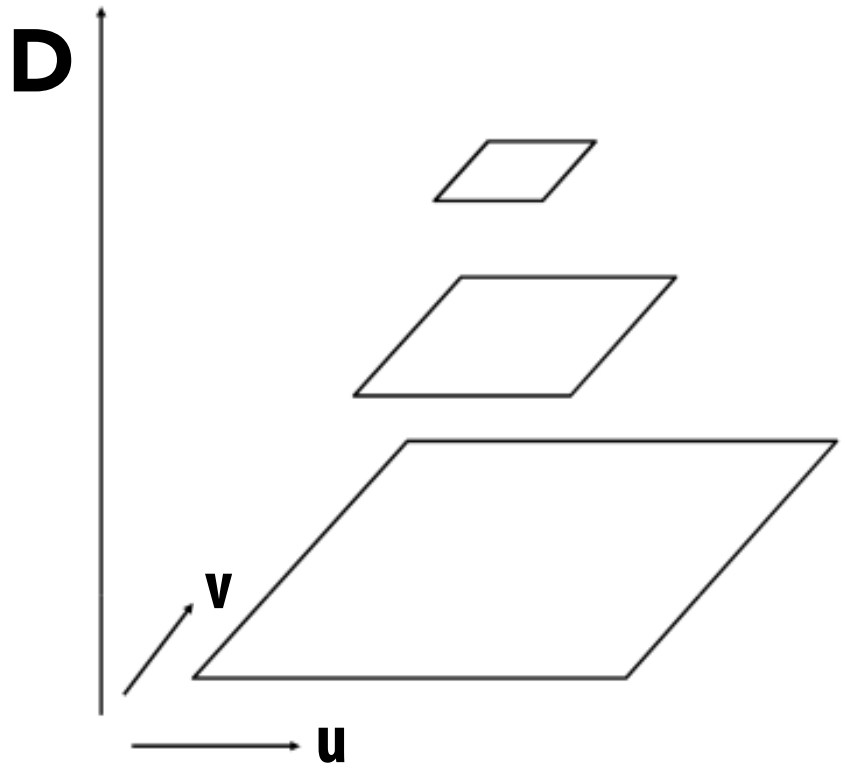
**“Mip” comes from the Latin “multum in parvo", meaning a multitude in a small space**

‘Much in little’，既“放置很多东西的小空间”

# Mipmap (L. Williams 83)



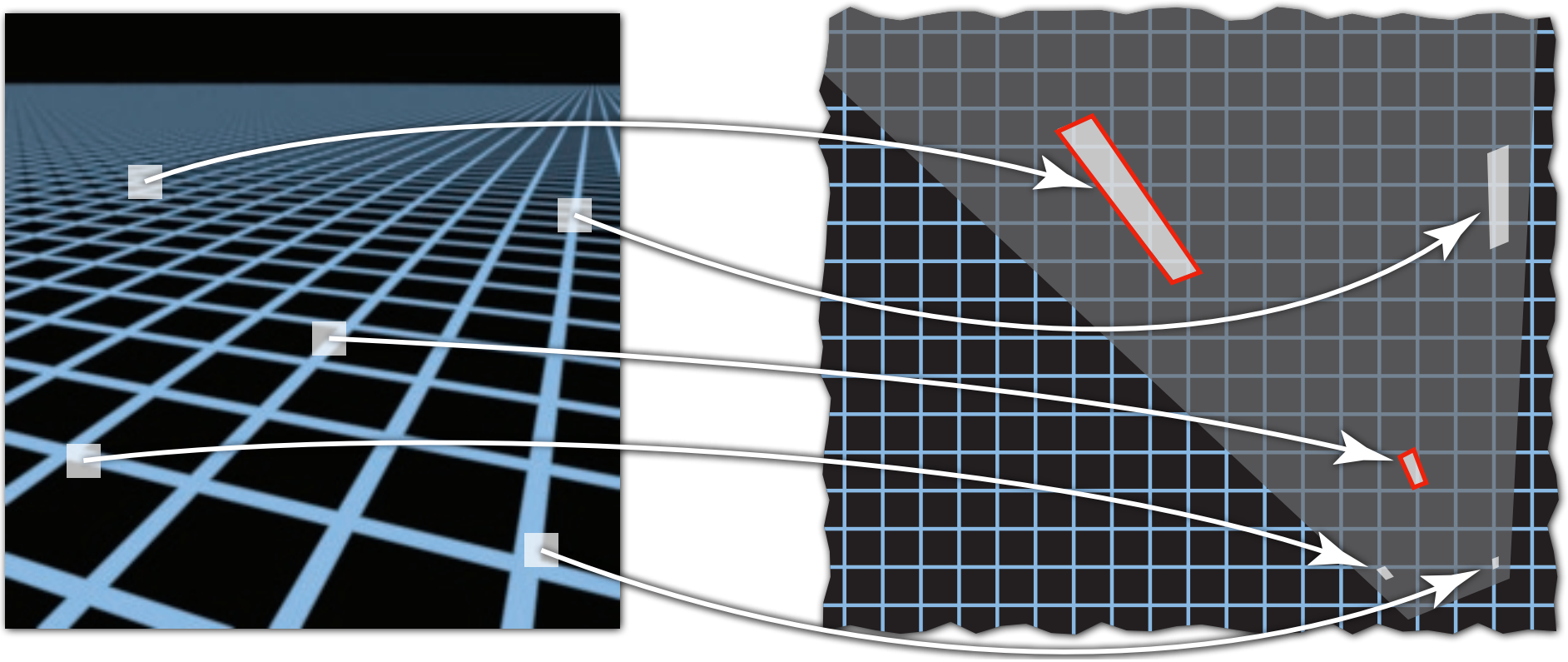

Williams' original
proposed mipmap layout




"Mip hierarchy"
level = D

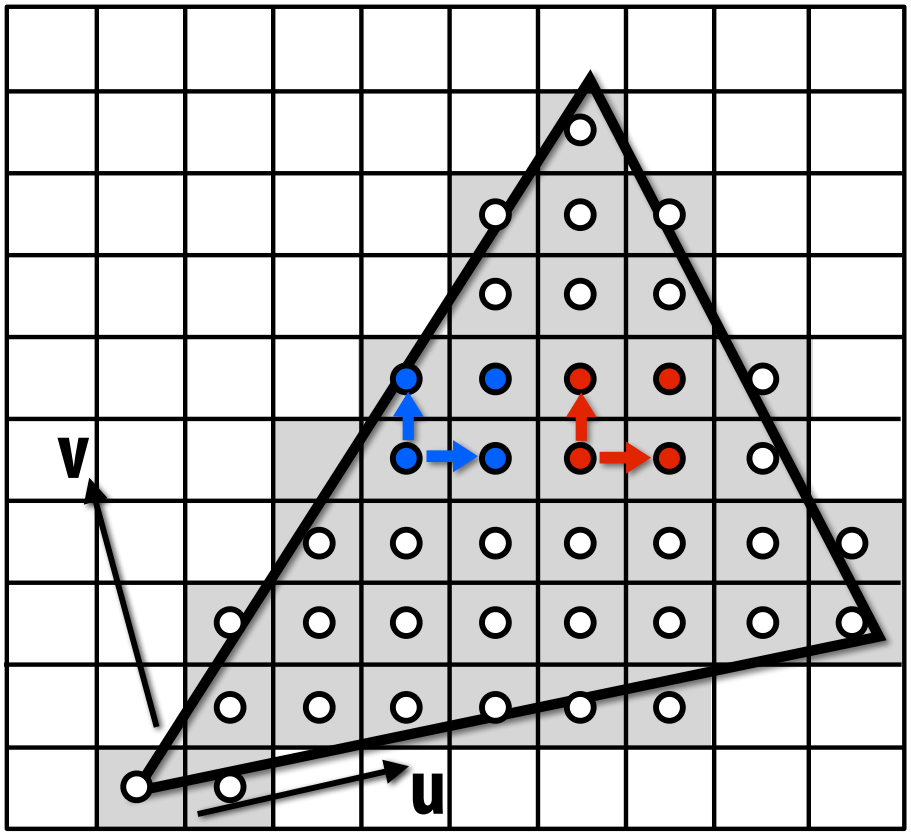
What is the storage overhead of a mipmap?

# How to compute Mipmap Level?

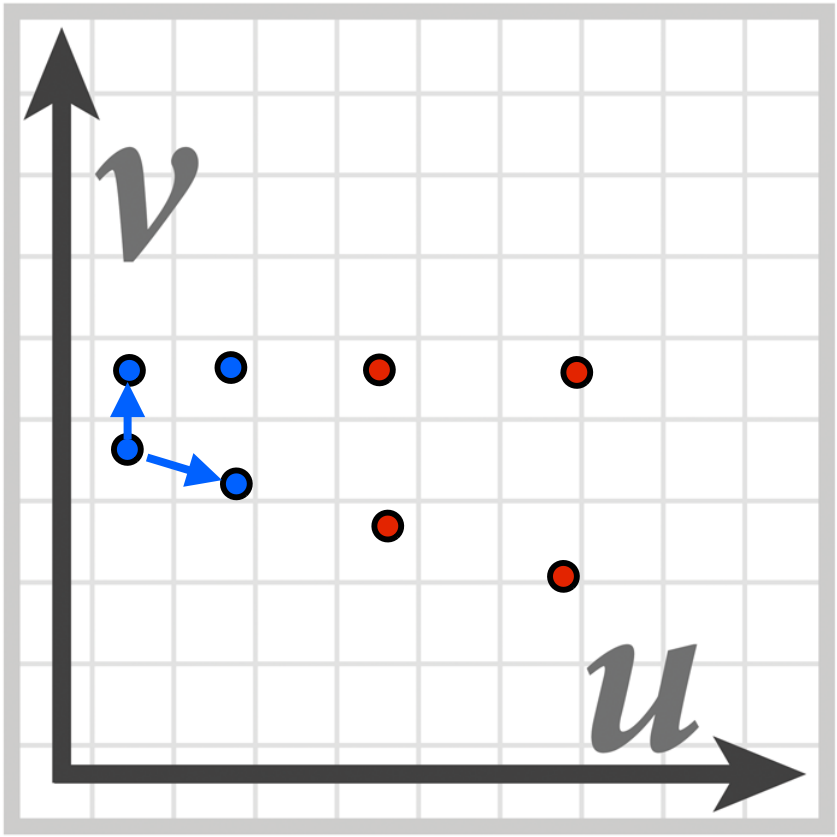


**Squared pixels from screen space map to irregular regions in texture space.**

# Computing Mipmap Level D



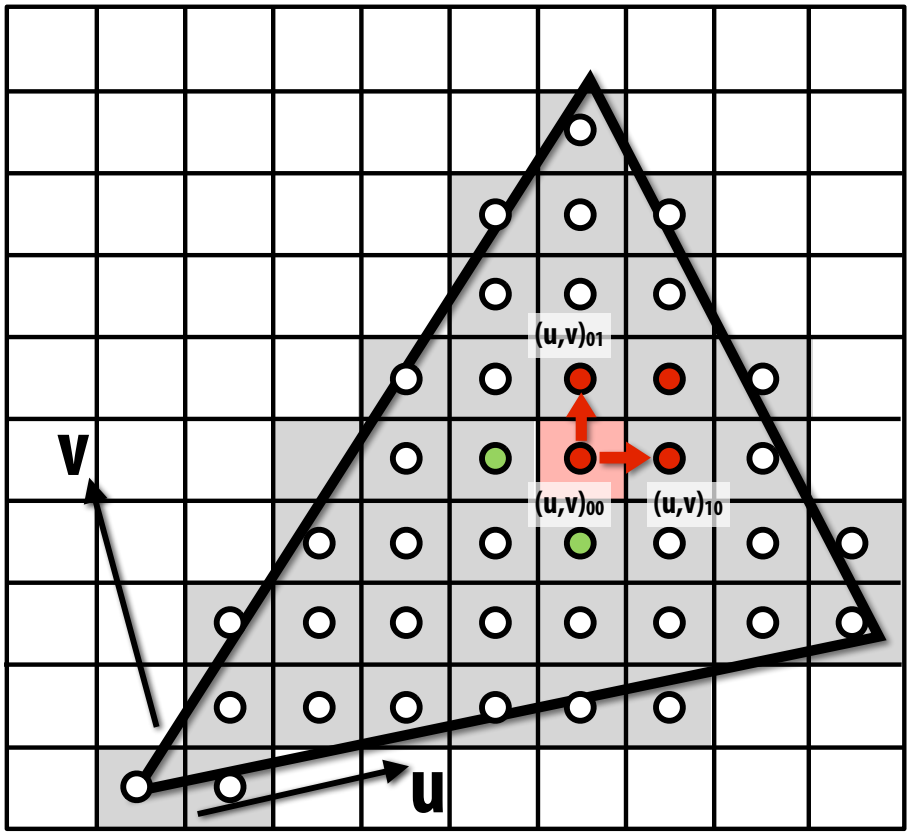

Screen space (x,y)

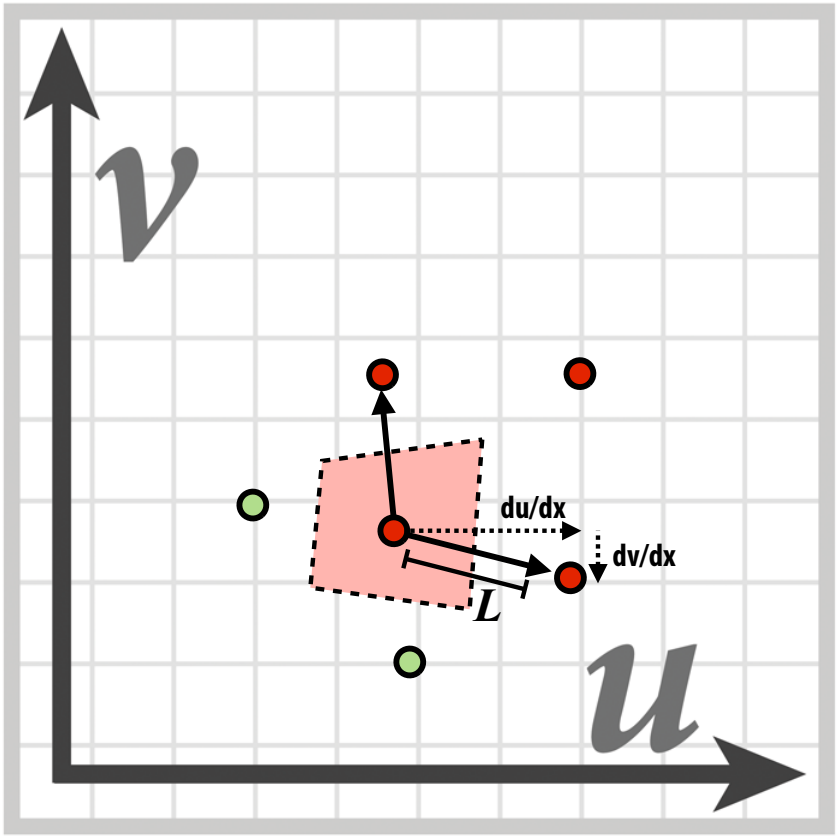



Texture space (u,v)

Estimate texture footprint using texture coordinates of neighboring screen samples

# Computing Mipmap Level D



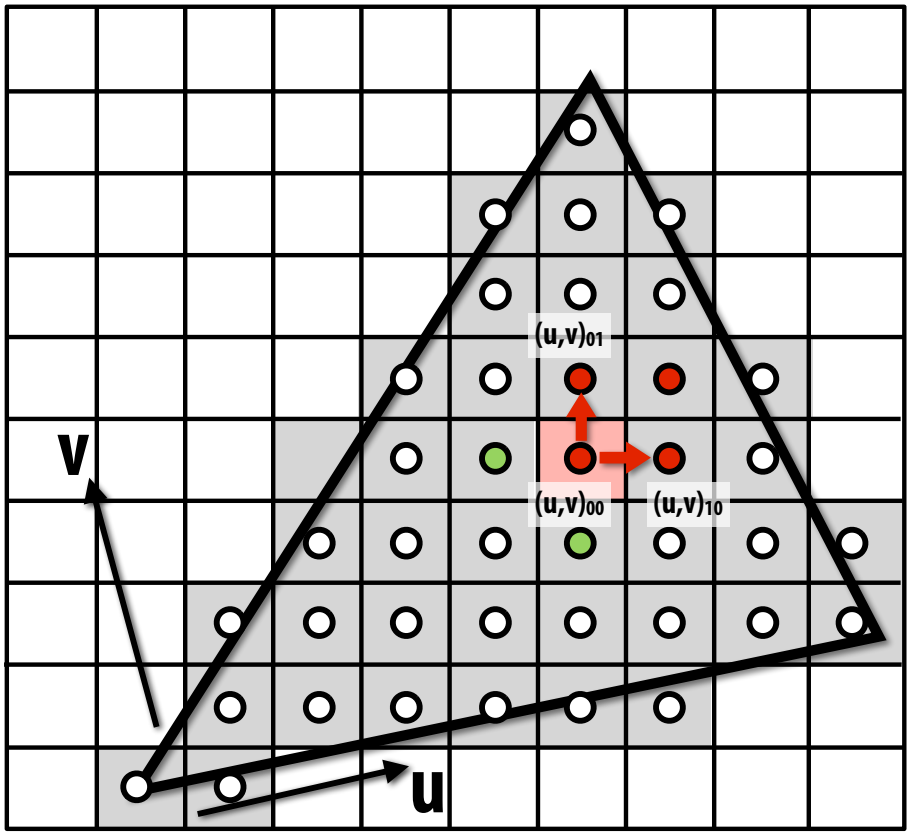



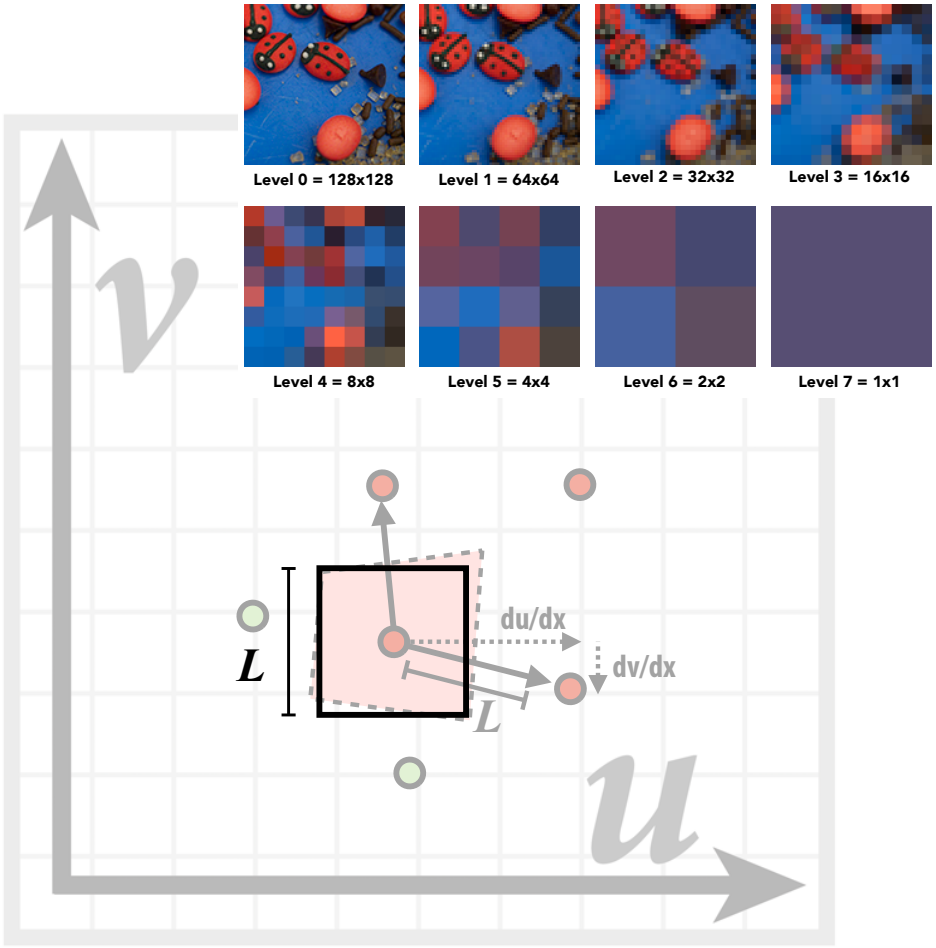

$$\frac{du}{dx} = u_{10} - u_{00} \qquad \frac{dv}{dx} = v_{10} - v_{00}$$

$$\frac{du}{dy} = u_{01} - u_{00} \qquad \frac{dv}{dy} = v_{01} - v_{00}$$

$$L = \max\left(\sqrt{\left(\frac{du}{dx}\right)^2 + \left(\frac{dv}{dx}\right)^2}, \sqrt{\left(\frac{du}{dy}\right)^2 + \left(\frac{dv}{dy}\right)^2}\right)$$

# Computing Mipmap Level D

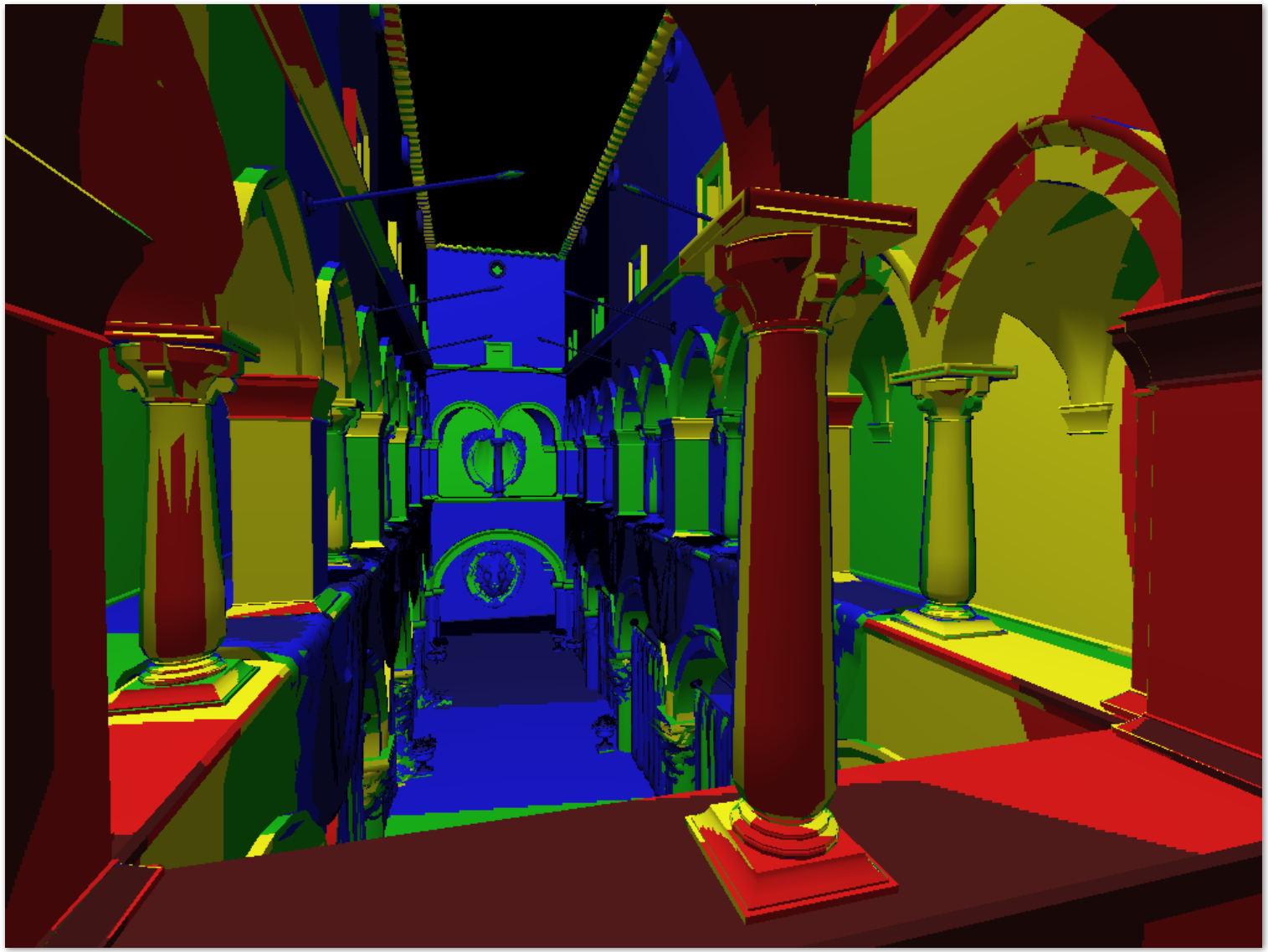



$$\text{Mip Map Level } D = \log_2 L$$

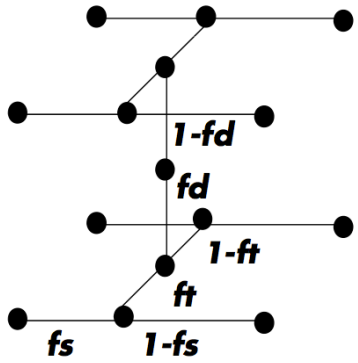
$$L = \max\left(\sqrt{\left(\frac{du}{dx}\right)^2 + \left(\frac{dv}{dx}\right)^2}, \sqrt{\left(\frac{du}{dy}\right)^2 + \left(\frac{dv}{dy}\right)^2}\right)$$

# Visualization of Mipmap Level

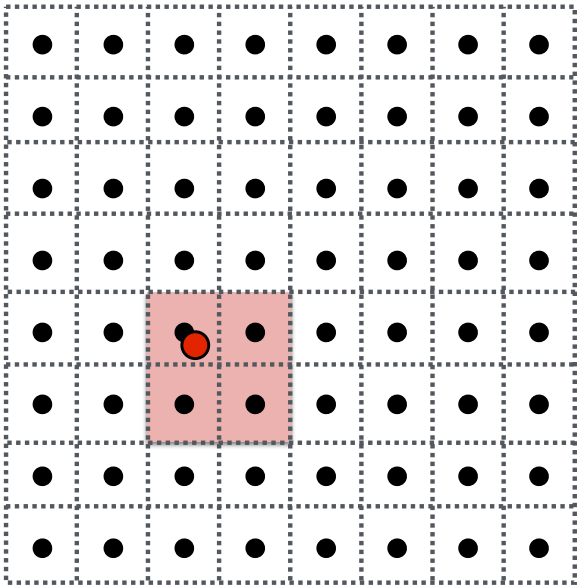


D rounded to nearest integer level

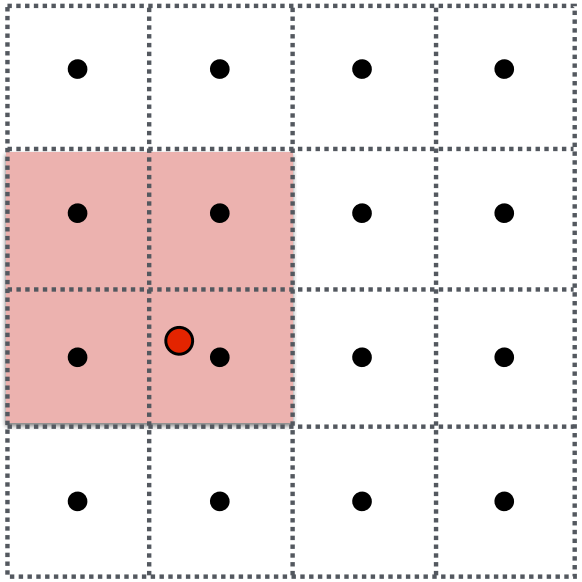
# Trilinear Filtering




$$rp(t, v_1, v_2) = v_1 + t(v_2 - v_1)$$



Mipmap Level D



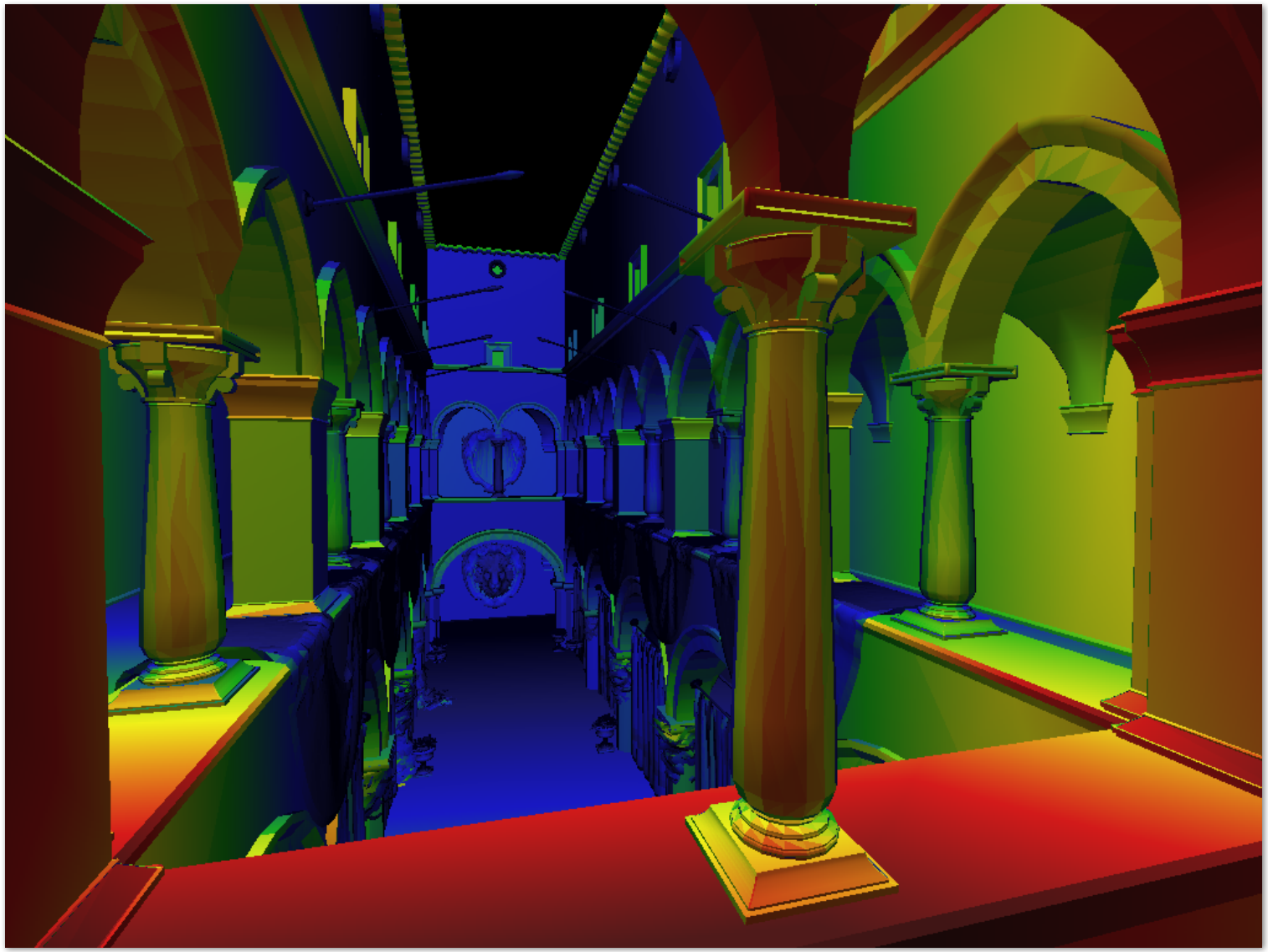
Mipmap Level D+1

Bilinear result

Bilinear result

Linear interpolation based on continuous D value

# Visualization of Mipmap Level



Trilinear filtering: visualization of continuous D

# Texture Filtering

Image resampling choices

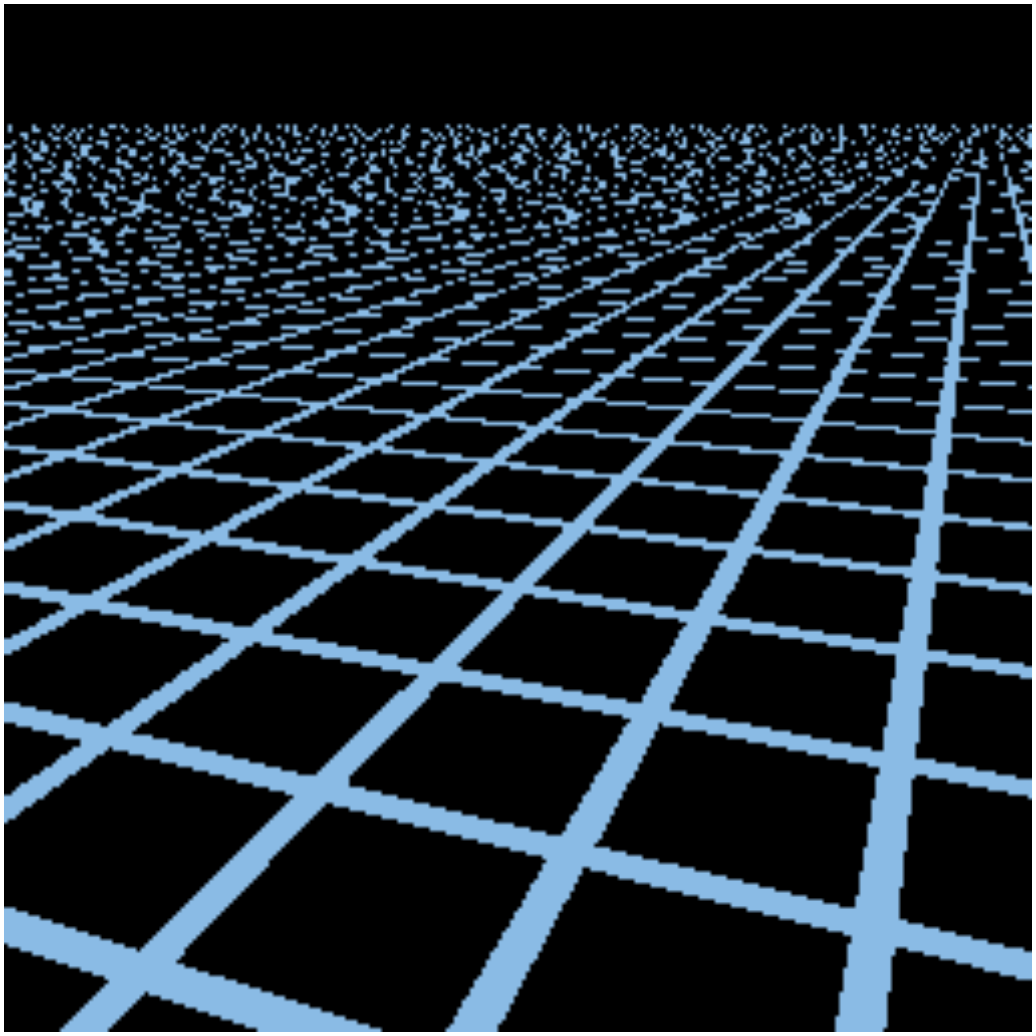
- Nearest
- Bilinear interpolation

Mipmap level resampling choices

- Always level 0
- Nearest D
- Linear interpolation

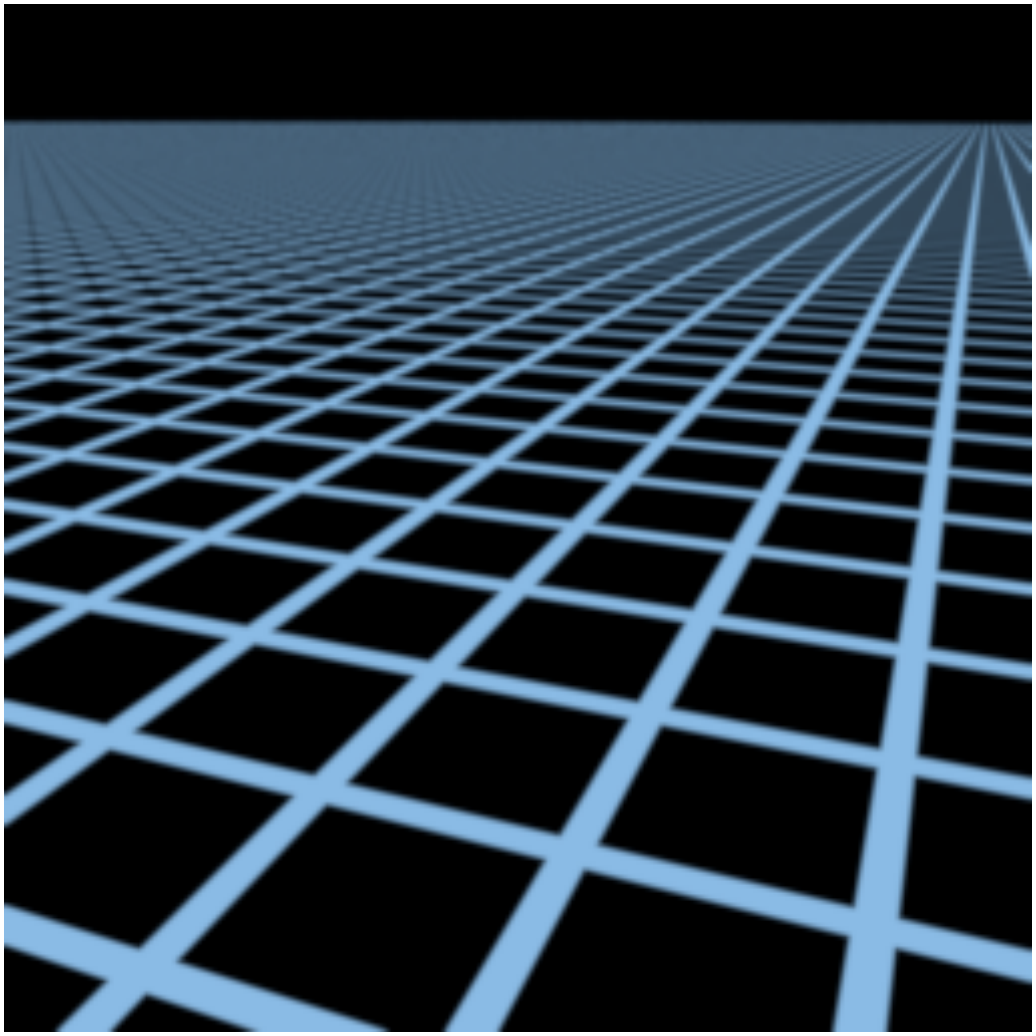
2 x 3 = 6 choices

# Mipmap Limitations



Point sampling

# Mipmap Limitations

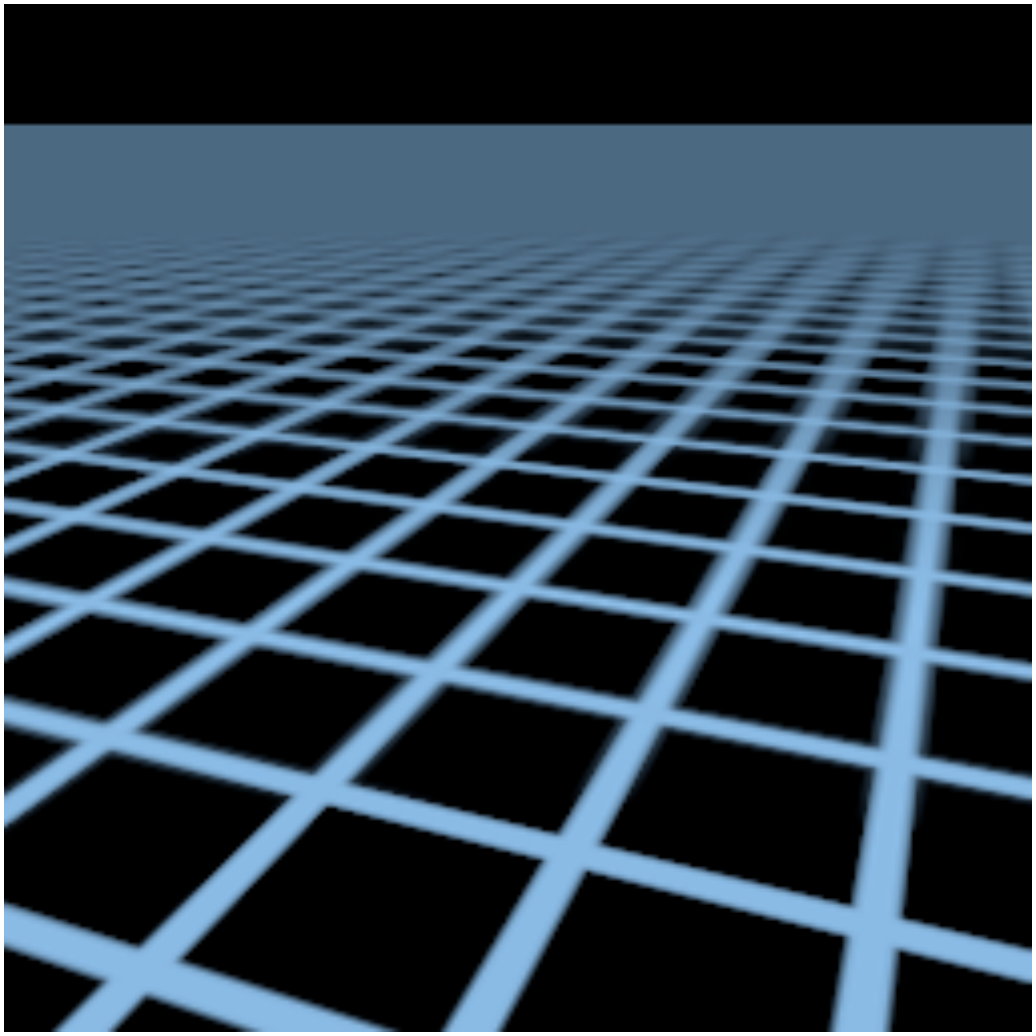


Supersampling 512x

# Mipmap Limitations

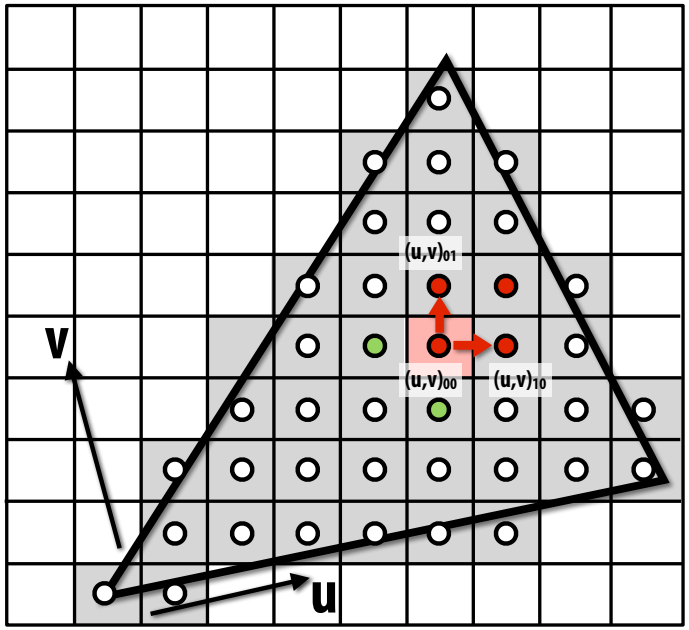
Overblur

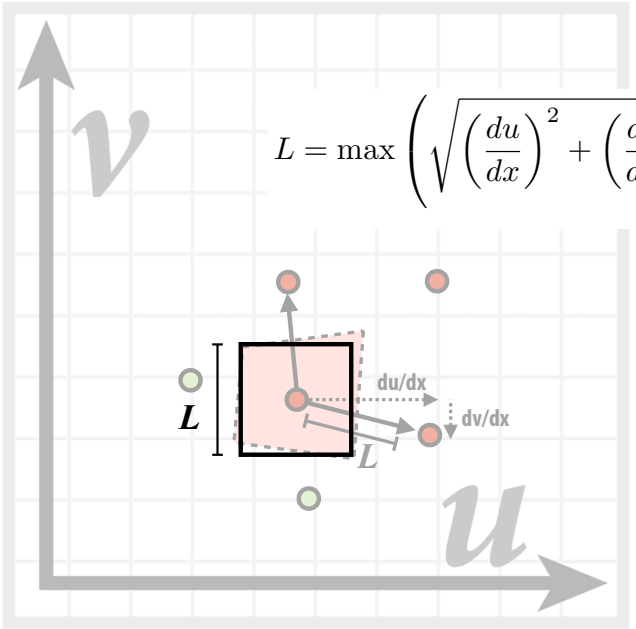
Why?



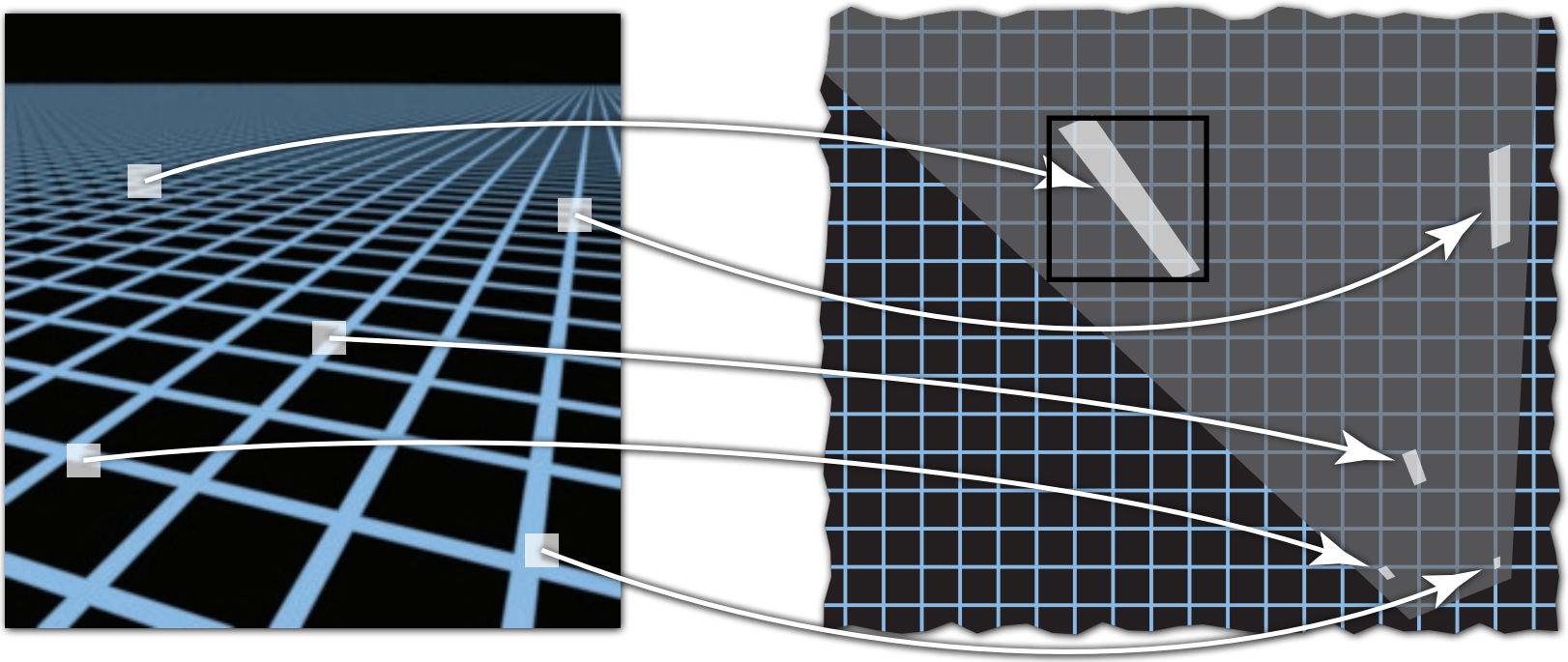
Mipmap trilinear sampling

# Mipmap Assumes Regular Footprint in Texture



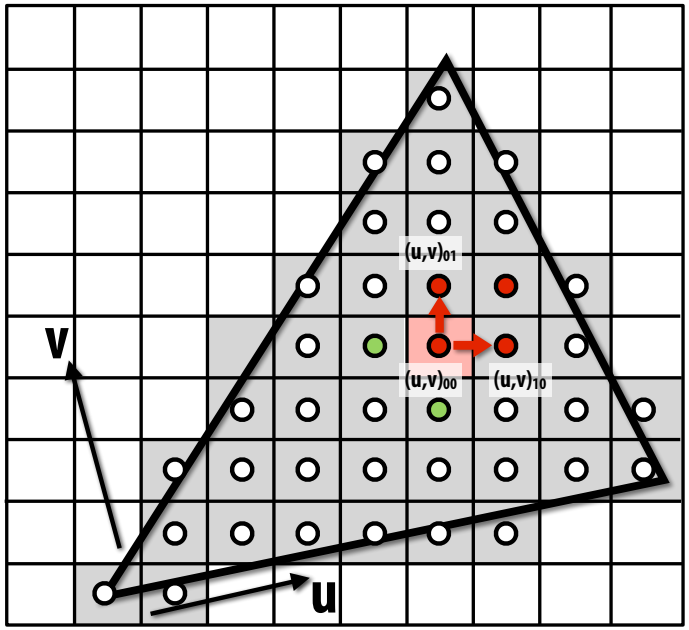



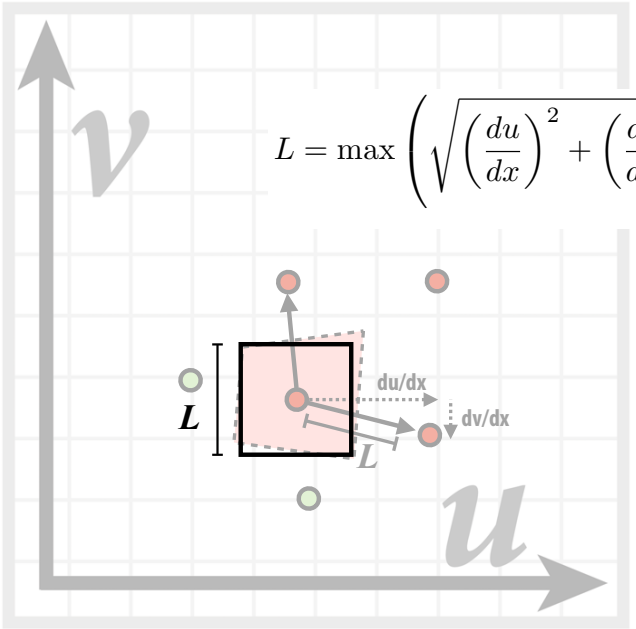

$$L = \max\left(\sqrt{\left(\frac{du}{dx}\right)^2 + \left(\frac{dv}{dx}\right)^2}, \sqrt{\left(\frac{du}{dy}\right)^2 + \left(\frac{dv}{dy}\right)^2}\right)$$



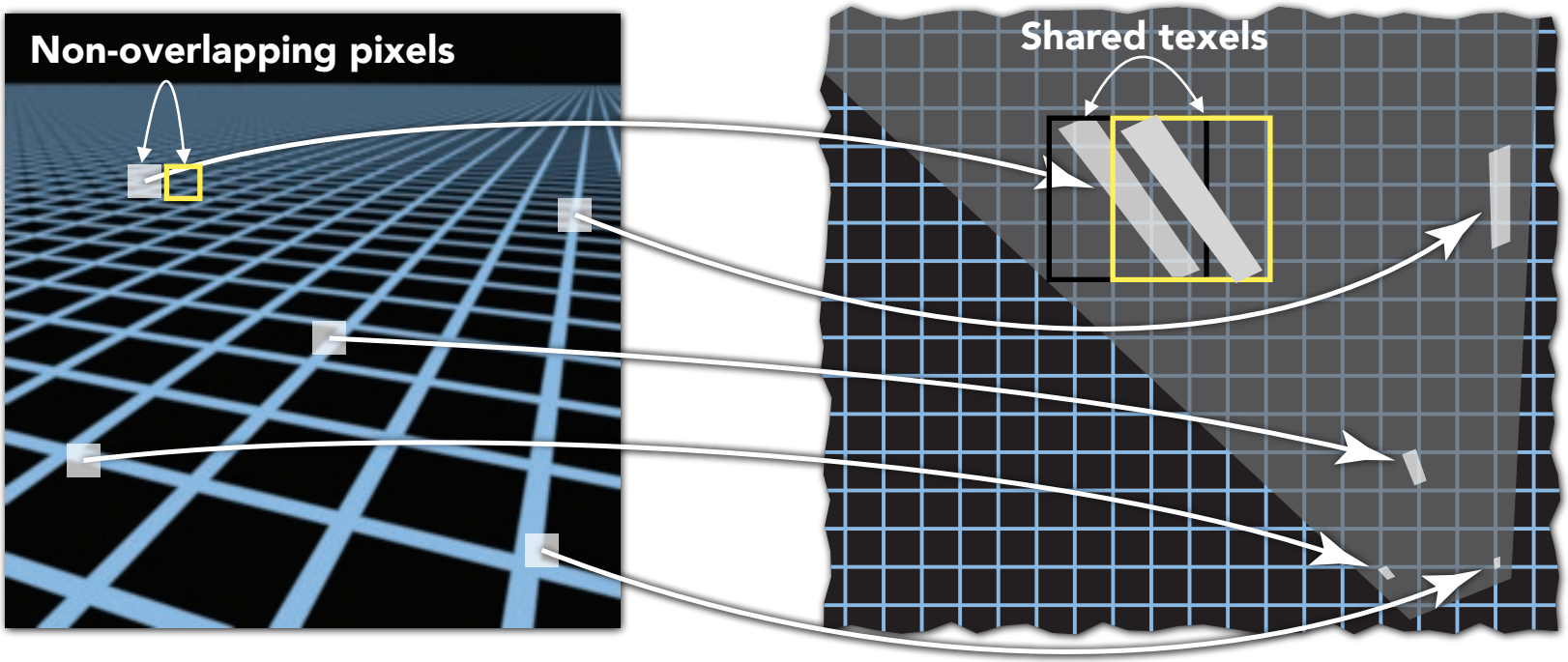
**Irregular pixel footprint!**

# Mipmap Assumes Regular Footprint in Texture



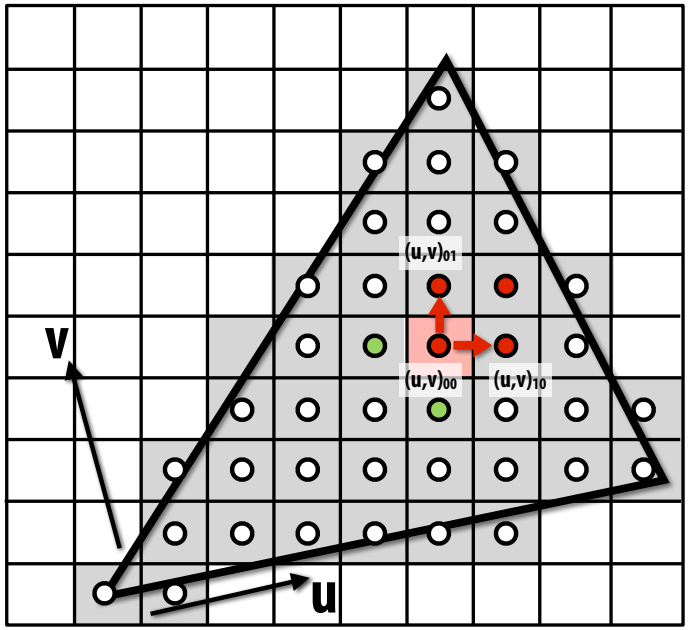



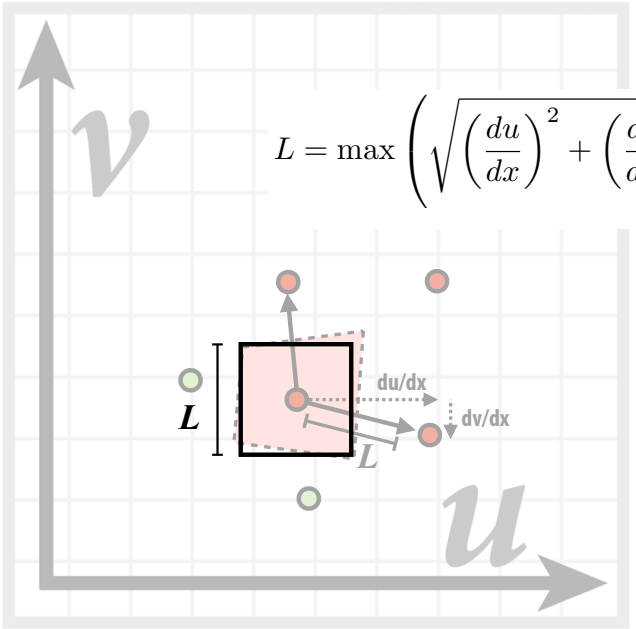

$$L = \max\left(\sqrt{\left(\frac{du}{dx}\right)^2 + \left(\frac{dv}{dx}\right)^2}, \sqrt{\left(\frac{du}{dy}\right)^2 + \left(\frac{dv}{dy}\right)^2}\right)$$



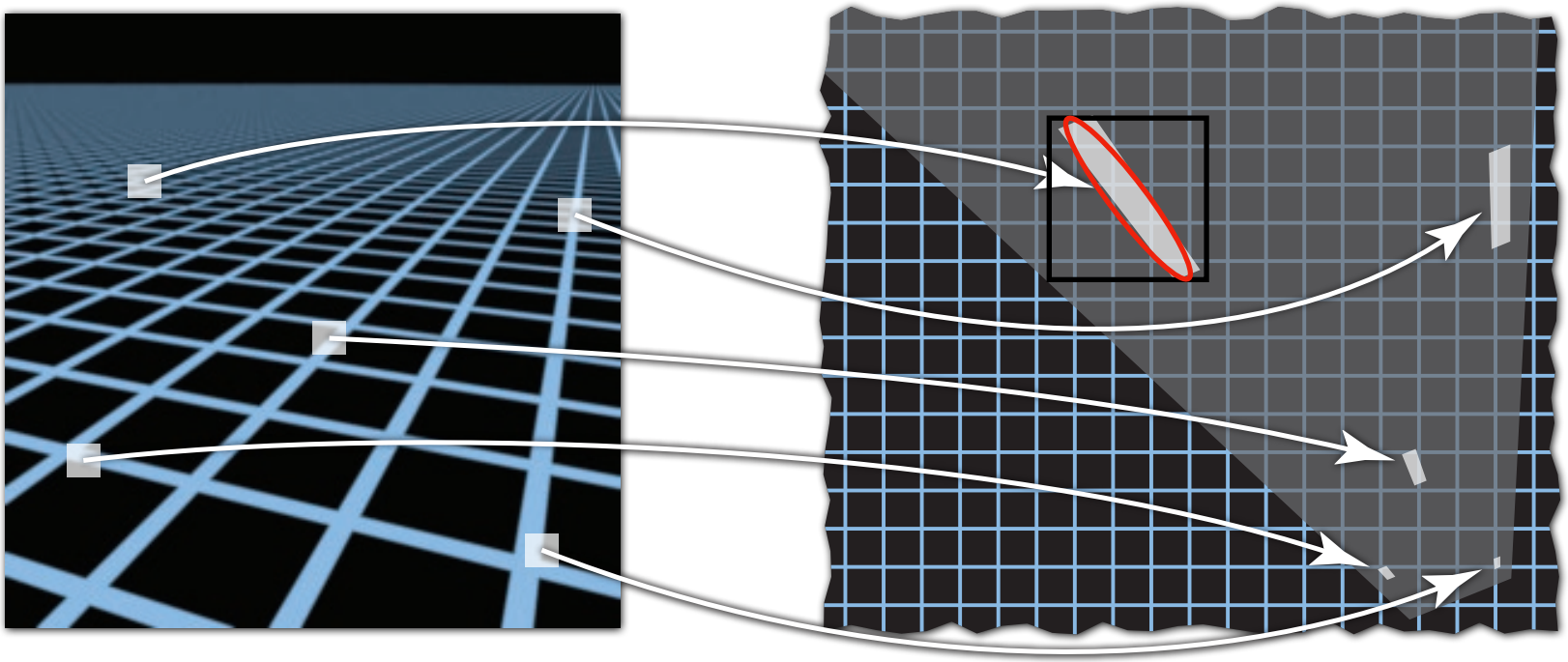

**Irregular pixel footprint!**

# Mipmap Assumes Regular Footprint in Texture




$$L = \max\left(\sqrt{\left(\frac{du}{dx}\right)^2 + \left(\frac{dv}{dx}\right)^2}, \sqrt{\left(\frac{du}{dy}\right)^2 + \left(\frac{dv}{dy}\right)^2}\right)$$



**Irregular pixel footprint!**

# Anisotropic Filtering

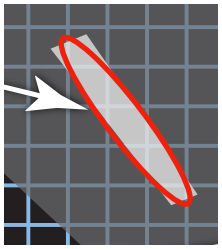
各向异性滤波：相对于Mipmap各向同性（isotropic）而言的改进

## Ripmaps and summed area tables

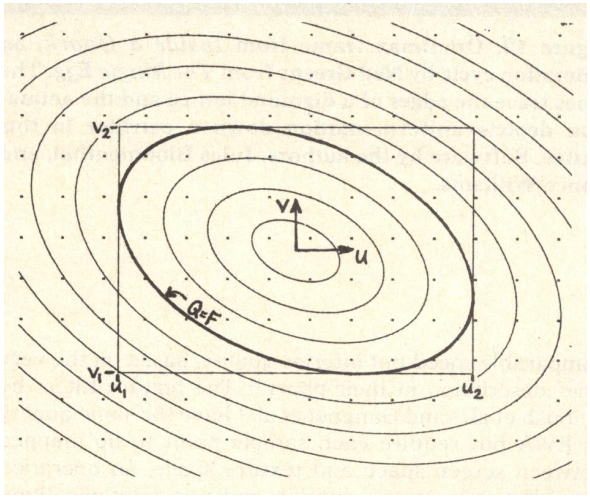
- Can look up axis-aligned rectangular zones
- Diagonal footprints still a problem

Wikipedia

## EWA filtering



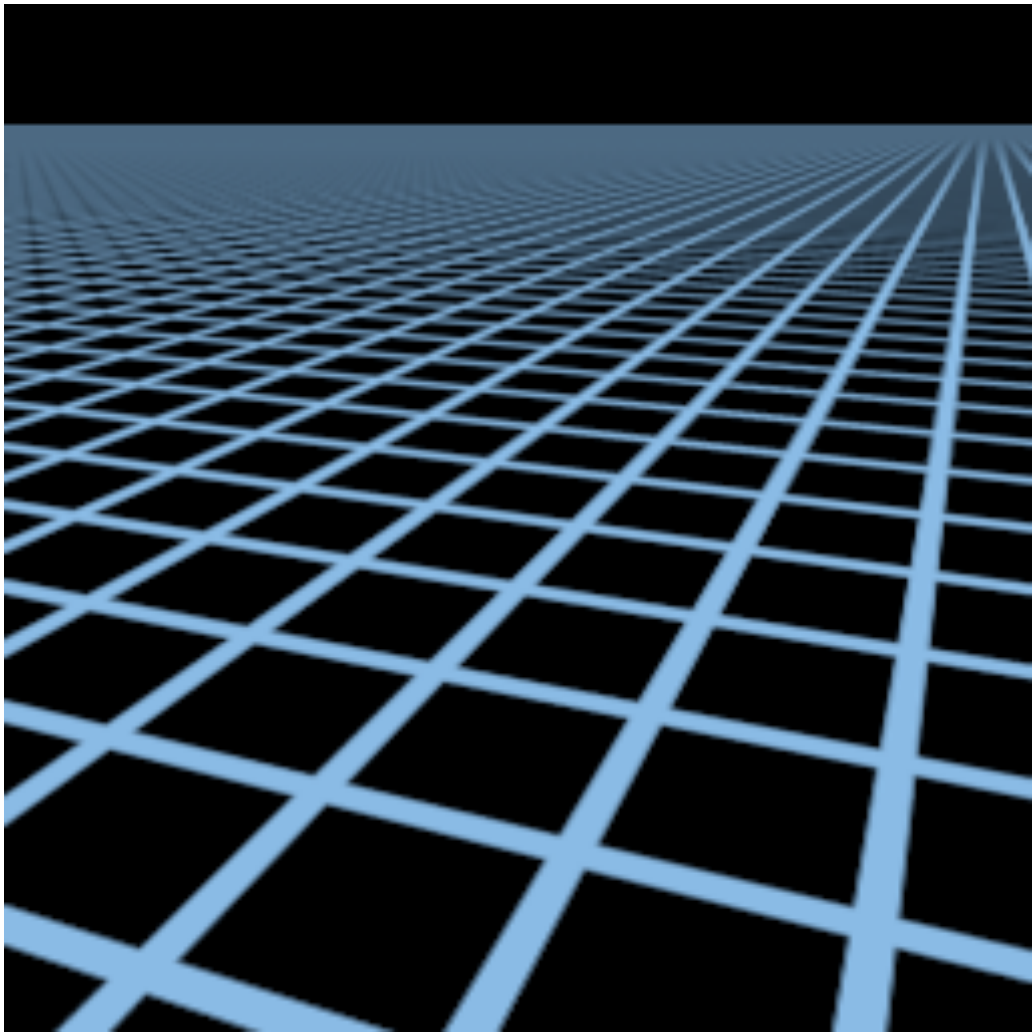
- Use multiple lookups
- Weighted average
- Mipmap hierarchy still helps




Greene & Heckbert ‘86

**Elliptical Weighted Average**

# Anisotropic Filtering



## Elliptical weighted average (EWA) filtering

算法实现原理介绍可参考：https://zhuanlan.zhihu.com/p/105167411

# Things to Remember

Many uses of texturing

- Bring high-resolution data to fragment calculations
- Colors, normals, lighting on sphere, volumetric data, …

How does texturing work?

- Texture coordinate parameterization
- Barycentric interpolation of coordinates
- Texture sampling pattern and frequency
- Mipmaps: texture filtering hierarchy, level calculation, trilinear interpolation
- Anisotropic sampling

# Course Roadmap

## Rasterization Pipeline

Core Concepts

- Sampling
- Antialiasing
- Transforms

Intro

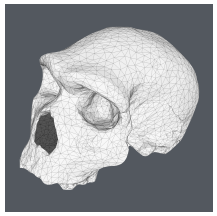
Rasterization

Transforms & Projection

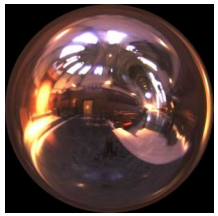
Visibility, Shading

Texture Mapping

## Geometric Modeling

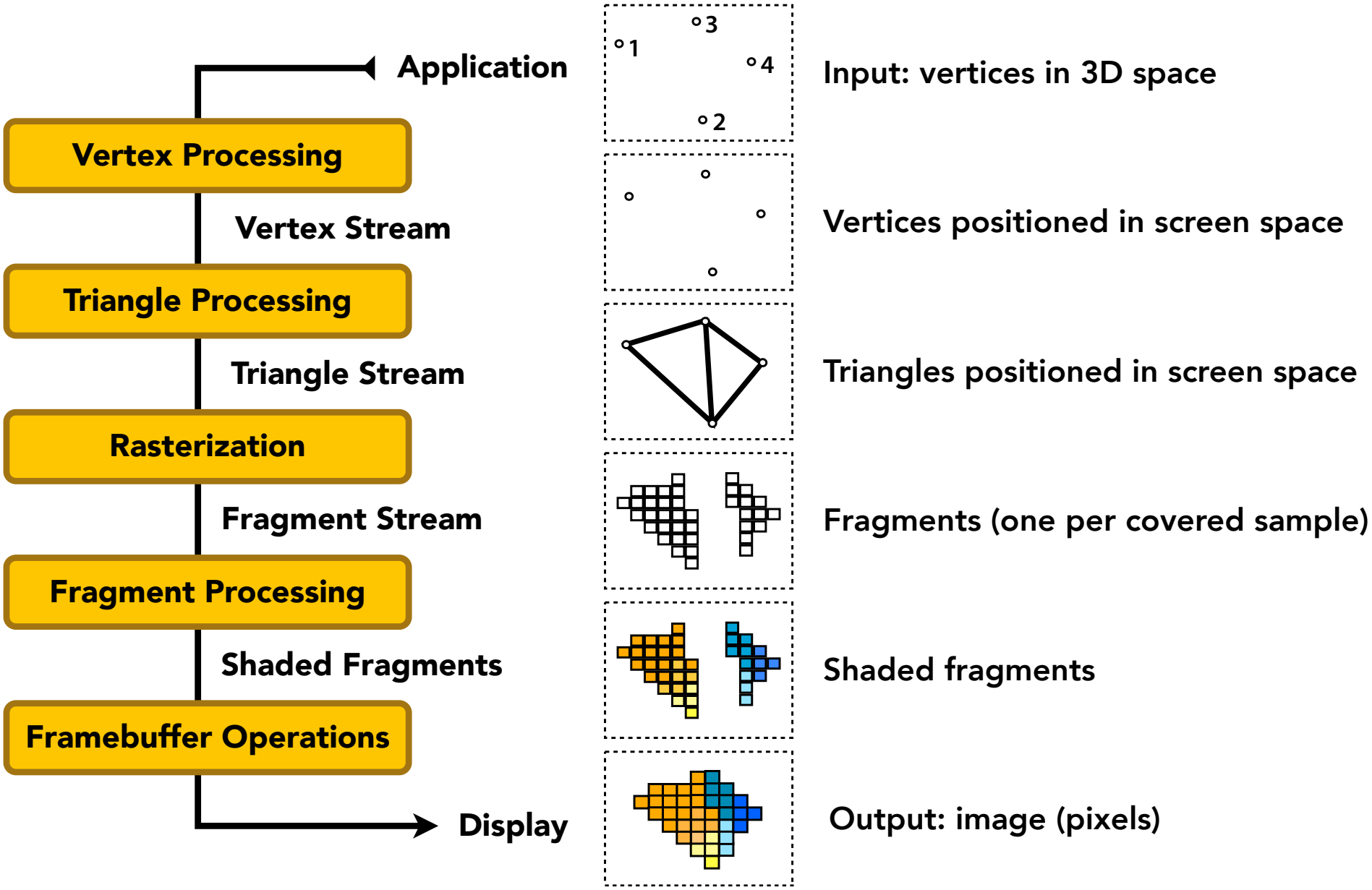


## Lighting & Materials

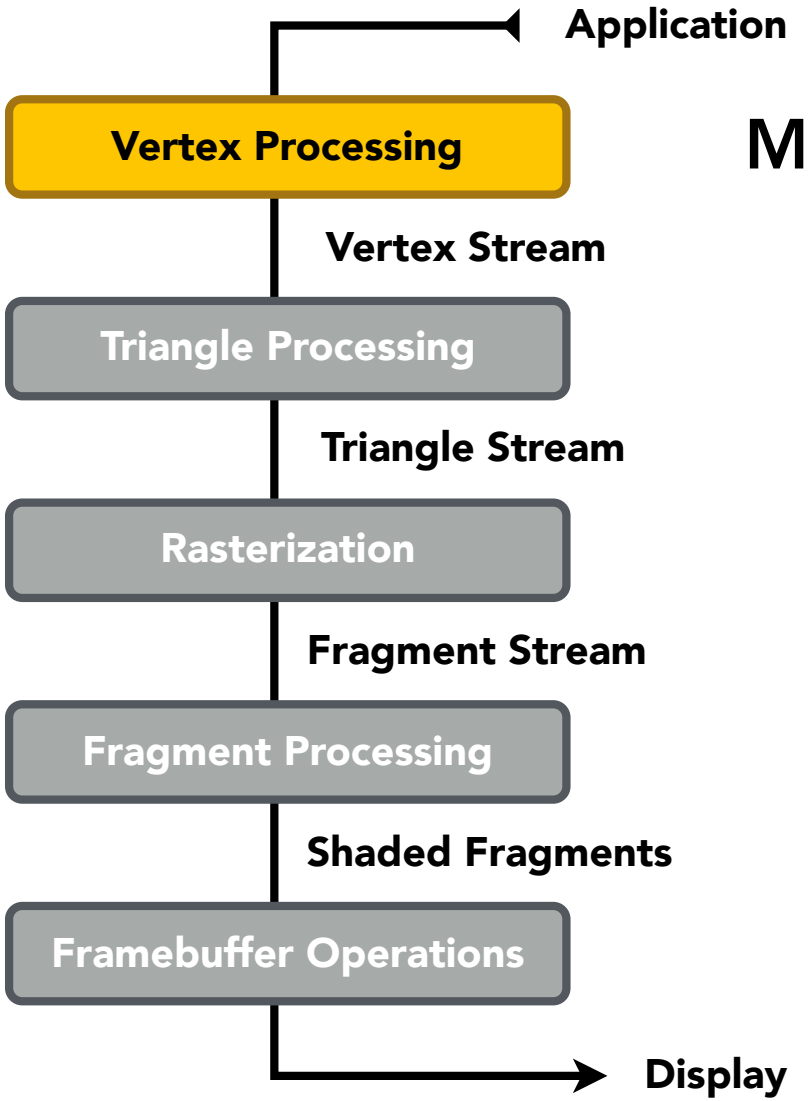


## Cameras & Imaging

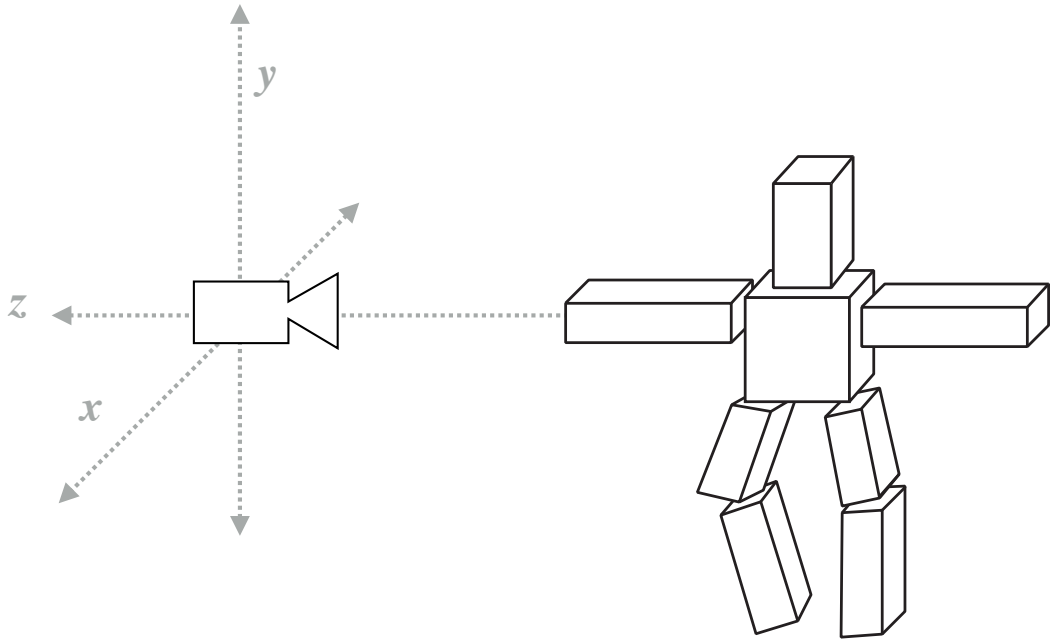
# Rasterization Pipeline



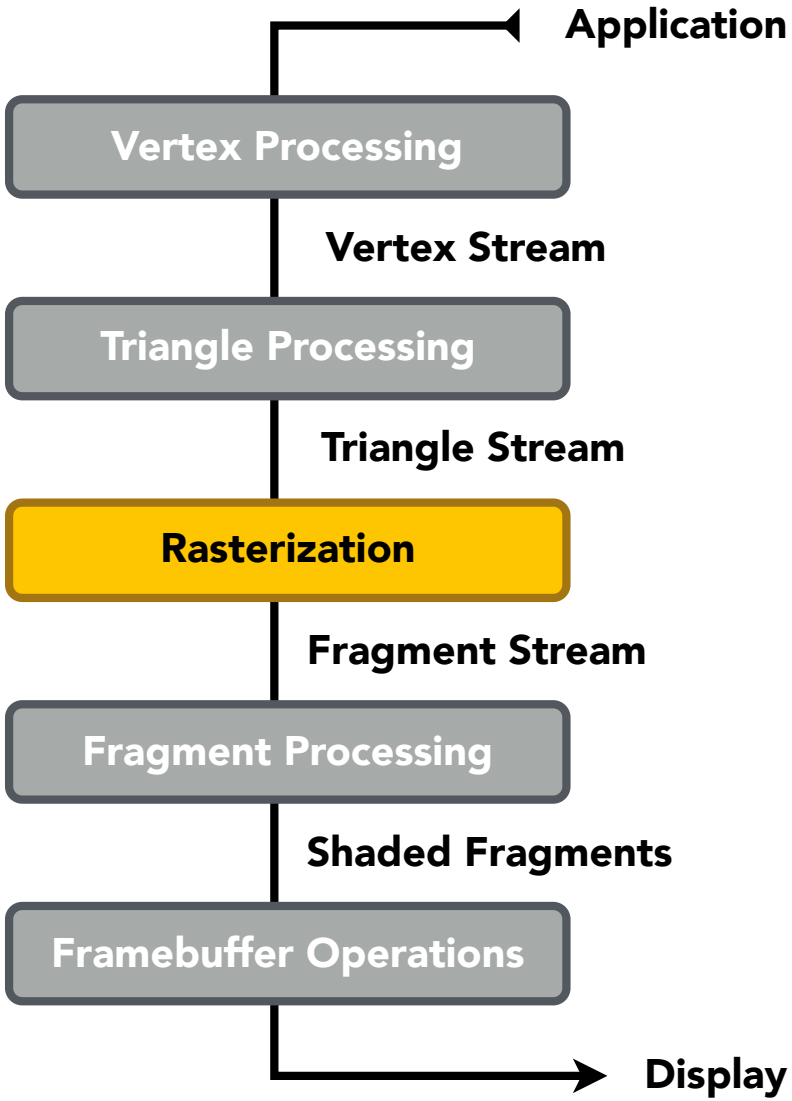
# Rasterization Pipeline



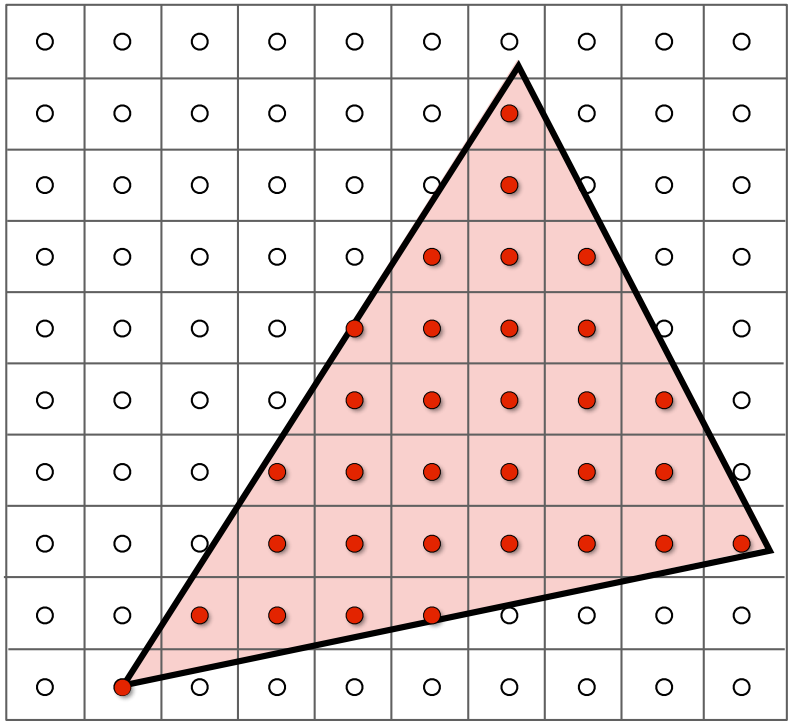

## Model, View, Projection transforms



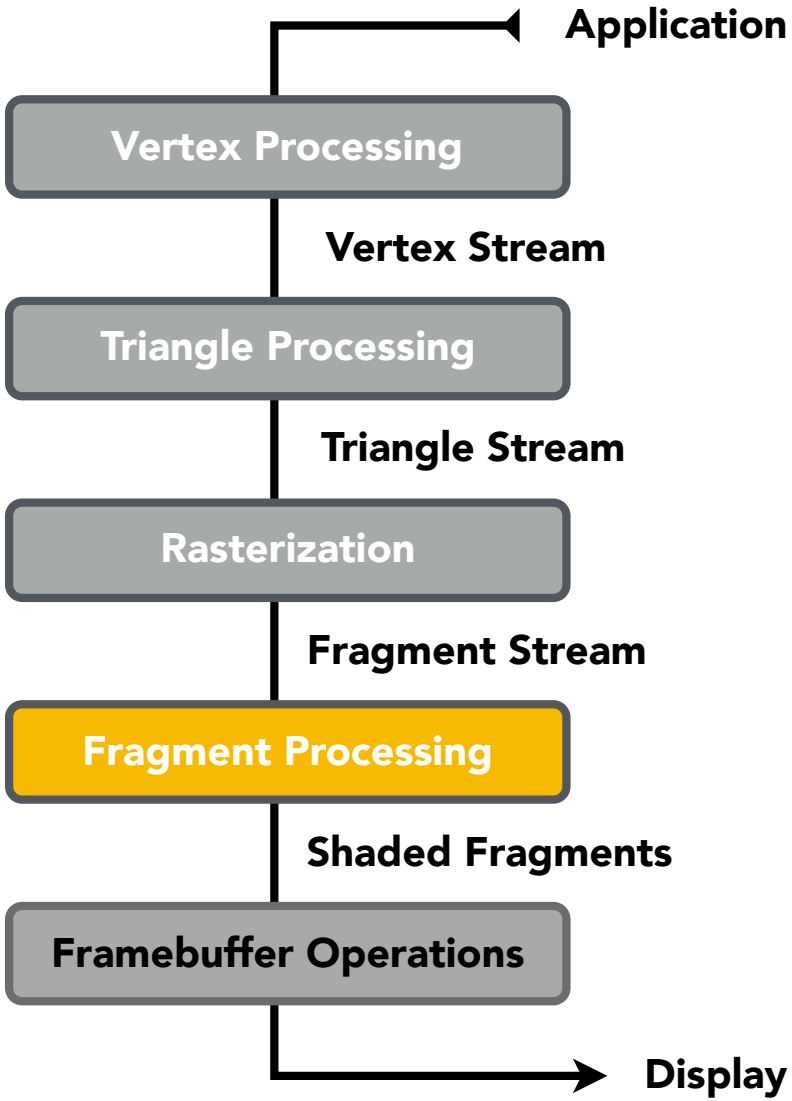
# Rasterization Pipeline




## Sampling triangle coverage

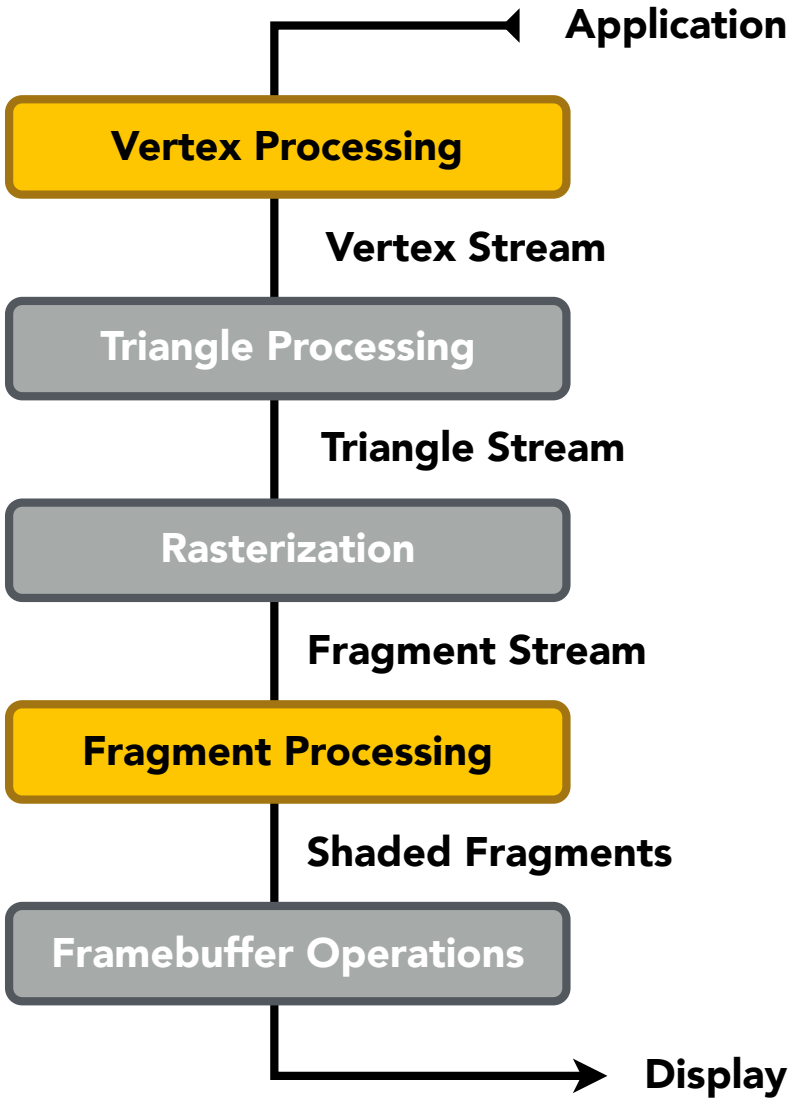


# Rasterization Pipeline

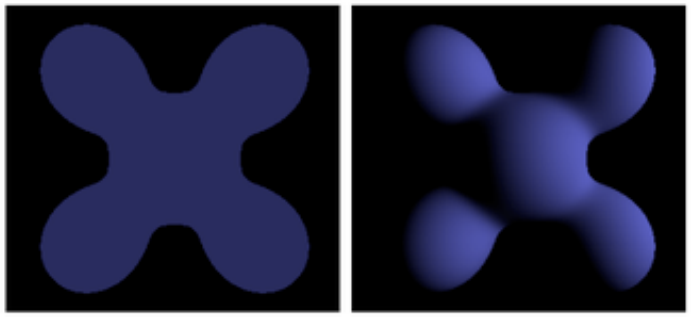



## Z-Buffer Visibility Tests

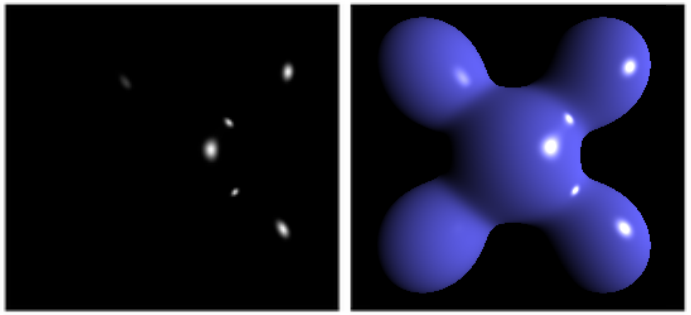
# Rasterization Pipeline




## Shading

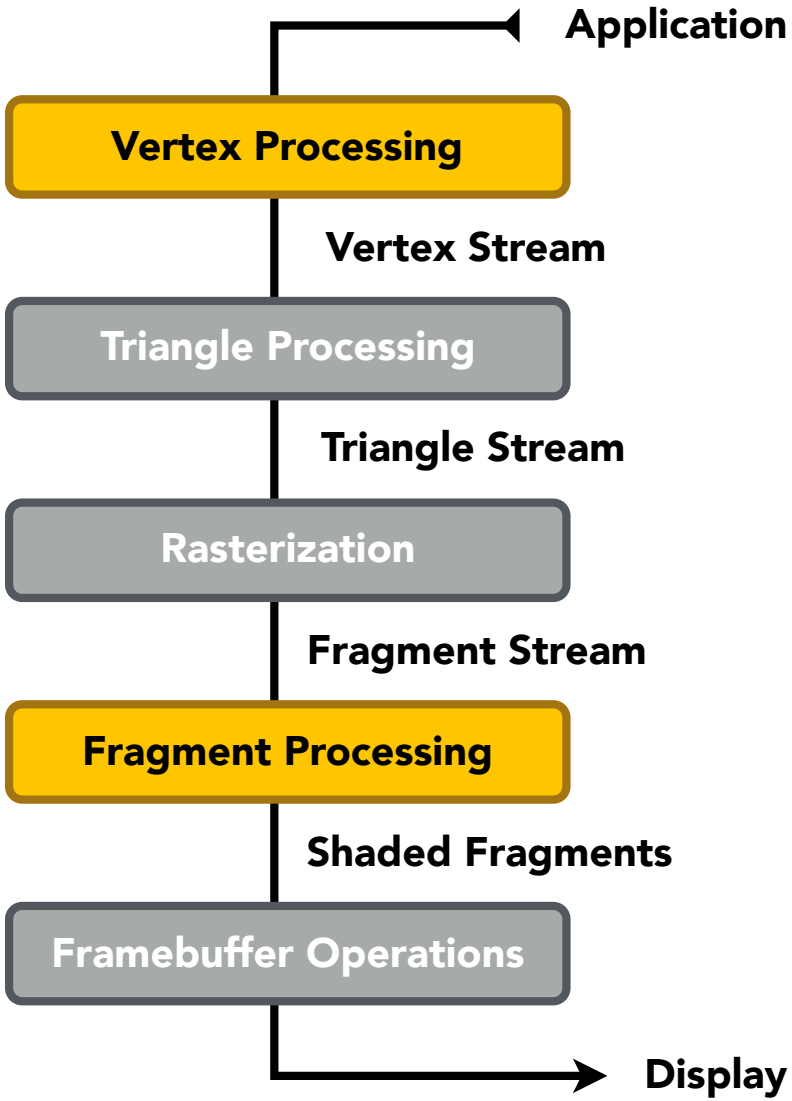


Ambient + Diffuse

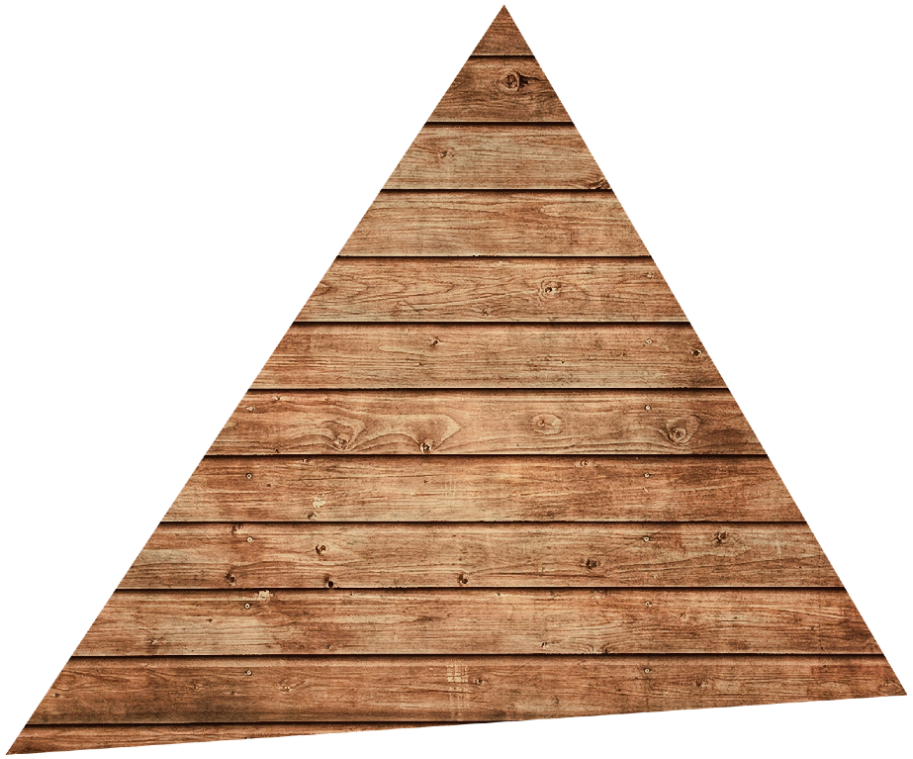


+ Specular = Blinn-Phong Reflectance Model

# Rasterization Pipeline




## Texture Mapping



# Shader Programs

- Shader programs are the programmable parts of the rendering pipeline
- Runs on the GPU (Graphics Processing Unit).
- Written in a shading language (like GLSL, HLSL, or Metal Shading Language)
- Responsible for controlling how vertices and pixels (fragments) are processed during rendering.
    - Vertex Shader: Runs once per vertex. Transforms vertex positions (e.g., from model space → world space → screen space). Passes data (like color, normals, texture coordinates) to the next stage.
    - Fragment Shader (Pixel Shader): Runs once per fragment (potential pixel). Determines the final color, texture, lighting, transparency, etc.

# Shader Programs

- Program vertex and fragment processing stages
- Describe operation on a single vertex (or fragment)

## Example GLSL fragment shader program

```
uniform sampler2D myTexture;
uniform vec3 lightDir;
varying vec2 uv;
varying vec3 norm;

void diffuseShader()
{
  vec3 kd;
  kd = texture2d(myTexture, uv);
  kd *= clamp(dot(–lightDir, norm), 0.0, 1.0);
  gl_FragColor = vec4(kd, 1.0);
}
```

- Shader function executes once per fragment.
- Outputs color of surface at the current fragment's screen sample position.
- This shader performs a texture lookup to obtain the surface's material color at this point, then performs a diffuse lighting calculation.

# Shader Programs

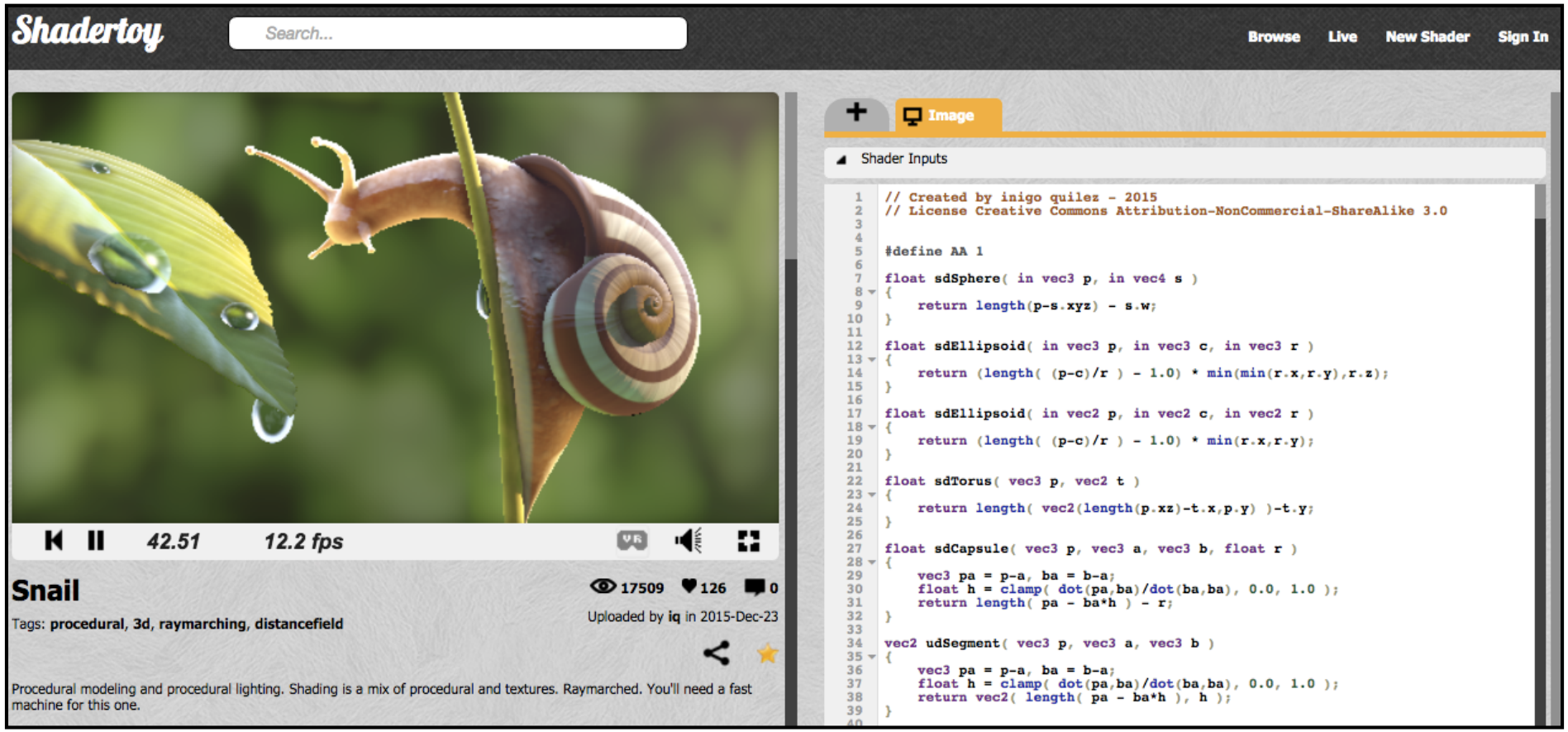
- Program vertex and fragment processing stages
- Describe operation on a single vertex (or fragment)

## Example GLSL fragment shader program

```
uniform sampler2D myTexture;      // program parameter
uniform vec3 lightDir;            // program parameter
varying vec2 uv;                  // per fragment value (interp. by rasterizer)
varying vec3 norm;                // per fragment value (interp. by rasterizer)

void diffuseShader()
{
  vec3 kd;
  kd = texture2d(myTexture, uv);                  // material color from texture
  kd *= clamp(dot(–lightDir, norm), 0.0, 1.0);    // Lambertian shading model
  gl_FragColor = vec4(kd, 1.0);                   // output fragment color
}
```

# Snail Shader Programs




## Inigo Quilez

Procedurally modeled, 800 line shader.

http://shadertoy.com/view/ld3Gz2

# Goal: Highly Complex 3D Scenes in Realtime

**Unreal Engine Kite Demo (Epic Games 2015)**

# Homework 1

- ## 实现一个光栅化器

  本次作业包括6个部分，覆盖了第 2-6 讲的内容。

  - 1. 绘制三角形
  - 2. 基于超采样的反走样算法
  - 3. 几何变换
  - 4. 质心坐标
  - 5. 基于像素采样的纹理映射
  - 6. MipMap纹理映射

- ## 作业完成日期：10.17

# What's More?

What is Texture Mapping?

Applying Textures

Texture Filtering

Advanced Texturing Methods

# Many, Many Uses for Texturing

In modern GPUs, texture = memory + filtering

- General method to bring data to fragment calculations

Many applications

- Environment lighting
- Store microgeometry
- Procedural textures
- Solid modeling
- Volume rendering
- …

# Environment Map 环境光贴图

## A function from the sphere to colors, stored as a texture.

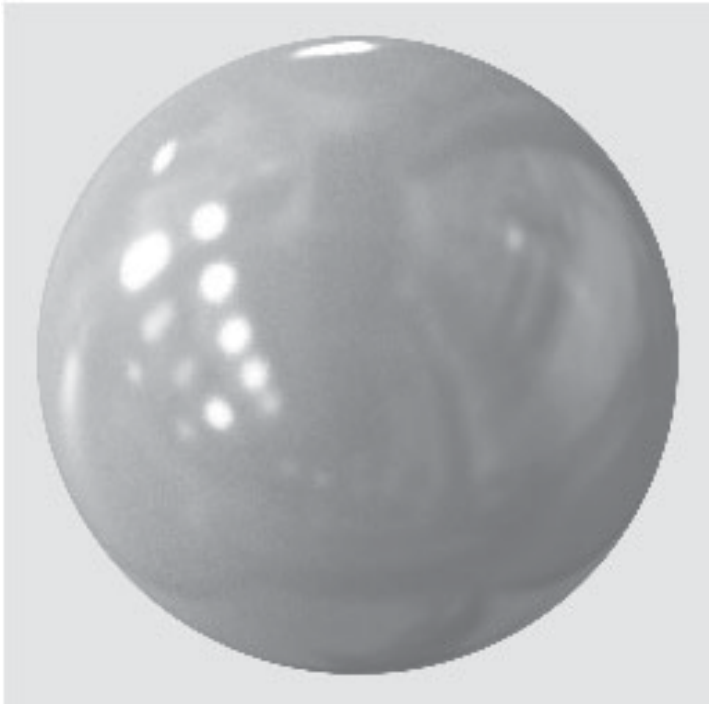
Lat / long texture map

Reflection vector indexes into texture map

[Blinn & Newell 1976]

模型烘焙：提供足够的细节同时保持性能。在游戏和VR中做为模型渲染的替代。

# Environment Map



With environment map

Without environment map

[Dror, Willsky, & Adelson 2004]

# Spherical Environment Map

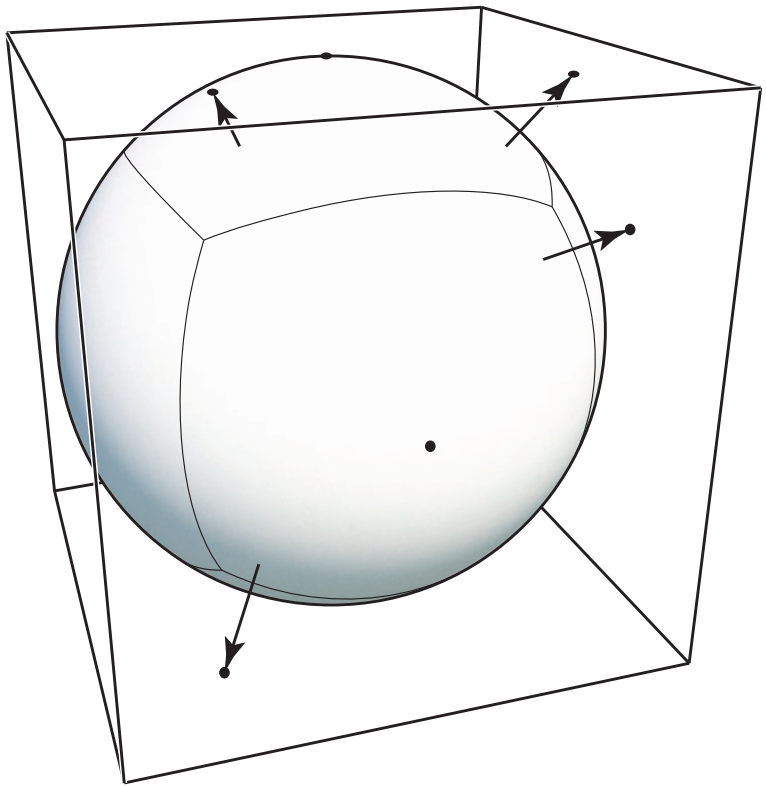
*Hand with Reflecting Sphere*. M. C. Escher, 1935. lithograph

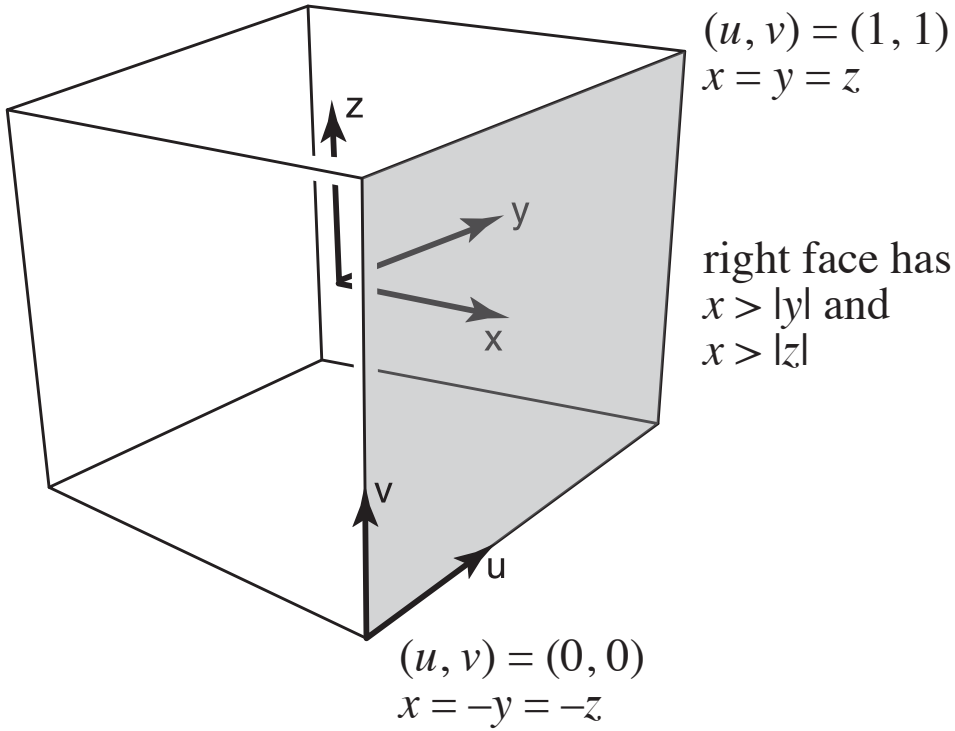
*Light Probes*, Paul Debevec

# Spherical Environment Map

Prone to distortion (top and bottom parts)!

# Cube Map





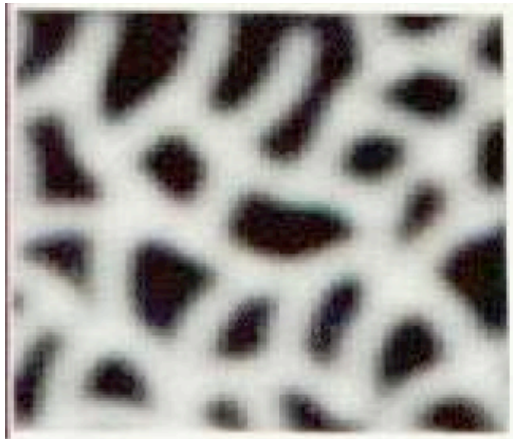

**A vector maps to cube point along that direction.**

**The cube is textured with 6 square texture maps.**

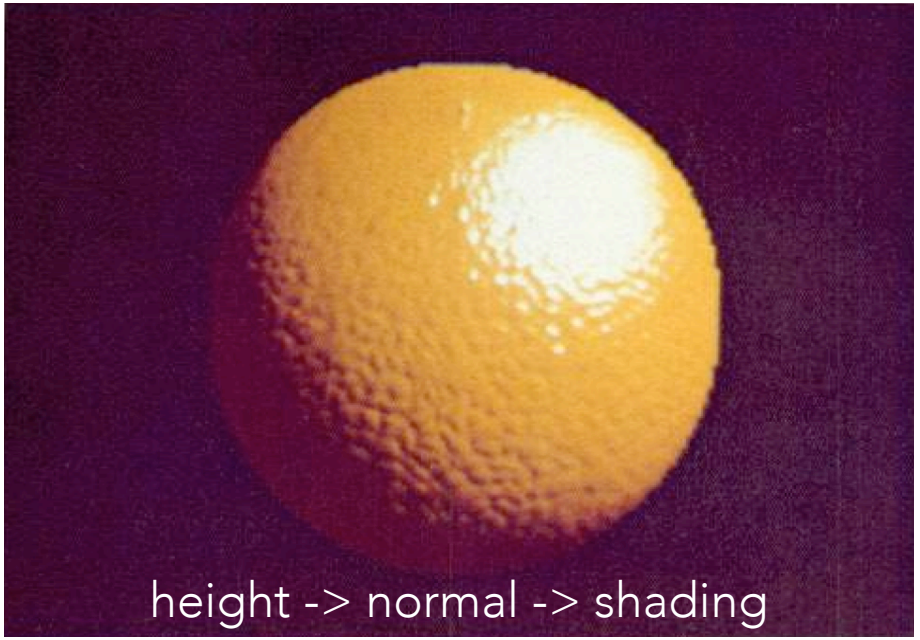
[Emil Persson]

# Textures can affect shading!

- Textures doesn't have to only represent colors
  - What if it stores the height / normal?
  - Bump / normal mapping
  - Fake the detailed geometry



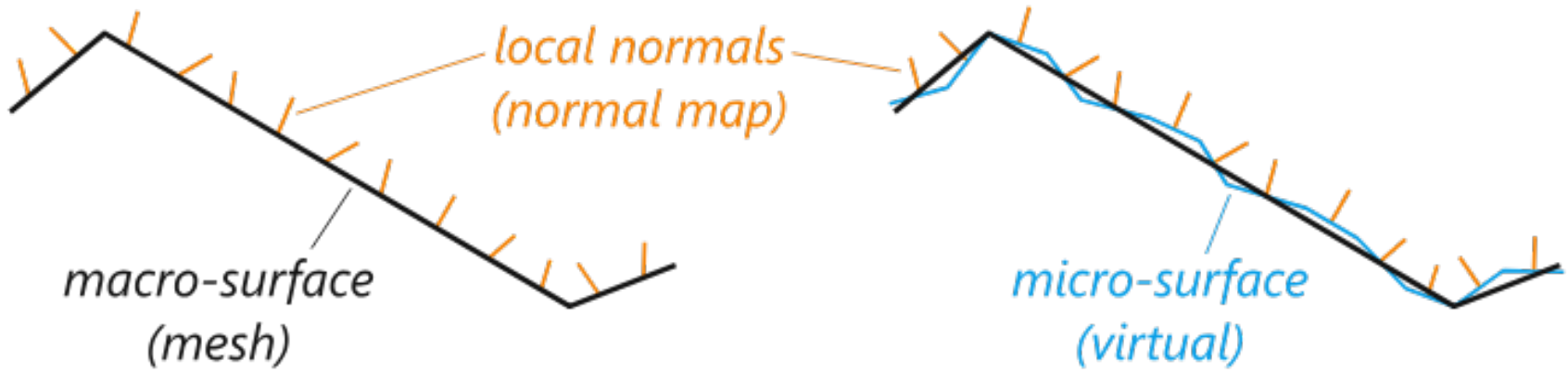
Relative height to the underlying surface



height -> normal -> shading

# Textures can affect shading!

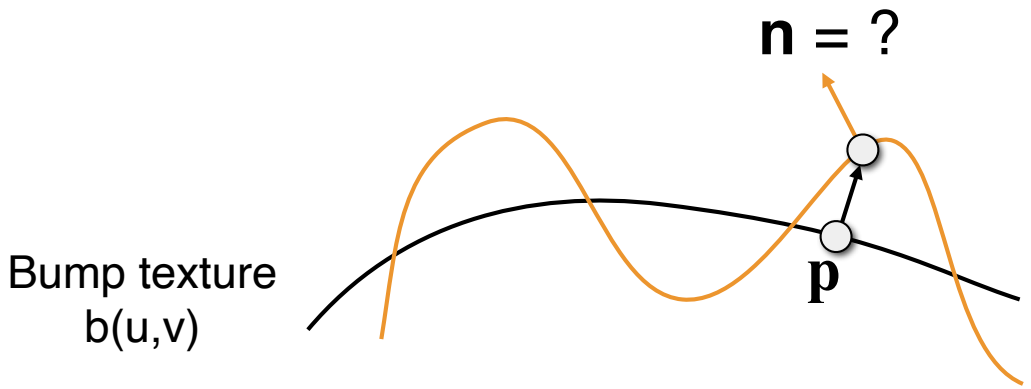
- Textures doesn't have to only represent colors
  - What if it stores the height / normal?
  - Bump / normal mapping
  - Fake the detailed geometry



法向贴图

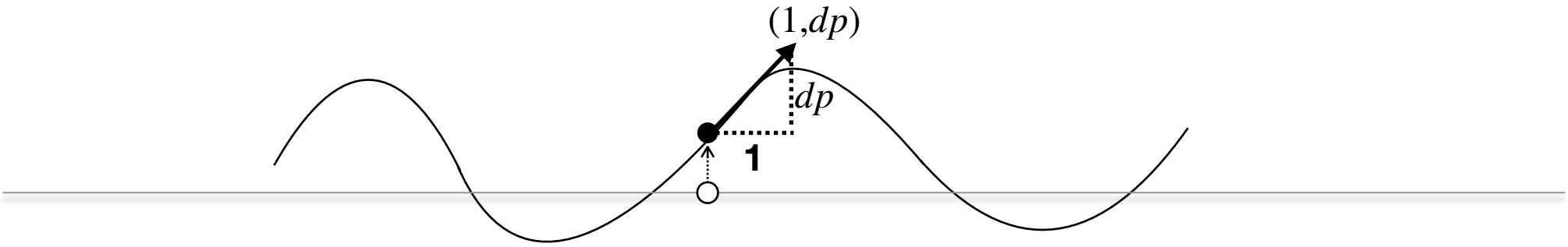
# Bump Mapping (Normal Mapping)

- Adding surface details without adding more triangles
  - Perturb surface normal per pixel (for shading computations only)
  - "Height shift" per texel defined by a texture
  - How to modify normal vector?



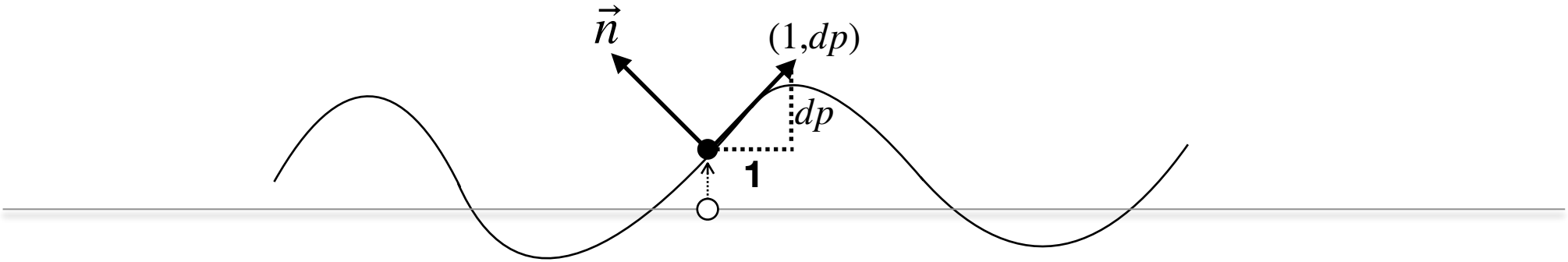
# How to perturb the normal (in flatland)

- Original surface normal $n(p) = (0,1)$
- Derivative at $p$ is $dp = c * [h(p+1) - h(p)]$



# How to perturb the normal (in flatland)

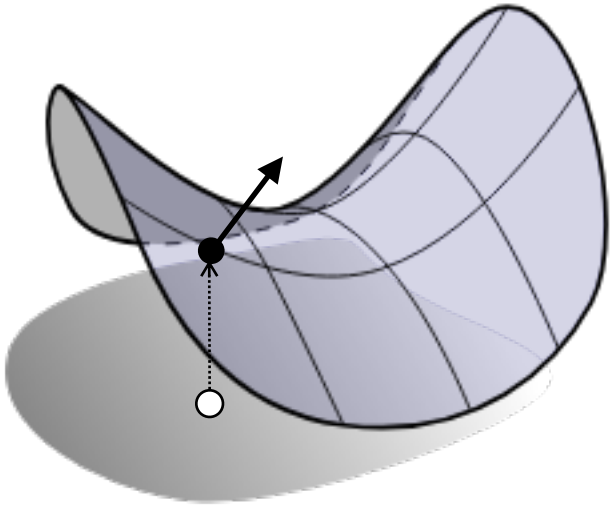
- Original surface normal $n(p) = (0,1)$
- Derivative at $p$ is $dp = c * [h(p+1) - h(p)]$
- Perturbed normal is then $n(p) = (-dp,1)$.normalized()



# How to perturb the normal (in 3D)

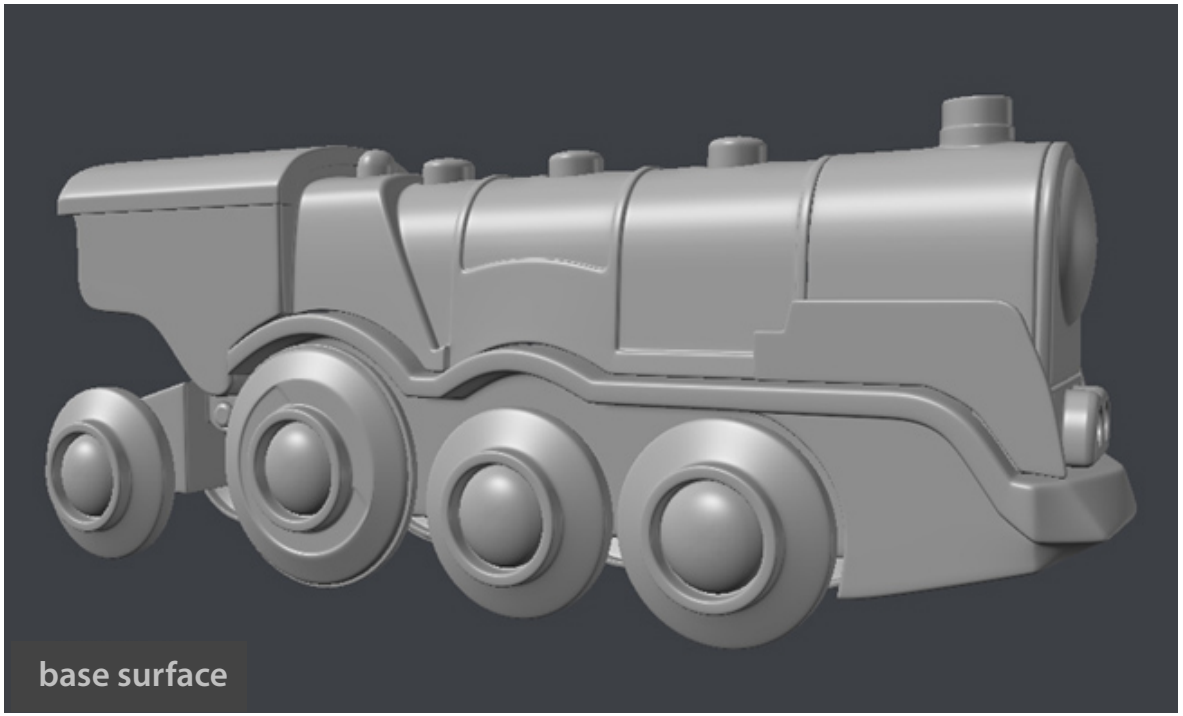
- Original surface normal $n(p) = (0,0,1)$
- Derivative at $p$ are
  - $\frac{dp}{du} = c_1 * [h(u+1) - h(u)]$
  - $\frac{dp}{dv} = c_2 * [h(v+1) - h(v)]$
- Perturbed normal is then

  $n(p) = (-dp/du, -dp/dv, 1)$.normalized()
- Note that this is in local coordinate!

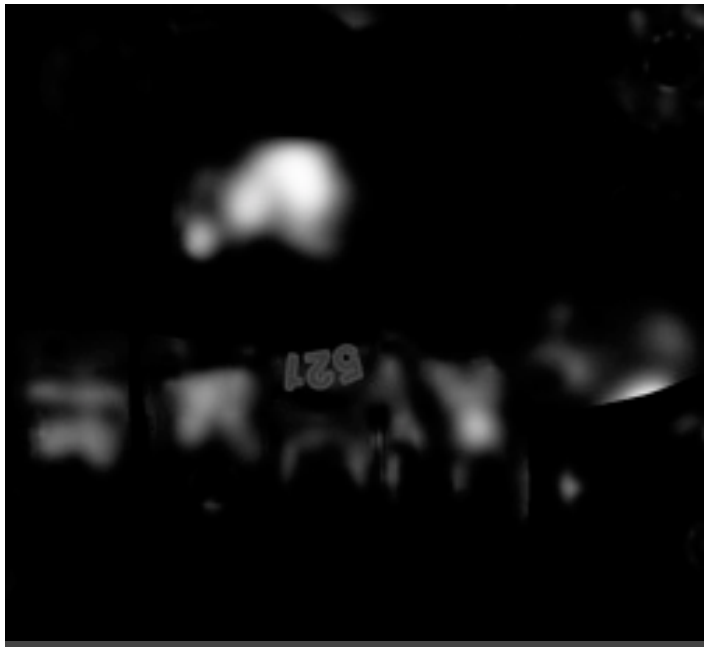


# Displacement Mapping 位移贴图

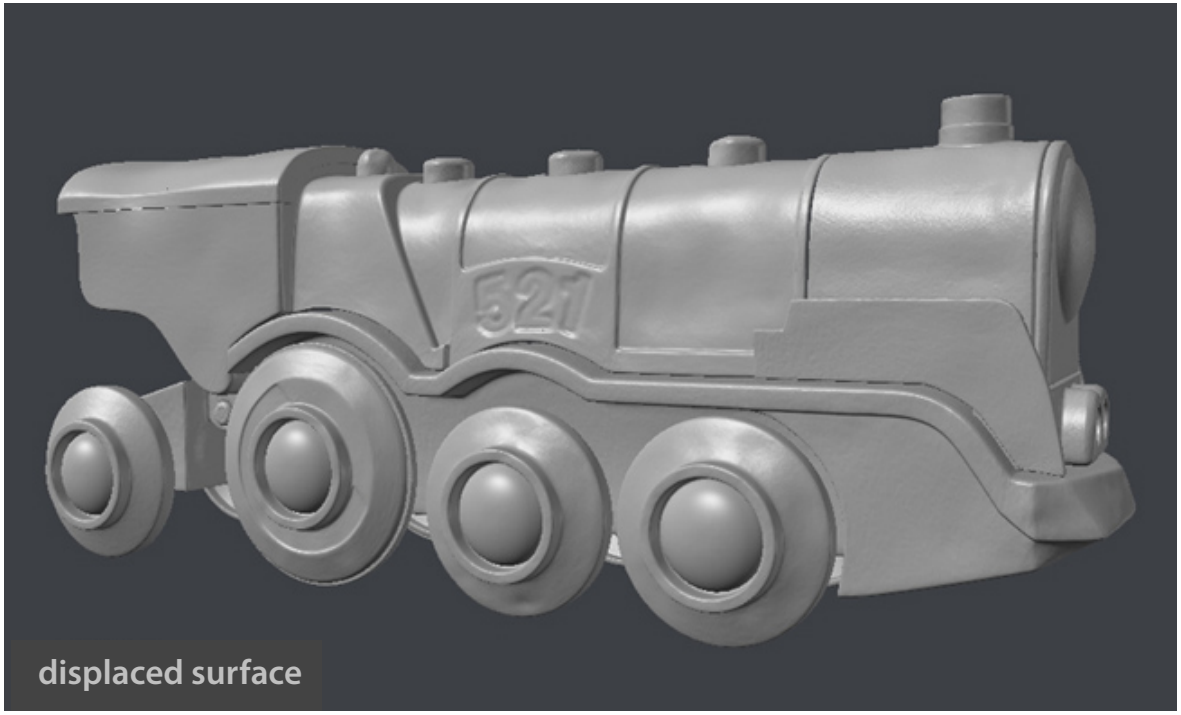
## Texture stores perturbation to surface position



base surface



hand-painted displacement map (detail)

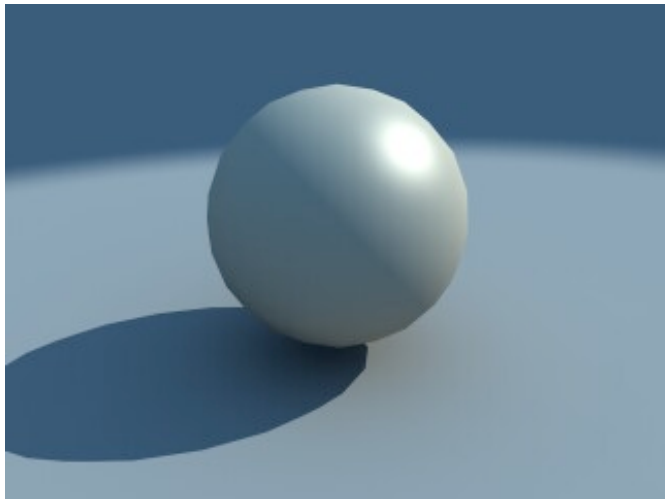



displaced surface

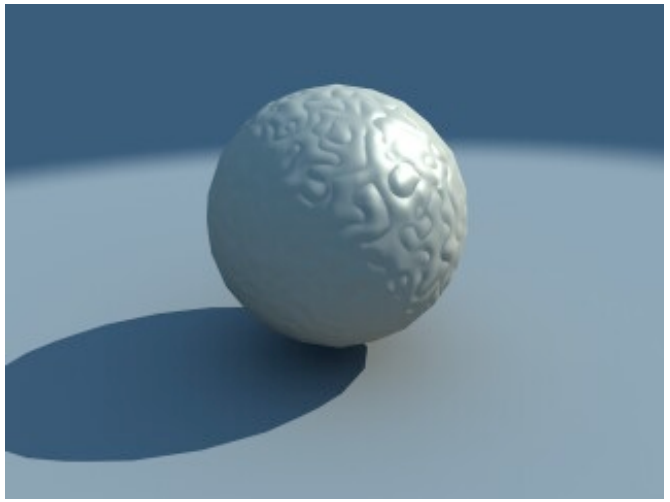
**Paweł Filip**
**tolas.wordpress.com**

# Textures can affect shading!

- Displacement mapping — a more advanced approach
  - Uses the same texture as in bumping mapping
  - Actually moves the vertices

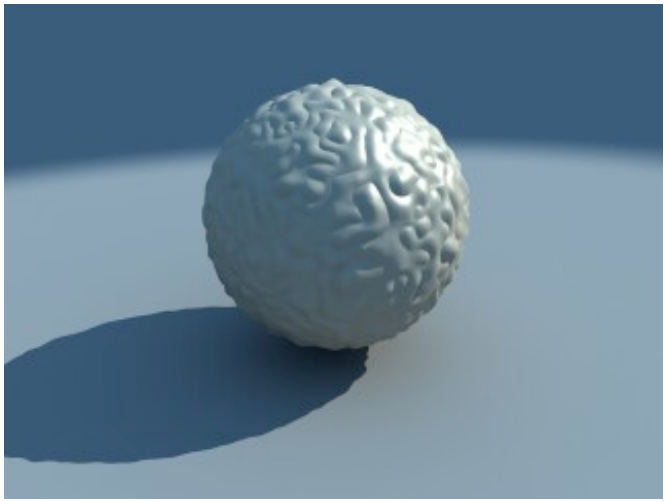


Geometry



Bump mapping

Perturbs normals



Displacement mapping

Perturbs positions

# Provide Precomputed Shading

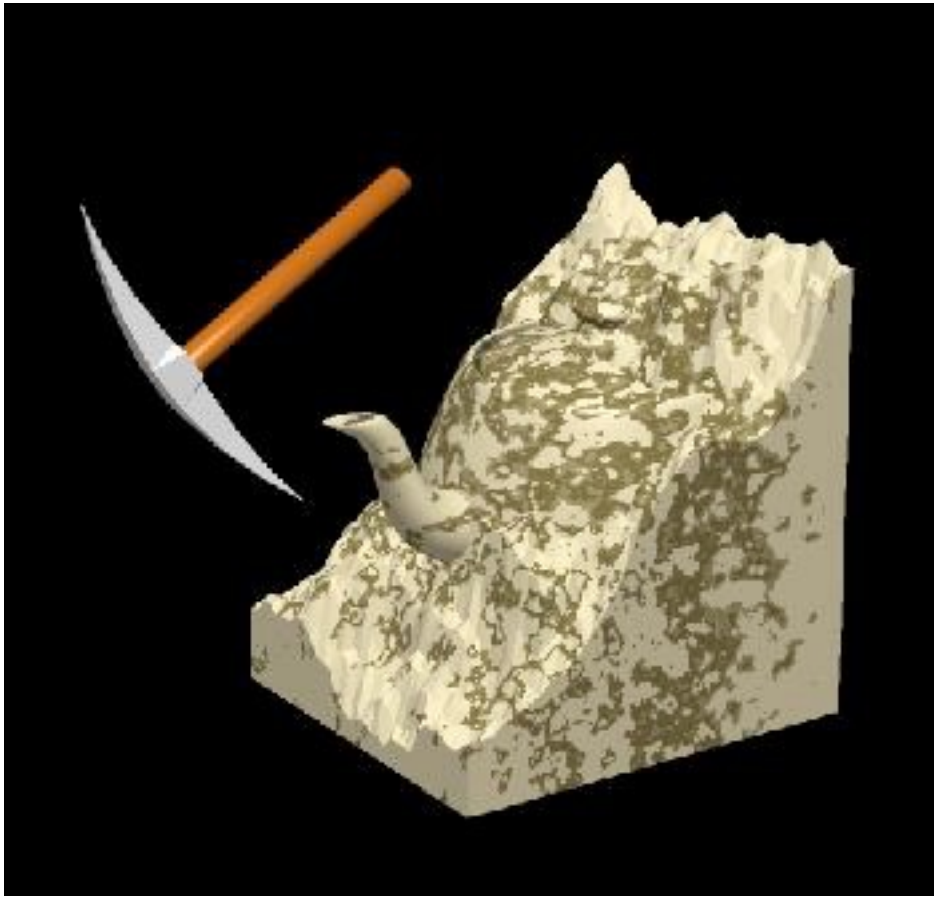
Simple shading

Ambient occlusion texture map

With ambient occlusion

Autodesk

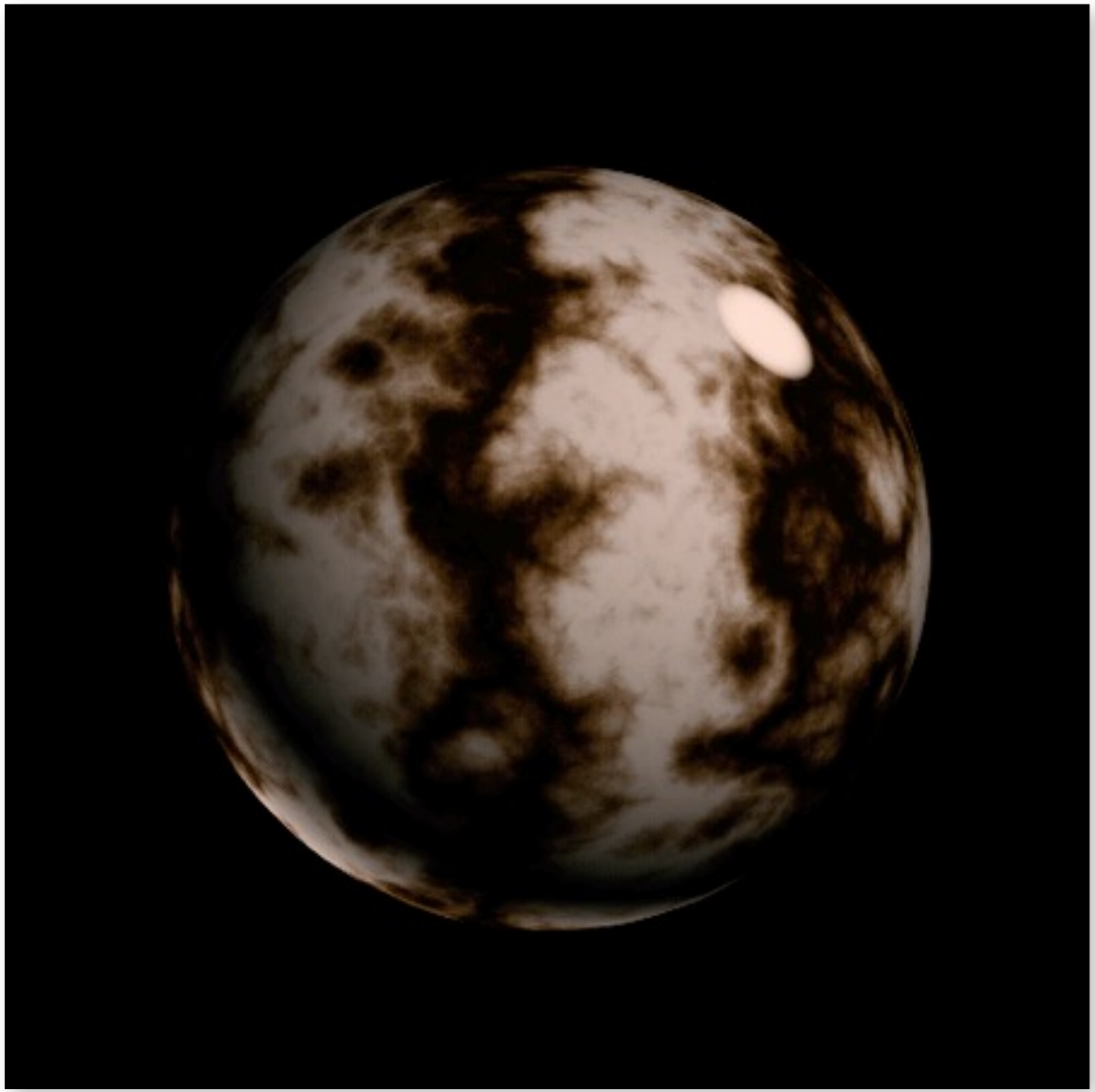
# 3D Textures

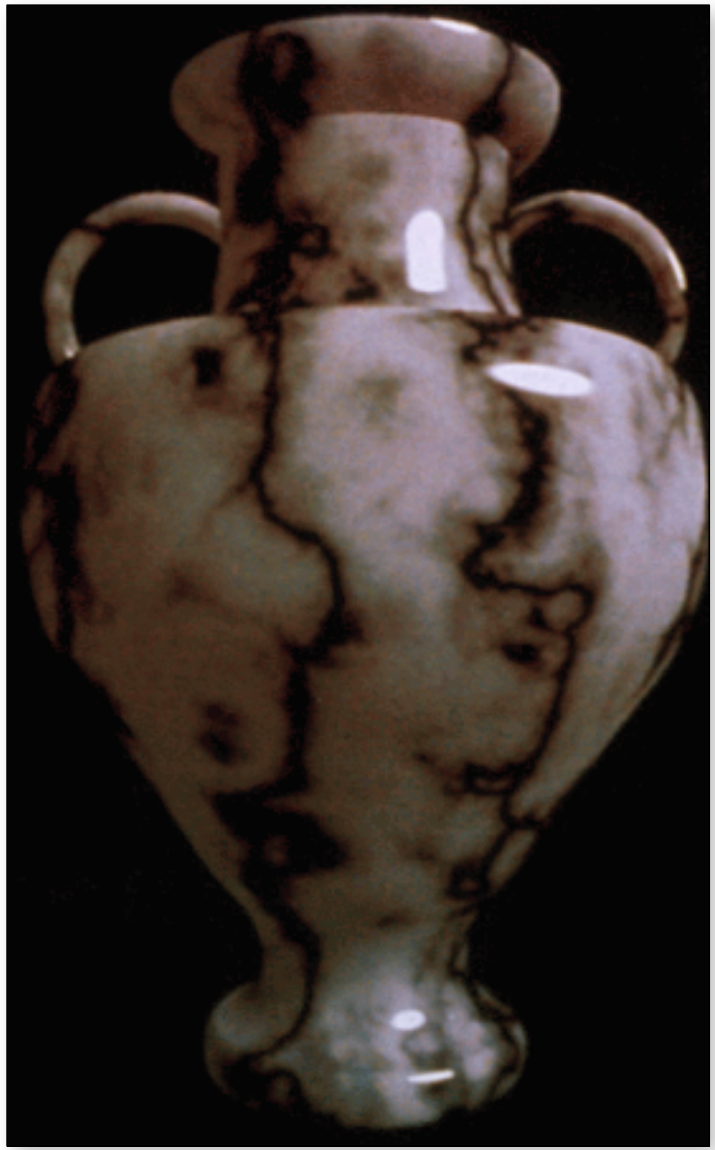


[Wolfe / SG97 Slide set]

Textures is a function of (u,v,w). Solid modeling.

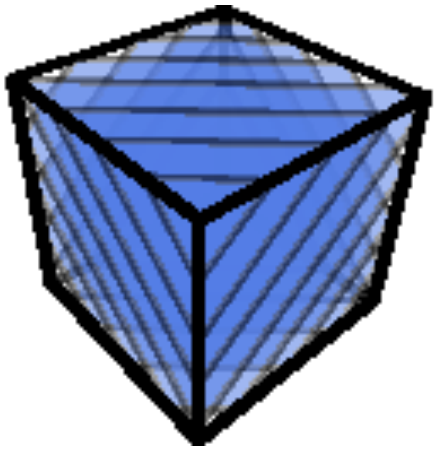
# 3D Procedural Noise + Solid Modeling

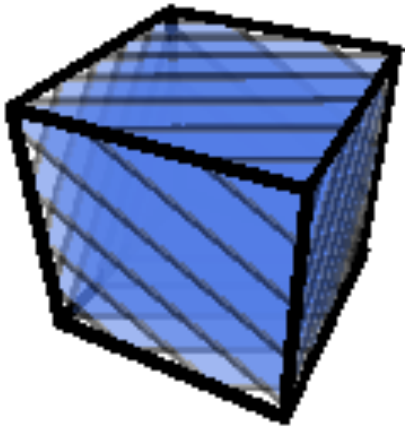




Perlin noise, Ken Perlin

# 3D Textures and Volume Rendering





Marc Levoy